# Two Rankings from One Posterior: A Bayesian Power Rating and a Wins-Above-Benchmark Resume for College Football

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**Abstract**

A college football ranking is asked two questions: which team is better, and which has earned more. The first is predictive and should use margin of victory; the second is retrodictive and should reward only results. We answer both from one Bayesian model. Team strength, in points on a neutral field, is observed through the regulation margin and an efficiency margin built from expected points added, modelled jointly so that one performance is not counted twice, pooled within conferences and divisional tiers, carried over between seasons, and allowed to drift from week to week as a random walk. The shape of the margin distribution is chosen among four families by leave-one-out cross-validation, and a parallel model for the total score splits strength into offence and defence. The power rating is the posterior of strength. The resume, Wins Above Benchmark, counts the wins a team has beyond those the twelfth-best team would expect against the same schedule, computed per posterior draw. We evaluate the system on every week of every season since 2014 with convergence gates, posterior predictive and prior-sensitivity checks, simulation-based calibration, ablations, and a rolling out-of-sample backtest against results, closing betting lines and totals, and baseline ratings.

# Table of contents


**1 Introduction** **4**
- 1.1 Notation . . . 6

**2 Data** **6**

**3 The Power Model** **8**
- 3.1 Likelihood . . . 8
- 3.2 Hierarchy, Preseason Information and the Weekly Walk . . . 10
- 3.3 Totals, Offence and Defence . . . 15
- 3.4 Priors . . . 16

**4 What the Efficiency Signal Adds** **18**

**5 The Resume** **20**
- 5.1 Wins Above Benchmark . . . 20
- 5.2 Strength of Record and Ranks . . . 21
- 5.3 Record-only Reference: the Colley Matrix . . . 22
- 5.4 Choosing the Benchmark . . . 23
- 5.5 How Margin Reaches the Resume . . . 24

**6 Computation and Pipeline** **24**
- 6.1 Sampling and Gates . . . 24
- 6.2 The Pipeline . . . 25

**7 Model Checking** **27**
- 7.1 Posterior Predictive Checks . . . 27
- 7.2 Prior Sensitivity . . . 28
- 7.3 Leave-one-out Cross-validation . . . 29
- 7.4 Choosing the Margin Family . . . 29

**8 Out-of-sample Evaluation** **32**
- 8.1 Design . . . 32
- 8.2 Accuracy . . . 32
- 8.3 Calibration . . . 34
- 8.4 Ablations and Baselines . . . 36
- 8.5 Priors Set Without the Season . . . 37
- 8.6 Preseason Ratings . . . 37
- 8.7 Tuned Choices . . . 38

**9 The 2025 Season** **38**
- 9.1 The 2025 Playoff Field . . . 45
- 9.2 The Resume Against the Committee . . . 47

**10 Discussion** **47**

**11 Appendix: Supplementary Posterior Checks** **48**
11.1 Sampler Behaviour Across All Snapshots . . . . . 49
11.2 Sampler Diagnostics . . . . . 49
11.3 Convergence . . . . . 50
11.4 Posterior Geometry . . . . . 58
11.5 Prior and Posterior . . . . . 59
11.6 Justification of the Informative Priors . . . . . 61
11.7 Induced Priors . . . . . 63
11.8 Posterior Predictive Checks . . . . . 65
11.9 Influence and PSIS Diagnostics . . . . . 68
11.10 Power-scaling Sensitivity . . . . . 69
11.11 Sensitivity to the Prior: Perturbation Refits . . . . . 70
11.12 Simulation-based Calibration . . . . . 73

**12 Appendix: The Stan Program** **74**

**13 Appendix: Reproducibility** **81**

**References** **82**

# 1 Introduction

Every autumn, some one hundred and thirty teams in the Football Bowl Subdivision (FBS) play about twelve games each, against opponents they largely do not choose, and at the end a committee selects twelve of them for a playoff. The schedule graph is sparse: a team plays fewer than a tenth of the other teams in its division, most of them from its own conference, and conferences are linked by a few dozen non-conference games played in September. Any ranking has to propagate information about strength through that graph, and the history of the problem is largely the history of ways to do so: least squares on scoreboard margin [1, 2], paired-comparison models for wins and losses [3, 4], eigenvector methods [5], and linear-algebraic record-only schemes such as the Colley matrix [6], one of the computer components of the Bowl Championship Series era [7, 8].

These methods disagree partly because they answer different questions. A *predictive* rating asks which team would win a game played next week, and by how much. Margin of victory is informative about that, so a predictive rating should use it, together with any other signal of how well a team played. A *retrodictive* ranking asks what a team has earned by its results. Here margin is usually excluded on purpose, so that a team is neither rewarded for running up a score nor penalised for resting its starters, and a win over a strong opponent is worth more than a win over a weak one. Combining the two into one number, as weighted composites of standardised scores do, gives a quantity that answers neither question and whose weights cannot be validated against anything observable.

This paper keeps the two apart but derives both from one posterior distribution. The contributions are four.

1. A **power model** (Section 3) for team strength in points on a neutral field, observed through the regulation scoreboard margin and through an efficiency margin built from expected points added per play [9]. The margin is modelled *conditionally* on the efficiency residual, so the two signals share one latent strength without being treated as independent evidence, and their relative weight is estimated rather than set. Strength is pooled within conferences and within three divisional tiers, carried over from the previous season by an estimated coefficient, and moves through the season as a random walk whose step size is estimated. The shape of the margin distribution is one of four families, chosen by cross-validation on every complete season (Section 7.4); the published one is the Student-$t$. A parallel model for the total score, observed through the final score and summed offensive efficiency, splits each rating into offence and defence without changing it (Section 3.3).
2. A result on how much the efficiency signal adds (Section 4). Under a Gaussian version of the model the gain in per-game Fisher information about strength has a closed form, and at the fitted parameters it is 0.4–5.1% — small, and explained by the estimated scale of efficiency being close to the value at which the gain vanishes. An ablation backtest measures the same thing out of sample (Section 8.4).
3. A **resume** measure, Wins Above Benchmark (Section 5): for each posterior draw, the number of wins a team has beyond those the draw's twelfth-best FBS team would be expected to earn against the same opponents at the same venues. It is additive over games, never rewards a loss, is centred at zero for a team exactly as strong as the benchmark, and inherits

uncertainty in the opponents, the benchmark and the ranks directly from the posterior.

4. A **validation** of the whole system (Section 6, Section 7, Section 8) on every week of every season from 2014 to 2026: 226 weekly posteriors fitted by the No-U-Turn sampler, of which 226 pass the convergence gates; posterior predictive checks, which reproduce the spread, the blowouts and home-field advantage but miss the share of one-score games in 13 of 13 seasons; prior-sensitivity diagnostics, which flag 23 of 26 structural parameters in at least one season; simulation-based calibration; ablations and baselines; and a rolling backtest of 18,958 games, each predicted from the posterior fitted through the week before, scored against the result and the closing betting line.

Figure 1 shows how these pieces fit together. Section 2 describes the data, Section 3 and Section 4 the power model, and Section 5 the resume and the reference rankings. Section 6 describes fitting and the pipeline, Section 7 model checking, Section 8 the out-of-sample evaluation, Section 9 the 2025 season as a worked example, and Section 10 the limitations.

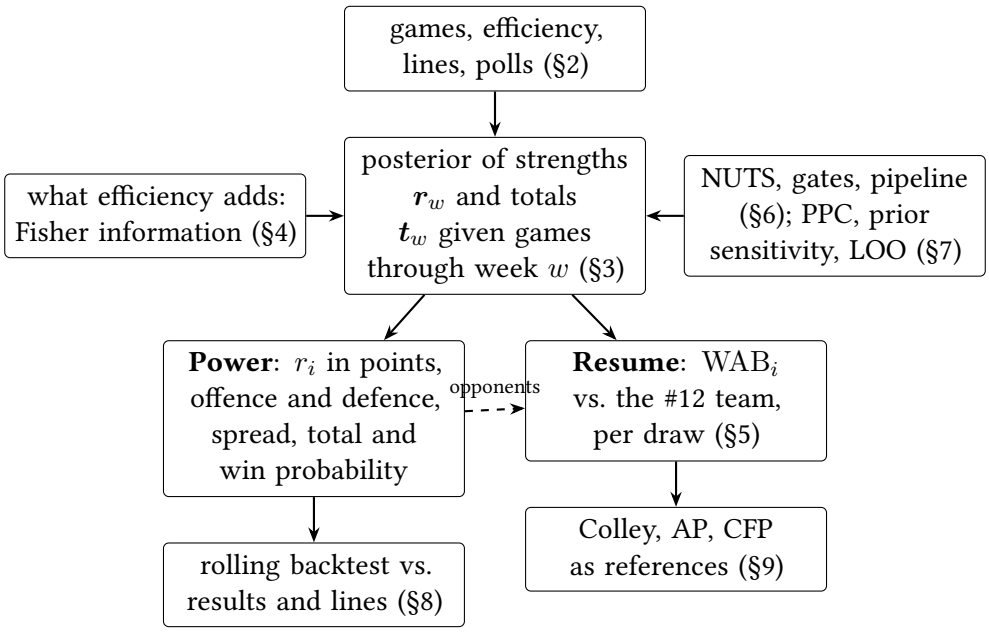



Figure 1: Structure of the paper. One posterior per week feeds both products; the dashed arrow is the only link between them: the Resume values results using opponent and benchmark strengths drawn from the Power posterior, but never uses margin. Each box names the section where it is developed.

**Related work.** The margin model descends from Harville [1] and Harville [10], who fit mixed linear models to scoreboard margin with random team effects, and from Stern [11], who showed that National Football League margins are close to normal around the betting line with a standard deviation near fourteen points, so that a spread determines a win probability. Glickman and Stern [12] made team strengths evolve over time in a Bayesian state-space model; the weekly random walk here is a version of theirs. Using the $t$ distribution for robustness to blowouts follows Lange et al. [13]. On the retrodictive side, Mease [4] and Colley [6] give record-only rankings; Stern [7] discusses why the Bowl Championship Series excluded margin from its computer rankings,

and Lopez et al. [14] quantify how much of a season's outcome is noise, which is the reason a resume should report uncertainty. Dynamic paired-comparison models [15, 16] let strengths drift between games, and Annis and Craig [17] combine win–loss and margin information for college football specifically. Among published football ratings, Elo systems [18, 19] update a rating after each game, SP+ [20] and ESPN's FPI [21] are proprietary efficiency-based power ratings, and Massey's composite [22] aggregates about a hundred computer rankings. Wins Above Benchmark is the football analogue of the "wins above bubble" measure on the team sheets of the college basketball selection committee [23], with the benchmark placed at the playoff boundary. The workflow follows Gelman et al. [24].

### 1.1 Notation

Table 1 collects the notation. Games are indexed by $g$, teams by $i$ and $j$, weeks by $w$, conferences by $c$ and divisional tiers by $\ell \in \{1, 2, 3\}$ for FBS, the Football Championship Subdivision (FCS) and all lower divisions. Game $g$ is played in week $w(g)$. Posterior draws are indexed by $s = 1, \ldots, S$, and a superscript $(s)$ marks a quantity evaluated at draw $s$. A team's rating without a week index, $r_i$, is its rating at the week of the snapshot being published. The code keeps its own variable names; Table 1 gives both, and the total model of Section 3.3 has its own, listed there.

## 2 Data

All data come from the CollegeFootballData.com API [25]: every game involving an FBS or FCS team from 2014 to 2026, with line scores, venue, divisional classification and conference of each team; per-game advanced statistics; closing betting lines (the spread and the over/under); and the AP, Coaches and College Football Playoff (CFP) polls. Teams are keyed by the provider's identifiers throughout. The following derived quantities enter the model.

**Regulation margin.** $y_g$ is the home score minus the away score after four quarters, computed from the line scores. A game decided in overtime (4.0% of games) therefore contributes a margin of zero. This is deliberate: overtime is a different game, with possession alternating from the opponent's 25-yard line, and its points say little about strength. The win still counts as a win in the resume.

**Efficiency margin.** For each side the provider reports the mean predicted points added (PPA) per play and the number of plays, both with garbage time excluded. PPA is the change in expected points across a play, where expected points are estimated from down, distance and field position [9]. We use the net total,

$$e_g = \overline{\mathrm{PPA}}_{h(g)}\, n_{h(g)} - \overline{\mathrm{PPA}}_{a(g)}\, n_{a(g)},$$

which is on the scale of points. It measures how well each offence moved the ball regardless of how drives ended, so it is less exposed than the scoreboard to turnovers returned for touchdowns, missed field goals and fourth-down variance. It is used only when both offences have at least 20 non-garbage plays, and is missing otherwise — in practice for most games against lower-division opponents, which the provider does not chart.

**Total.** For the total model (Section 3.3), $x_g$ is the final combined score and $e_g^{+} = \overline{\mathrm{PPA}}_{h(g)}\, n_{h(g)} +$

Table 1: Notation. The last column names the corresponding variable in `stan/power.stan` or `R/03_rankings.R`.

| Symbol | Meaning | Code |
|---|---|---|
| $h(g), a(g)$ | home and away team of game $g$ | `home`, `away` |
| $\delta_g$ | 1 for a true home game, 0 at a neutral site | `home_adv` |
| $y_g$ | regulation scoreboard margin, home minus away | `margin` |
| $e_g$ | efficiency margin: net expected points added | `eff` |
| $r_{i,w}$ | strength of team $i$ in week $w$, points on a neutral field | `r_state`, `r` |
| $r_i^0$ | preseason (week-1) level of team $i$ | `r0` |
| $\phi_r$ | weekly random-walk step sd of strength | `tau_r` |
| $\eta$ | home-field advantage, points | `hfa` |
| $\mu_g$ | $r_{h(g),w(g)} - r_{a(g),w(g)} + \eta\delta_g$, expected margin | `mu` |
| $\lambda, \sigma_e$ | scale and noise of the efficiency margin | `lambda`, `sigma_eff` |
| $\kappa, \sigma_m$ | loading and scale of the margin | `kappa`, `sigma_m` |
| $\nu, \gamma, p_w, \sigma_w$ | shape parameters of the margin family | `nu`, `gamma_m`, `p_wide`, `sigm` |
| $\bar{\sigma}$ | $(\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2}$, marginal margin scale | `sigma_marg` |
| $F(\cdot \mid \mu)$ | predictive distribution function of the margin at expected margin $\mu$ | `margin_cdf` |
| $m_\ell, \omega$ | tier mean ($m_1 = 0$) and conference spread | `mu_tier`, `sigma_conf` |
| $\alpha_c$ | mean strength of conference $c$ | `mu_conf` |
| $\tau_\ell$ | within-conference spread of tier $\ell$ | `sigma_team` |
| $d_i, s_i, \beta$ | last season's relative rating, its posterior sd, and the carry-over | `prev_dev`, `prev_sd`, `b_pre` |
| $B^{(s)}$ | strength of the benchmark team in draw $s$ | `bench` |
| $\pi_g$ | probability that the benchmark wins game $g$'s fixture | `p_bench` |
| $R_i^{(s)}$ | rank of team $i$ within its tier in draw $s$ | `rank_draws` |

$\overline{\mathrm{PPA}}_{a(g)}\, n_{a(g)}$ the summed offensive efficiency, available exactly when $e_g$ is. The total of a game decided in overtime is inflated by the extra possessions, so for those games $x_g$ is treated as missing and only $e_g^+$ is used.

**Tiers and conferences.** Each team is assigned to FBS, FCS or a single pooled tier for Division II, III and NAIA opponents, and to its conference in that season. An independent is not a conference, so each independent (8 FBS programmes over the period, among them Notre Dame, Army, BYU, Massachusetts) gets a conference effect of its own, drawn from its tier like any conference's; the lower-division opponents share one.

**Missing efficiency.** The efficiency margin is missing for 32% of games, almost all of them involving an FCS or lower-division team that the provider does not chart. Missingness therefore depends on the tiers of the two teams, which are observed, and not on how the game went, so the data are missing at random in the sense of Rubin [26] and the likelihood of the observed data (Equation 2 for these games) is the right one to use without modelling why efficiency is missing.

**Weeks.** Postseason games are relabelled as one week after the last regular-season week. In the season being played, a week counts as concluded once at least 90% of its games are final and six hours have passed since its last kickoff; in a finished season, every week with at least one result counts, since the remaining games were cancelled. Only a contiguous prefix of weeks can be concluded. A posterior fitted "through week $w$" uses exactly the games of concluded weeks $\leq w$.

**The 2020 season.** The 2020 season was shortened and partly cancelled, and most FCS conferences moved their season to the spring of 2021. Those 257 spring games belong to the 2020 season in the provider's data but are numbered from week 1 again; left as they are, a posterior "through week 3" of 2020 would contain games played five months later. They are therefore placed after the fall postseason in date order, as spring weeks 1 to 11 followed by a spring postseason, and fitted and backtested like any other weeks.

Table 2 summarises the data.

* In progress: through week 5.

# 3 The Power Model

## 3.1 Likelihood

For game $g$ let

$$\mu_g = r_{h(g),w(g)} - r_{a(g),w(g)} + \eta\, \delta_g$$

be the expected regulation margin: the difference between the two teams' strengths in the week of the game, plus the home-field advantage when the game is not at a neutral site. Both observed signals depend on the teams only through $\mu_g$. How strengths move from week to week is the subject of Section 3.2.

*Power model likelihood.* For a game with an efficiency margin,

$$e_g \mid \mu_g \sim \mathcal{N}\big(\lambda\mu_g,\ \sigma_e^2\big), \qquad y_g \mid e_g, \mu_g \sim \mathcal{D}\big(\mu_g + \kappa(e_g - \lambda\mu_g),\ \sigma_m;\ \mu_g\big), \tag{1}$$

Table 2: Completed games by season. FBS–FBS games carry most of the information about FBS strength; FBS–FCS games link the two tiers. *Eff.* is the share of games with an efficiency margin; *Snap.* is the number of weekly posteriors fitted, including the preseason one from 2015.

| Season | Games | FBS–FBS | FBS–FCS | FCS–FCS | Other | Eff. | Neutral | OT | Snap. |
|---|---|---|---|---|---|---|---|---|---|
| 2014 | 1,584 | 760 | 108 | 639 | 77 | 54% | 8% | 4% | 17 |
| 2015 | 1,536 | 765 | 105 | 611 | 55 | 56% | 8% | 4% | 17 |
| 2016 | 1,547 | 760 | 113 | 620 | 54 | 55% | 5% | 5% | 17 |
| 2017 | 1,549 | 776 | 98 | 629 | 46 | 56% | 5% | 4% | 17 |
| 2018 | 1,554 | 772 | 112 | 611 | 59 | 57% | 5% | 3% | 17 |
| 2019 | 1,621 | 774 | 114 | 666 | 67 | 55% | 4% | 4% | 18 |
| 2020 | 850 | 534 | 36 | 259 | 21 | 67% | 5% | 4% | 30 |
| 2021 | 1,570 | 770 | 117 | 635 | 48 | 56% | 4% | 4% | 17 |
| 2022 | 1,598 | 776 | 120 | 645 | 57 | 91% | 5% | 5% | 17 |
| 2023 | 1,603 | 792 | 118 | 635 | 58 | 93% | 5% | 5% | 17 |
| 2024 | 1,669 | 798 | 121 | 692 | 58 | 96% | 6% | 4% | 18 |
| 2025 | 1,687 | 808 | 126 | 691 | 62 | 98% | 5% | 4% | 18 |
| 2026* | 712 | 278 | 119 | 255 | 60 | 99% | 2% | 3% | 6 |

Table 3: The four margin families. $t_\nu(\xi, \sigma)$ is the Student-$t$ law with $\nu$ degrees of freedom, location $\xi$ and scale $\sigma$. The published family, chosen by cross-validation (Section 7.4), is the Student-$t$.

| Family | $\mathcal{D}(\xi, \sigma; \mu)$ | Extra parameters |
|---|---|---|
| Normal | $\mathcal{N}(\xi, \sigma^2)$ | none |
| Student-$t$ | $t_\nu(\xi, \sigma)$ | $\nu$ |
| Scale mixture | $(1 - p_w)\,\mathcal{N}(\xi, \sigma^2) + p_w\,\mathcal{N}(\xi, \sigma_w^2)$, $\sigma_w > \sigma$ | $p_w, \sigma_w$ |
| Scale-$t$ | $t_\nu\big(\xi, (\sigma^2 + \gamma^2\mu^2)^{1/2}\big)$ | $\nu, \gamma$ |

where $\mathcal{D}(\xi, \sigma; \mu)$ is a law with location $\xi$ and scale $\sigma$ from one of the four families of Table 3. For a game without one,

$$y_g \mid \mu_g \sim \mathcal{D}\big(\mu_g,\ \bar{\sigma};\ \mu_g\big), \qquad \bar{\sigma}^2 = \sigma_m^2 + \kappa^2\sigma_e^2, \tag{2}$$

with the mixture's wide scale widened in the same way, $\bar{\sigma}_w^2 = \sigma_w^2 + \kappa^2\sigma_e^2$. Games are conditionally independent given the parameters.

Three features of Equation 1 matter. First, both signals load on the same $\mu_g$, so strength is measured in points of scoreboard margin: $\lambda$ converts it to the efficiency scale, and it is estimated. Second, the margin is modelled *given* the efficiency residual $e_g - \lambda\mu_g$. A team that outplays its expected efficiency usually outscores its expected margin too, and the loading $\kappa$ absorbs that shared component. Were the two modelled as conditionally independent given $\mu_g$, one good performance would be counted twice. Third, the margin is not normal. The Student-$t$ lets a 63–0 result inform strength without dominating it [13]. The scale mixture puts most games in a narrow component and a minority in a wide one, which bunches margins near the expected value more

than the $t$ does. The scale-$t$ lets the noise grow with the expected margin $|\mu_g|$: a mismatch is less predictable in size than an even game, because the favourite may empty its bench at any score. The family is a modelling choice made once, by leave-one-out cross-validation on every complete season (Section 7.4), and its parameters are estimated in every fit.

The scale $\bar{\sigma}$ in Equation 2 is chosen so that the two cases agree for the Gaussian families.

*Marginal Margin Lemma.* Replace $\mathcal{D}$ in Equation 1 by the normal law with the same location and scale. Then, marginally over $e_g$, $y_g \mid \mu_g \sim \mathcal{N}(\mu_g, \bar{\sigma}^2)$, and $(e_g, y_g)$ is bivariate normal with mean $(\lambda\mu_g, \mu_g)$ and covariance

$$\Sigma = \begin{pmatrix} \sigma_e^2 & \kappa\sigma_e^2 \\ \kappa\sigma_e^2 & \sigma_m^2 + \kappa^2\sigma_e^2 \end{pmatrix}.$$

*Proof.* Write $e_g = \lambda\mu_g + \varepsilon_e$ and $y_g = \mu_g + \kappa\varepsilon_e + \varepsilon_m$ with $\varepsilon_e \sim \mathcal{N}(0, \sigma_e^2)$ and $\varepsilon_m \sim \mathcal{N}(0, \sigma_m^2)$ independent. Then $(e_g, y_g)$ is a linear image of a Gaussian vector, with the stated mean, $\mathrm{Var}(y_g) = \kappa^2\sigma_e^2 + \sigma_m^2$ and $\mathrm{Cov}(e_g, y_g) = \kappa\sigma_e^2$. □

The same argument applied to each component shows that Equation 2 is exact for the normal and the scale mixture. Under the two $t$ families the marginal of $y_g$ is not exactly of the form Equation 2, so there it is an approximation that matches the location and the scale; it affects only the games without an efficiency margin, which are mostly lopsided games against lower-division teams. The same marginal law defines the predictive win probability. Write $F(\cdot \mid \mu)$ for the distribution function of $\mathcal{D}(\mu, \bar{\sigma}; \mu)$, the predictive law of a margin whose expected value is $\mu$.

*Win probability.* The probability that team $i$ beats team $j$ at venue $v \in \{-1, 0, 1\}$ (away, neutral, home for $i$) is the posterior mean

$$p_{ij}(v) = \frac{1}{S}\sum_{s=1}^{S}\Big[1 - F^{(s)}\big(0 \;\big|\; \mu_{ij}^{(s)}(v)\big)\Big], \qquad \mu_{ij}^{(s)}(v) = r_i^{(s)} - r_j^{(s)} + \eta^{(s)}v,$$

where $F^{(s)}$ uses the family's parameters in draw $s$. The predicted spread is the posterior mean of $\mu_{ij}(v)$.

For the Student-$t$ family $1 - F(0 \mid \mu) = F_\nu(\mu/\bar{\sigma})$, with $F_\nu$ the distribution function of the standard $t_\nu$ law: the rule of Stern [11] with the normal replaced by the fitted $t$ and averaged over the posterior, which widens the predictive distribution by the uncertainty in the ratings. For the scale-$t$ it is $F_\nu\big(\mu/(\bar{\sigma}^2 + \gamma^2\mu^2)^{1/2}\big)$, which still increases with $\mu$ but approaches $F_\nu(1/\gamma)$ rather than one as $\mu$ grows: no favourite is certain. A game $h$ weeks after the snapshot adds the random-walk drift of both teams over those weeks (Section 3.2) to $\mu_{ij}^{(s)}$ before the average is taken.

## 3.2 Hierarchy, Preseason Information and the Weekly Walk

A team's strength starts the season at a preseason level, pooled at two levels — teams within conferences and conferences within tiers — and then moves from week to week.

*Strength hierarchy.* For team $i$ in conference $c(i)$ of tier $\ell(i)$, the preseason (week-1) level is

$$r_i^0 = \alpha_{c(i)} + \beta\, d_i + \left(\tau^2_{\ell(i)} + \beta^2 s_i^2\right)^{1/2} z_i, \qquad \alpha_c = m_{\ell(c)} + \omega\, u_c, \qquad z_i, u_c \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{3}$$

with $m_1 = 0$. Here $d_i$ is team $i$'s final power rating in the previous season minus the mean final rating of its current tier, $s_i$ is the posterior standard deviation of that rating, and $d_i = s_i = 0$ when there is no previous rating (the first season, and teams new to the data).

*Weekly random walk.* Let $w_1 < w_2 < \cdots$ be the weeks in which team $i$ plays, and $w_0 = 1$. Its strength in those weeks is

$$r_{i,w_k} = r_{i,w_{k-1}} + \phi_r\,(w_k - w_{k-1})^{1/2}\,\varepsilon_{i,k}, \qquad r_{i,w_0} = r_i^0, \qquad \varepsilon_{i,k} \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{4}$$

so a team that does not play in a week accumulates that week's variance at its next game. Its rating at the week $w$ of a snapshot is its last state plus the drift since, $r_i = r_{i,w_K} + \phi_r\,(w - w_K)^{1/2}\varepsilon_i$, with $w_K$ its last game week up to $w$.

Only the weeks in which a team plays carry a state, so a season adds about a dozen parameters per team rather than one per week. The step $\phi_r$ is shared by all teams and estimated. It is the model's answer to how quickly a rating should forget: with $\phi_r = 0$ every game of the season counts equally, as in a static model, and the larger $\phi_r$ is relative to the noise of a game, the more the latest games dominate. A prediction for a game $h$ weeks after the snapshot adds independent drift for both teams, so the variance of the expected margin grows by $2h\phi_r^2$.

The parameterisation is non-centred [27] throughout, in $z_i$, $u_c$ and the steps $\varepsilon_{i,k}$: in early weeks each team has one or two games and the data barely constrain $z_i$, which is the regime in which a centred hierarchy produces the funnel geometry that defeats Hamiltonian samplers [28]. The random walk has a funnel of its own, between $\phi_r$ and the steps, and the data identify $\phi_r$ only weakly, which is why its prior is informative and keeps it away from zero (Section 3.4).

The carry-over $\beta$ does the work of a preseason rating. In week 0 no game has been played and the posterior of $r_i$ is centred on $\beta d_i$ plus its conference and tier means; as games accumulate, the team effect $z_i$ and the steps take over. Because $d_i$ is measured against the tier and not the conference, it carries conference strength over as well, so preseason predictions of non-conference games are informative from the first week.

Figure 2 shows the estimated step by season. At the posterior mean of a typical complete season, $\phi_r = 0.83$ points per week: over a twelve-week regular season a team's strength drifts by about $2.9$ points (one sd), small next to the spread of FBS strengths but not negligible next to the margin of a close game. Whether that drift improves predictions over static ratings is tested directly in Section 8.4.

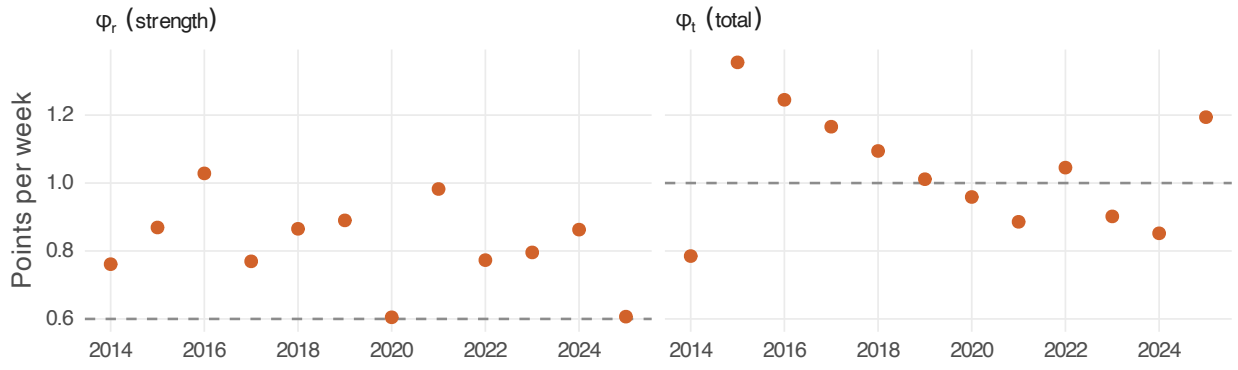



Figure 2: Weekly random-walk step sd by season, posterior mean of the published final fit: $\phi_r$ for strength and $\phi_t$ for the total tendency of Section 3.3. The dashed lines are the prior means.

**Uncertainty in last season's rating.** $d_i$ is built from a posterior mean, and last season's rating is itself uncertain: the final posterior standard deviation of an FBS rating is about 4.3 points, against a spread of 12 points in $d_i$. Treating the true previous rating as $\mathcal{N}(d_i, s_i^2)$ and integrating it out adds $\beta^2 s_i^2$ to the team's variance, which is the extra term in Equation 3; a team whose previous rating was poorly determined is shrunk further towards its conference. Using the posterior mean as the covariate is regression calibration [29]: the posterior mean is already the best predictor of the true rating given last season's data, so the regression on it is not attenuated in the way a regression on a noisy measurement would be, and $\beta$ needs no correction. The reliability $1 - \bar{s}^2 / \mathrm{Var}(d)$ is 0.78–0.91 across seasons, so even a classical measurement-error correction would change $\beta$ by only a few percent.

**Teams that change tier.** A programme moving from FCS to FBS (11 team-seasons since 2014, among them Charlotte, Coastal Carolina, Liberty, James Madison, Jacksonville State) has a previous rating on the FBS scale far below the FBS mean, so its $d_i$ is strongly negative and its preseason rating keeps its absolute level rather than jumping to an average FBS team's. Its conference effect and $\tau_1$ are the new tier's. In weeks 1–4 of a transition season the out-of-sample error on these teams' FBS games is 14.6 points (31 games), against 13.0 for other FBS games in the same weeks: larger, as expected for a team whose schedule changes completely, but not badly biased (mean error -0.0 points).

**One conference spread for all tiers.** The team spreads $\tau_\ell$ differ by tier, but the conference spread $\omega$ is shared. FBS and FCS each have only about a dozen conferences a season, and the lower divisions are one node, so a tier-specific $\omega_\ell$ would be estimated from a dozen values and would lean on its prior. The fitted conference means support pooling: at the end of 2025 their standard deviation within FBS is 1.1 points and within FCS 0.9.

**One home-field advantage.** $\eta$ is one number per season for every true home game, FBS or FCS, near or far. Travel, altitude and crowd size surely vary, but the data identify one average well and many small effects poorly. Figure 3 shows the estimate by season: it is stable at about 2.5 points, except in 2020, when most games were played without crowds and it falls to 1.4.

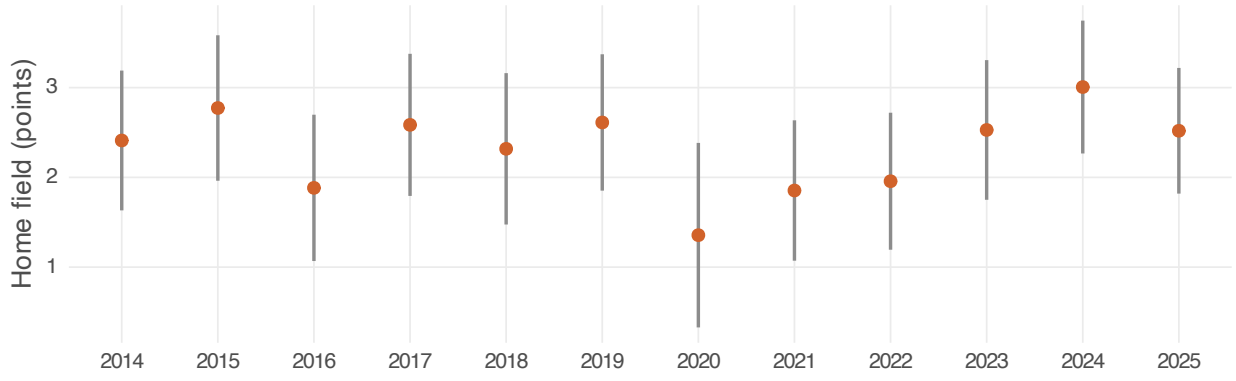



Figure 3: Home-field advantage $\eta$ by season: posterior mean and $\pm 2$ sd of the full-season fit under vague priors (Section 3.4). The 2020 season was played largely without crowds.

*Remark (identification).* The likelihood depends on the strengths only through differences $r_{i,w} - r_{j,w}$ within a connected component of the schedule graph, so it is flat along one direction per component. Every prior in the model is proper (Table 5), so the posterior is proper regardless, and the published FBS-centred ratings $\tilde{r}_i = r_i - \bar{r}_{\text{FBS}}$ do not depend on the anchor $m_1 = 0$. Between components the relative strengths come from the prior alone — from the carry-over and the tier and conference structure — which is what happens in the first weeks.

Table 4 counts the components. In week 1 the FBS teams fall into dozens of pieces; the graph of teams that have played becomes connected within the FBS by week 3 in a typical season, and by the end of every season except 2020 the whole graph, all three tiers, is one component. In 2020 a few teams (1 small component(s)) played only among themselves, and their ratings relative to everyone else rest on the prior. Figure 4 draws one season's graph.

Table 4: Connected components of the schedule graph among teams that have played, by season: after week 1, at the first snapshot in which all FBS teams are connected, and at the end of the season (all tiers).

| Season | Week 1 | FBS connected from week | Final, all tiers |
|---|---|---|---|
| 2014 | 84 FBS | 3 | 1 |
| 2015 | 85 FBS | 3 | 1 |
| 2016 | 86 FBS | 3 | 1 |
| 2017 | 83 FBS | 3 | 1 |
| 2018 | 82 FBS | 3 | 1 |
| 2019 | 85 FBS | 3 | 1 |
| 2020 | 9 FBS | 17 | 2 |
| 2021 | 79 FBS | 3 | 1 |
| 2022 | 78 FBS | 3 | 1 |
| 2023 | 81 FBS | 3 | 1 |
| 2024 | 93 FBS | 3 | 1 |
| 2025 | 87 FBS | 3 | 1 |
| 2026 | 74 FBS | 2 | 1 |

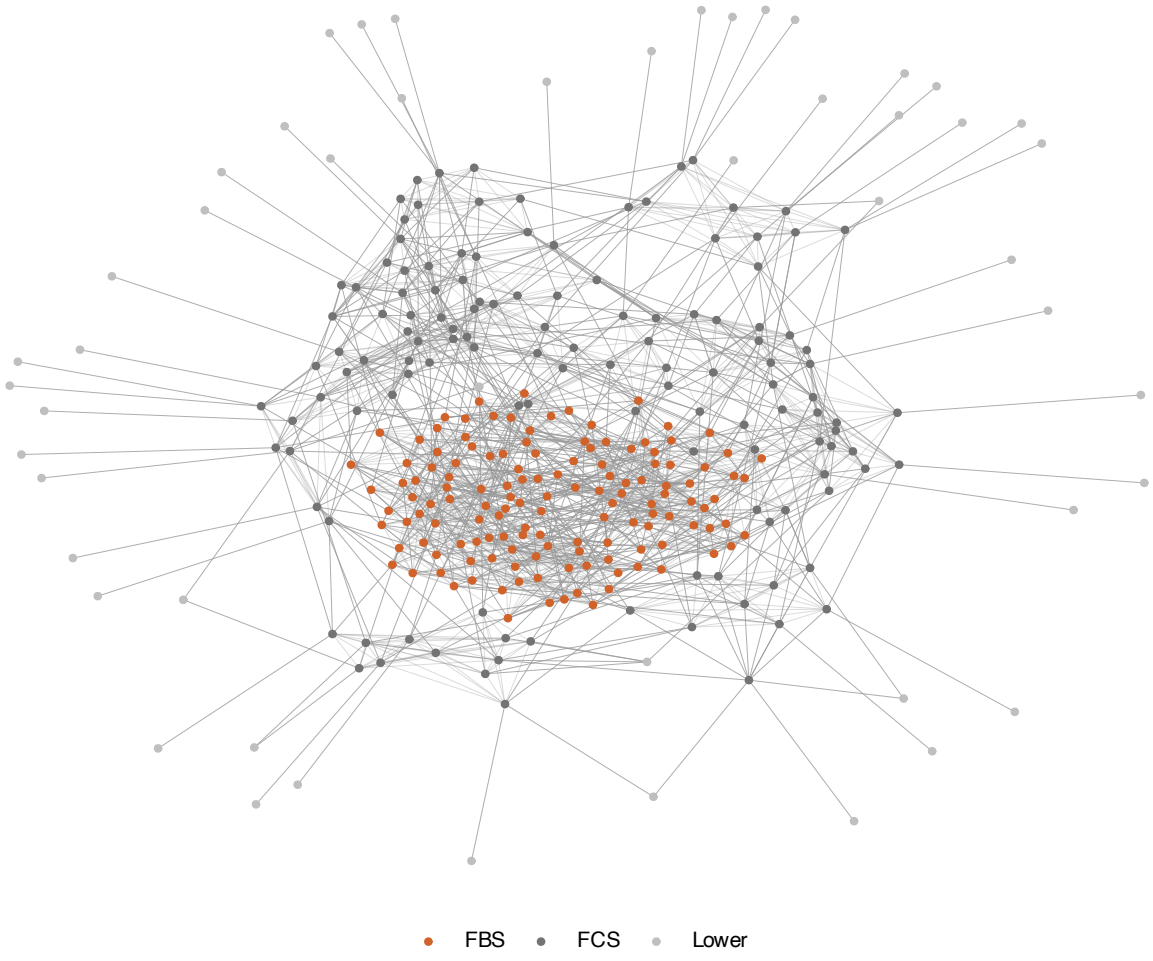



Figure 4: The 2025 schedule graph, all completed games. Points are teams (FBS in orange, FCS in grey, lower divisions in light grey), placed by a force-directed layout so that conferences form clusters; lines are games. Non-conference games are the few long edges that tie the clusters together.

Published ratings are the centred $\tilde{r}_i$, computed draw by draw, so a rating of $+10$ means ten points better than an average FBS team on a neutral field.

Figure 5 draws the full model as a directed acyclic graph.

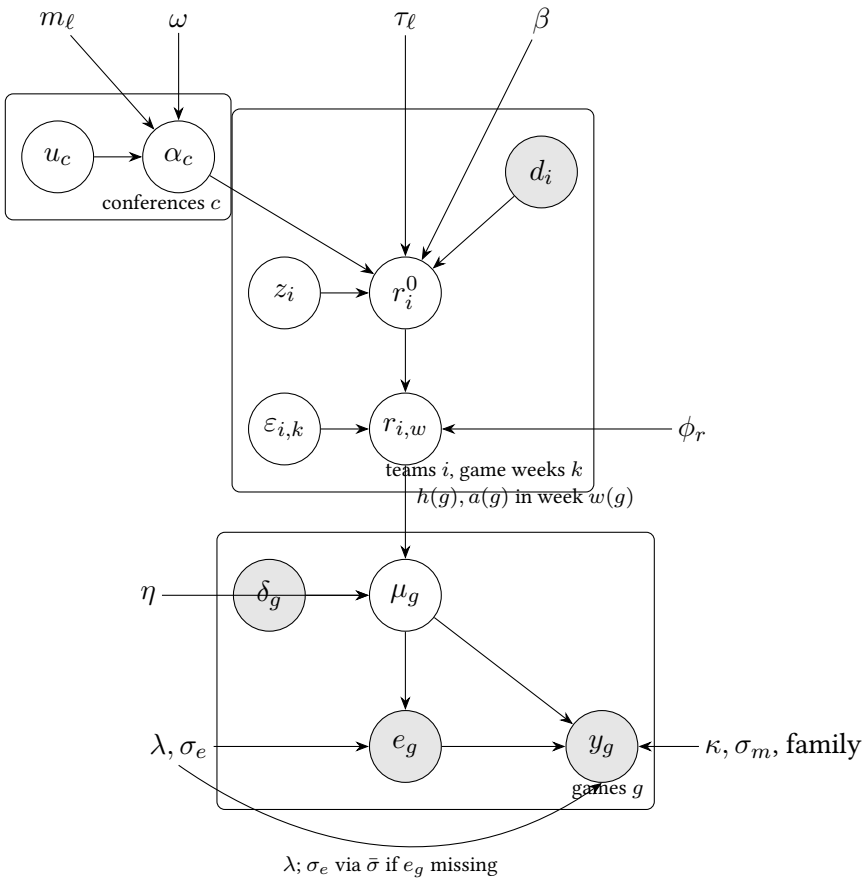



Figure 5: The power model. Shaded nodes are observed; plates repeat over conferences, teams (and, for $\varepsilon_{i,k}$ and $r_{i,w}$, the weeks in which a team plays) and games; $s_i$, the posterior sd of $d_i$, enters $r_i^0$ alongside $d_i$ and is not drawn. "Family" stands for the shape parameters of the margin family ($\nu$, $\gamma$, or $p_w$ and $\sigma_w$). The total model of Section 3.3 has the same structure with its own parameters and is not drawn. The efficiency margin $e_g$ is a parent of the scoreboard margin $y_g$, which is how the conditional likelihood of Equation 1 avoids counting one performance twice. The curved edge carries two dependences: the residual $e_g - \lambda\mu_g$ enters $y_g$ through $\lambda$, and when $e_g$ is missing the edge $e_g \to y_g$ is marginalised as in Equation 2, so $\sigma_e$ enters $y_g$ through $\bar{\sigma}$.

## 3.3 Totals, Offence and Defence

The margin says which team is better but not how many points a game will produce. A second latent quantity per team supplies that: a *total tendency* $t_{i,w}$, how far above or below the average the scores of team $i$'s games run. It has the same structure as strength — a preseason level $t_i^0$ pooled within conferences (spread $\omega_t$) and tiers (offsets $m_2^t, m_3^t$), a within-conference spread $\tau^t$, a carry-over $\beta_t$ of last season's final total tendency measured against its tier, and a weekly random

walk with its own step $\phi_t$ — and is observed in the same way as strength, through a scoreboard quantity conditioned on an efficiency one.

*Total model.* With $T_g = t_{h(g),w(g)} + t_{a(g),w(g)}$, for a game with an efficiency margin

$$e_g^+ \mid T_g \sim \mathcal{N}\big(\alpha_s + \lambda_t T_g,\ \sigma_s^2\big), \qquad x_g \mid e_g^+, T_g \sim \mathcal{N}\big(2\theta + T_g + \kappa_t(e_g^+ - \alpha_s - \lambda_t T_g),\ \sigma_x^2\big), \quad (5)$$

and for a game without one $x_g \mid T_g \sim \mathcal{N}(2\theta + T_g, \sigma_x^2 + \kappa_t^2\sigma_s^2)$. Here $\theta$ is the points each team scores in an average game.

Strength and total tendency together give each team an offence and a defence. Writing $\mathrm{off}_i = (r_i + t_i)/2$ and $\mathrm{def}_i = (r_i - t_i)/2$, the expected scores of game $g$ are

$$\text{home: } \theta + \tfrac{1}{2}(\mu_g + T_g), \qquad \text{away: } \theta + \tfrac{1}{2}(T_g - \mu_g),$$

whose difference is the expected margin $\mu_g$ and whose sum is the expected total $2\theta + T_g$; the home-field advantage, part of $\mu_g$, adds $\eta/2$ to the home score and takes $\eta/2$ from the away score. The published **Off** and **Def** are the points a team would score and allow against an average FBS team on a neutral field, $\mathrm{Off}_i = \theta + \frac{1}{2}(\tilde{r}_i + t_i + \bar{t})$ and $\mathrm{Def}_i = \theta + \frac{1}{2}(t_i + \bar{t} - \tilde{r}_i)$, with $\bar{t}$ the mean FBS total tendency, computed draw by draw. By construction $\mathrm{Off}_i - \mathrm{Def}_i = \tilde{r}_i$, the power rating.

The total model shares no parameter with the margin model of Equation 1, and its likelihood involves only the total tendencies, so the joint posterior factorises: adding it changes neither the power rating nor the resume, only what is reported alongside them. The price of that separation is that the noise of the margin and the noise of the total are modelled as independent, and that overtime games inform the total only through $e_g^+$. The total is scored against the closing over/under in Section 8.2.

## 3.4 Priors

The structural parameters — tier offsets, spreads, the carry-over, home field and the likelihood scales — describe the sport rather than any team. In the first weeks of a season, with about one game per team, the data cannot estimate them, and a vague prior leaves the sampler in a funnel. They are therefore given informative priors, set from history. Each complete season's final snapshot (about 1,550 games) is refitted under deliberately vague priors (Section 11.6), and each published prior is centred at the mean of those full-season posterior means, with standard deviation twice their between-season standard deviation. The refits are the `prior_refits` target of the pipeline. The twelve priors of Table 5 were pooled in this way under an earlier version of the model, with static ratings and a Student-$t$ margin, and have not yet been re-pooled; the table's refit columns are under the published model, so they show directly how far the pooled values have moved since.

Two seasons are left out of the pooling. In 2014, the first season of the data, there is no carry-over ($d_i = 0$ for every team), so the conference effects absorb team strength that later seasons attribute to $\beta d_i$, and $\omega$ is 10.0 points against 2.8 in a typical later season; it measures a different quantity. And 2020, played largely without crowds, is left out for $\eta$ (Figure 3). Table 5 lists the priors with

Table 5: Priors (normal, truncated to the support where a parameter is constrained) and the range over complete seasons (2014–2025) of the full-season posterior mean, refitted under vague priors and as published. The last column is the median posterior sd of the vague-prior refits. The FBS tier offset is fixed at zero.

| Parameter | Prior | Vague priors | Published | Post. sd |
|---|---|---|---|---|
| $\eta$ (home field) | $\mathcal{N}(2.40, 0.80^2)$ | 1.36 to 3.01 | 1.62 to 2.91 | 0.39 |
| $\beta$ (carry-over) | $\mathcal{N}(0.78, 0.09^2)$ | 0.50 to 0.83 | 0.70 to 0.81 | 0.05 |
| $\lambda$ | $\mathcal{N}(0.76, 0.05^2)$ | 0.67 to 0.80 | 0.68 to 0.80 | 0.02 |
| $\kappa$ | $\mathcal{N}(0.77, 0.06^2)$ | 0.72 to 0.82 | 0.72 to 0.81 | 0.02 |
| $\sigma_e$ | $\mathcal{N}(14.26, 0.68^2)$ | 12.62 to 14.83 | 12.89 to 14.74 | 0.33 |
| $\sigma_m$ | $\mathcal{N}(8.01, 0.81^2)$ | 7.31 to 8.95 | 7.38 to 8.79 | 0.29 |
| $m_2$ (FCS) | $\mathcal{N}(-29.20, 3.45^2)$ | -31.84 to -26.47 | -31.17 to -27.85 | 2.10 |
| $m_3$ (lower) | $\mathcal{N}(-61.20, 7.31^2)$ | -67.13 to -55.36 | -65.59 to -58.10 | 5.05 |
| $\omega$ | $\mathcal{N}(2.95, 2.53^2)$ | 1.70 to 10.02 | 1.94 to 8.17 | 1.17 |
| $\tau_1$ (FBS) | $\mathcal{N}(6.73, 1.45^2)$ | 4.93 to 9.05 | 5.34 to 8.60 | 0.75 |
| $\tau_2$ (FCS) | $\mathcal{N}(6.32, 1.56^2)$ | 3.56 to 11.16 | 4.85 to 10.14 | 0.86 |
| $\tau_3$ (lower) | $\mathcal{N}(13.60, 9.35^2)$ | 7.43 to 21.55 | 8.40 to 20.56 | 3.72 |

the range of the full-season posterior means under the vague priors and under the published priors.

Some parameters vary between seasons and others hardly at all, and the table cannot distinguish real change from estimation noise by itself. Comparing the two standard deviations can: for 9 of the 12 parameters ($\beta$, $\eta$, $m_2$, $m_3$, $\omega$, $\sigma_e$, $\tau_1$, $\tau_2$ and $\tau_3$) the between-season standard deviation of the posterior means is less than 1.3 times the typical within-season posterior standard deviation, which is about what sampling noise alone would produce. For these the rule "prior sd = twice the between-season sd" measures noise, not season-to-season change, and makes the priors conservative: wider than the sport's true variation.

The remaining parameters were not pooled (Table 6). The shape parameters of the margin family keep weakly informative priors: $\nu$ a $\mathrm{Gamma}(2, 0.1)$ prior truncated to $\nu > 1$ [30], which lets the data choose between heavy and nearly normal tails. The random-walk steps get informative priors set from a preliminary full-season fit, for the reason given in Section 3.2: the data identify $\phi_r$ only weakly, and a prior with much mass near zero produces a funnel between the step size and the steps. The total model's priors were set from the FBS games of 2022–2025 (about 26.5 points per team), and are wide next to what a season of data determines. The vague-prior refits estimate these parameters too, and the last column of Table 6 gives their range.

The priors are empirical Bayes at the level of the sport, and two consequences need checking. First, a season's own data help set its prior, so saying that its posterior lies inside that prior would be circular. Section 11.6 therefore also builds, for each season, the prior pooled from the *other* seasons only, and asks whether the season's own estimate lies inside it. Second, the early-week backtest of Section 8 for, say, 2016 uses priors informed by later seasons. Section 8.5 refits those weeks under leave-one-season-out priors and measures the difference. Section 11.7 gives the priors they

Table 6: Priors not pooled from the refits: normal, truncated to the support where a parameter is constrained, unless stated. The last column is the range over complete seasons of the full-season posterior mean under the vague priors of Section 11.6; the random-walk steps keep their published priors there. Shape parameters of families other than the published Student-$t$ are omitted.

| Parameter | Meaning | Prior | Vague-prior refits |
|---|---|---|---|
| $\nu$ | degrees of freedom | $\mathrm{Gamma}(2, 0.1)$, $\nu > 1$ | 8.31 to 26.70 |
| $\phi_r$ | weekly step of strength | $\mathcal{N}(0.60, 0.25^2)$ | 0.61 to 1.06 |
| $\phi_t$ | weekly step of the total tendency | $\mathcal{N}(1.00, 0.30^2)$ | 0.76 to 1.36 |
| $\theta$ | points per team, average game | $\mathcal{N}(26.50, 2.00^2)$ | 26.27 to 28.78 |
| $m_2^t$ | FCS total offset | $\mathcal{N}(0.00, 6.00^2)$ | -3.64 to 0.22 |
| $m_3^t$ | lower-division total offset | $\mathcal{N}(0.00, 10.00^2)$ | -12.14 to 6.94 |
| $\omega_t$ | conference spread of totals | $\mathcal{N}(2.00, 1.50^2)$ | 0.52 to 1.19 |
| $\tau^t$ | team spread of totals | $\mathcal{N}(4.00, 2.00^2)$ | 2.49 to 6.68 |
| $\beta_t$ | carry-over of totals | $\mathcal{N}(0.50, 0.25^2)$ | 0.35 to 0.79 |
| $\lambda_t$ | scale of summed efficiency | $\mathcal{N}(0.80, 0.30^2)$ | 0.86 to 1.28 |
| $\alpha_s$ | intercept of summed efficiency | $\mathcal{N}(13.50, 8.00^2)$ | 12.80 to 20.42 |
| $\sigma_s$ | noise of summed efficiency | $\mathcal{N}(17.00, 4.00^2)$ | 15.34 to 17.85 |
| $\kappa_t$ | loading of the total on it | $\mathcal{N}(0.55, 0.20^2)$ | 0.42 to 0.58 |
| $\sigma_x$ | noise of the total | $\mathcal{N}(12.50, 2.00^2)$ | 12.15 to 13.39 |

induce on team strengths, win probabilities and margins, and Section 11.11 refits the model under widened and shifted priors.

# 4 What the Efficiency Signal Adds

The efficiency margin is expensive — it needs play-by-play data — and a natural question is how much it sharpens the estimate of strength once the scoreboard is known. The Gaussian version of the model answers it in closed form.

*Information Proposition.* In the Gaussian model of the Marginal Margin Lemma, the Fisher information about $\mu_g$ carried by one game is

$$\mathcal{I}_y = \frac{1}{\Sigma_{22}} = \frac{1}{\sigma_m^2 + \kappa^2 \sigma_e^2}$$

from the margin alone, and

$$\mathcal{I}_{e,y} = \frac{\lambda^2}{\sigma_e^2} + \frac{(1 - \kappa\lambda)^2}{\sigma_m^2} = \mathcal{I}_y + \frac{(\lambda - \lambda^*)^2}{\Sigma_{11} - \Sigma_{12}^2/\Sigma_{22}}, \qquad \lambda^* = \frac{\Sigma_{12}}{\Sigma_{22}} = \frac{\kappa\sigma_e^2}{\sigma_m^2 + \kappa^2\sigma_e^2}, \tag{6}$$

from both signals. Hence $\mathcal{I}_{e,y} \geq \mathcal{I}_y$, with equality exactly when $\lambda = \lambda^*$.

*Proof.* For a Gaussian vector with mean $\boldsymbol{a}\mu$ and known covariance $\Sigma$ the information about $\mu$ is $\boldsymbol{a}^\top \Sigma^{-1} \boldsymbol{a}$. With $\boldsymbol{a} = (\lambda, 1)^\top$, factor the joint density as $p(y)\, p(e \mid y)$: the margin contributes

Table 7: Per-game Fisher information about the expected margin under the Gaussian model, at each season's final posterior means. $R^2$ is the share of the margin's variance explained by the efficiency residual, $\kappa^2\sigma_e^2/\bar{\sigma}^2$.

| Season | $\lambda$ | $\lambda^*$ | $R^2$ | $10^3\mathcal{I}_y$ | $10^3\mathcal{I}_{e,y}$ | Gain |
|---|---|---|---|---|---|---|
| 2014 | 0.682 | 0.730 | 0.55 | 5.87 | 5.90 | 0.5% |
| 2015 | 0.765 | 0.831 | 0.64 | 5.38 | 5.44 | 1.2% |
| 2016 | 0.733 | 0.810 | 0.63 | 5.24 | 5.33 | 1.6% |
| 2017 | 0.764 | 0.806 | 0.63 | 5.58 | 5.60 | 0.5% |
| 2018 | 0.730 | 0.828 | 0.67 | 5.02 | 5.16 | 2.9% |
| 2019 | 0.744 | 0.849 | 0.64 | 5.82 | 5.98 | 2.7% |
| 2020 | 0.785 | 0.870 | 0.68 | 5.40 | 5.51 | 2.0% |
| 2021 | 0.763 | 0.804 | 0.62 | 5.35 | 5.37 | 0.4% |
| 2022 | 0.772 | 0.920 | 0.66 | 5.87 | 6.17 | 5.1% |
| 2023 | 0.801 | 0.890 | 0.70 | 5.50 | 5.62 | 2.3% |
| 2024 | 0.773 | 0.879 | 0.66 | 5.74 | 5.90 | 2.8% |
| 2025 | 0.758 | 0.854 | 0.62 | 5.98 | 6.10 | 2.0% |

$1/\Sigma_{22}$, and $e \mid y$ is normal with mean $\lambda\mu + (\Sigma_{12}/\Sigma_{22})(y-\mu)$ and variance $\Sigma_{11} - \Sigma_{12}^2/\Sigma_{22}$, whose derivative in $\mu$ is $\lambda - \lambda^*$. Adding the two informations gives the second form of Equation 6. The first form follows by factoring instead as $p(e)\,p(y \mid e)$: $e$ contributes $\lambda^2/\sigma_e^2$ and $y \mid e$ has mean $(1-\kappa\lambda)\mu + \kappa e$ and variance $\sigma_m^2$. □

The proposition says that the efficiency margin helps exactly insofar as its scale $\lambda$ differs from $\lambda^*$, the slope of the regression of $e$ on $y$. If efficiency were nothing but a noisy copy of the scoreboard, its mean would move with $\mu$ exactly as the regression predicts from $y$, and it would add nothing.

*Example.* At the end of 2025 the posterior means are $\lambda = 0.758$, $\kappa = 0.722$, $\sigma_e = 14.06$ and $\sigma_m = 8.00$. The margin alone then has scale $\bar{\sigma} = (\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2} = 12.93$ points and carries $\mathcal{I}_y = 1/\bar{\sigma}^2 = 5.98 \times 10^{-3}$ per game. The slope at which efficiency would add nothing is $\lambda^* = \kappa\sigma_e^2/\bar{\sigma}^2 = 0.854$, 0.096 from the fitted $\lambda$. The second term of Equation 6 is $(\lambda - \lambda^*)^2/(\sigma_e^2 - \kappa^2\sigma_e^4/\bar{\sigma}^2) = 0.122 \times 10^{-3}$, so both signals together carry $\mathcal{I}_{e,y} = 6.10 \times 10^{-3}$: a gain of 2.0%, or about 0.24 extra games' worth of information over a twelve-game season.

Table 7 evaluates Equation 6 at the posterior means of each complete season's final fit.

The efficiency residual explains 55%–70% of the margin's variance, so the two signals are strongly related, and the fitted $\lambda$ lies within about 0.15 of $\lambda^*$. The per-game information gain is correspondingly small: 0.4% to 5.1%, about 0.24 games' worth over a twelve-game season. Three caveats limit how far this should be read. The calculation is Gaussian, while the published margin law is the Student-$t$, with fitted $\nu$ from 9 to 28; under a heavy-tailed likelihood the efficiency residual also helps decide *which* large margins are outliers, which a Fisher-information calculation at the mean does not capture. It holds $\lambda$, $\kappa$ and the scales fixed, whereas in the fit they are estimated jointly with strength. And it is per game: it does not measure any robustness the efficiency margin brings

to the early weeks. The backtest is the arbiter. Section 8.4 refits the model without the efficiency likelihood on every week of 2022–2025: on the same 6,557 games, the margin-only model's mean absolute error is -0.01 points (standard error 0.01) lower and its Brier score -0.3 $\times 10^{-3}$ (se 0.2) lower. At each season's final posterior the expected log predictive density of the margin itself (Section 7.3) favours the full model by 6 in a typical season (full minus margin-only : 0 to 15 across seasons, standard errors about 3).

# 5 The Resume

## 5.1 Wins Above Benchmark

A resume should value results, not performances, and should value them relative to a standard that matters for selection. The playoff takes twelve teams, so the natural standard is the twelfth-best team.

*Benchmark.* In posterior draw $s$, the benchmark strength $B^{(s)}$ is the twelfth largest of the FBS strengths $\{r_i^{(s)} : \ell(i) = 1\}$.

The benchmark is a strength, not a team: in different draws it is a different team, and its uncertainty is part of the posterior.

*Wins Above Benchmark.* Let team $i$ have played games $g \in G_i$, with result $W_g \in \{0, 1\}$ (overtime wins count), opponent $o(g)$ and venue $v_g \in \{-1, 0, 1\}$ from $i$'s side. The probability that the benchmark would have won the same fixture is

$$\pi_g^{(s)} = 1 - F^{(s)}\big(0 \mid B^{(s)} - r_{o(g)}^{(s)} + \eta^{(s)} v_g\big),$$

the win probability of Section 3.1 with the benchmark in team $i$'s place, and

$$\mathrm{WAB}_i^{(s)} = \sum_{g \in G_i} \big(W_g - \pi_g^{(s)}\big). \tag{7}$$

The published value is the posterior mean $\mathrm{WAB}_i = S^{-1} \sum_s \mathrm{WAB}_i^{(s)}$ with an 80% interval.

Equation 7 uses the scoreboard only through $W_g$: margin enters the resume solely by having informed the strengths of the opponents and the benchmark. Both are taken at the week of the snapshot, not at the week of the game: an opponent is valued at what the posterior now believes it to be, so a win over a team that improves later in the season gains value as it does. With the random walk the two can differ, by about $\phi_r \, |w - w(g)|^{1/2}$ points in sd. The following properties are elementary but they are the reason for the choice.

*WAB Properties.* For every draw $s$:

(i) *Additivity.* $\mathrm{WAB}_i^{(s)}$ is a sum of per-game contributions $W_g - \pi_g^{(s)}$, each in $(-1, 1)$.

(ii) *Monotonicity.* A win contributes $1 - \pi_g^{(s)} > 0$ and a loss $-\pi_g^{(s)} < 0$; with the draw held fixed, changing any loss to a win raises $\mathrm{WAB}_i^{(s)}$ by exactly one, whatever the opponent.

(iii) *Schedule response.* The contribution of a win is increasing in the opponent's strength and that of a loss is increasing too (a loss to a strong team costs less), and both are decreasing in the venue $v_g$: road wins count for more than home wins.

(iv) *Calibration.* If the model is correct and team $i$ has strength $r_i^{(s)} = B^{(s)}$, then $\mathbb{E}[\mathrm{WAB}_i^{(s)} \mid \text{parameters}] = 0$ for every schedule.

*Proof.* Parts (i) and (ii) are immediate from Equation 7, since $0 < \pi_g^{(s)} < 1$. For (iii), $1 - F(0 \mid \mu)$ is increasing in $\mu$ for every family of Table 3 (for the scale-$t$ because $\mu/(\bar{\sigma}^2 + \gamma^2\mu^2)^{1/2}$ is increasing), so $\pi_g^{(s)}$ is decreasing in $r_{o(g)}$ and increasing in $v_g$, and both contributions have the form const $- \pi_g^{(s)}$. For (iv), under the model $\Pr(W_g = 1) = 1 - F(0 \mid r_i - r_{o(g)} + \eta v_g) = \pi_g^{(s)}$ when $r_i = B^{(s)}$, so each term has expectation zero. □

Property (ii) holds draw by draw. The *published* $\mathrm{WAB}_i$ changes by a slightly different amount when a result flips, because the result is also data for the posterior: a win moves the opponent's strength down, and the benchmark with it, so the new draws differ from the old ones. Section 5.5 measures a related effect of the margin. Property (ii) excludes what the old composite resume did, rewarding a team for *playing* strong opponents regardless of result; (iii) is the precise sense in which the schedule is counted once; and (iv) gives the scale its meaning: a positive WAB is evidence of having done better than a bubble team would have with the same games. Summing over games makes WAB a count, not a rate, so a team that plays a thirteenth game can gain from it. That is intended: a conference championship game is a real opportunity to earn or lose standing.

*Example.* Table 8 works through WAB for Oklahoma, 10–3 and ranked 12 in the final 2025 resume. For each game, $\bar{\pi}_g$ is the posterior mean probability that the benchmark would have won the same fixture at the same venue; the contribution is $W_g - \bar{\pi}_g$, and the published WAB is their sum, 0.66. (The mean of the per-draw sums equals the sum of the per-game means, so the table reproduces the published value exactly.) The strength of record of Section 5.2 is the probability that a benchmark team with these independent chances wins at least 10 of the 13 games; the Poisson-binomial recursion over the column $\bar{\pi}_g$ gives 0.471.

## 5.2 Strength of Record and Ranks

Two further quantities are reported with the resume. The **strength of record** is the probability that the benchmark would do at least as well as team $i$ did, $\mathrm{SOR}_i = \Pr\big(\sum_{g \in G_i} B_g \geq \sum_{g \in G_i} W_g\big)$ with independent $B_g \sim \mathrm{Bernoulli}(\bar{\pi}_g)$, $\bar{\pi}_g$ the posterior mean of $\pi_g^{(s)}$. The sum of independent Bernoulli variables with unequal probabilities is Poisson-binomial, and its distribution is computed exactly by the recursion

$$P_0^{(0)} = 1, \qquad P_k^{(t)} = (1 - \bar{\pi}_t) P_k^{(t-1)} + \bar{\pi}_t P_{k-1}^{(t-1)}, \quad k = 0, \ldots, t,$$

over a team's $t = 1, \ldots, |G_i|$ games, in $O(|G_i|^2)$ operations [31]. Smaller is more impressive. **Schedule strength** is the benchmark's expected loss rate, $1 - |G_i|^{-1} \sum_g \bar{\pi}_g$.

Table 8: Wins Above Benchmark, game by game, for Oklahoma in 2025. H, A, N: home, away, neutral site. $\bar{\pi}_g$: probability that the twelfth-best team would win the same fixture.

| Week | Opponent | Venue | Score | $W_g$ | $\bar{\pi}_g$ | $W_g - \bar{\pi}_g$ | Running WAB |
|---|---|---|---|---|---|---|---|
| Week 1 | Illinois State | H | 35–3 | 1 | 0.987 | 0.013 | 0.01 |
| Week 2 | Michigan | H | 24–13 | 1 | 0.708 | 0.292 | 0.31 |
| Week 3 | Temple | A | 42–3 | 1 | 0.953 | 0.047 | 0.35 |
| Week 4 | Auburn | H | 24–17 | 1 | 0.757 | 0.243 | 0.60 |
| Week 6 | Kent State | H | 44–0 | 1 | 0.996 | 0.004 | 0.60 |
| Week 7 | Texas | N | 6–23 | 0 | 0.558 | -0.558 | 0.04 |
| Week 8 | South Carolina | A | 26–7 | 1 | 0.707 | 0.293 | 0.34 |
| Week 9 | Ole Miss | H | 26–34 | 0 | 0.538 | -0.538 | -0.20 |
| Week 10 | Tennessee | A | 33–27 | 1 | 0.574 | 0.426 | 0.22 |
| Week 12 | Alabama | A | 23–21 | 1 | 0.446 | 0.554 | 0.78 |
| Week 13 | Missouri | H | 17–6 | 1 | 0.752 | 0.248 | 1.03 |
| Week 14 | LSU | H | 17–13 | 1 | 0.781 | 0.219 | 1.25 |
| Postseason | Alabama | H | 24–34 | 0 | 0.590 | -0.590 | 0.66 |

Two choices in this definition need defending. First, SOR plugs in the posterior mean $\bar{\pi}_g$ rather than averaging the per-draw probabilities $\Pr(\sum B_g^{(s)} \geq \sum W_g)$ as WAB does. The two differ because the tail probability is non-linear in $\pi_g$; refitting each complete season's final week with every draw kept, the largest difference over all FBS teams and seasons is 0.017 and the rank correlation of the two versions is at least 0.999, so the published, cheaper version loses nothing. Second, "at least as well" counts ties as the benchmark matching the team. The strict version, $\Pr(\sum B_g > \sum W_g)$, is smaller for every team, but orders the teams almost identically (rank correlation at least 0.986); the weak inequality is kept because it makes $\mathrm{SOR} = 1$ for a winless team rather than for no team, which reads naturally as "anyone would have done at least this well".

Uncertainty in ranks is reported by ranking within each draw.

*Posterior rank summaries.* Let $R_i^{(s)}$ be the rank of $\mathrm{WAB}_i^{(s)}$ among the teams of $i$'s tier in draw $s$ (1 is best). The published rank orders the posterior means $\mathrm{WAB}_i$; the 80% rank interval is the 10% and 90% quantiles of $\{R_i^{(s)}\}_s$; and $\Pr(\text{top } K) = S^{-1} \sum_s \mathbf{1}\{R_i^{(s)} \leq K\}$ for $K = 1, 4, 12$. The power rank is defined in the same way from $\tilde{r}_i^{(s)}$.

The published rank orders posterior means rather than taking the median rank, because the median ranks of different teams need not form a permutation.

## 5.3 Record-only Reference: the Colley Matrix

The model's resume uses opponent strengths from a model that sees margins. As a reference that sees only wins and losses we compute the Colley rating [6, 8].

*Colley rating.* Let $n_i$ be team $i$'s games, $w_i$ and $l_i$ its wins and losses, and $n_{ij}$ the number of games between $i$ and $j$. The Colley ratings $\boldsymbol{c}$ solve $C\boldsymbol{c} = \boldsymbol{f}$ with

$$C_{ii} = 2 + n_i, \qquad C_{ij} = -n_{ij}\ (i \neq j), \qquad f_i = 1 + \tfrac{1}{2}(w_i - l_i).$$

*Colley Lemma.* $C$ is symmetric positive definite, so $\boldsymbol{c}$ exists and is unique, and $\frac{1}{n}\sum_i c_i = \frac{1}{2}$.

*Proof.* $C = 2I + L$, where $L$ is the Laplacian of the schedule multigraph, which is positive semidefinite; so $C$ is positive definite. Every column of $L$ sums to zero, so $\mathbf{1}^\top C = 2\mathbf{1}^\top$ and $2\,\mathbf{1}^\top\boldsymbol{c} = \mathbf{1}^\top C\boldsymbol{c} = \mathbf{1}^\top\boldsymbol{f} = n + \frac{1}{2}\sum_i(w_i - l_i) = n$, since every game produces one win and one loss. □

The rating is Laplace's rule of succession, $(1 + w_i)/(2 + n_i)$, adjusted for the average rating of the opponents. We compute it on the same games as the posterior and rank within tier.

*Example.* Four teams play four games: A beats B and C, B beats C, and D beats C; A and D do not meet. Then A is 2–0, B 1–1, C 0–3 and D 1–0, and

$$C = \begin{pmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & -1 & 0 \\ -1 & -1 & 5 & -1 \\ 0 & 0 & -1 & 3 \end{pmatrix}, \qquad \boldsymbol{f} = \begin{pmatrix} 2 \\ 1 \\ -0.5 \\ 1.5 \end{pmatrix}, \qquad \boldsymbol{c} = C^{-1}\boldsymbol{f} = \begin{pmatrix} 0.683 \\ 0.483 \\ 0.250 \\ 0.583 \end{pmatrix}.$$

The ratings average 0.50, as the lemma requires. D is unbeaten but rated 0.583, below A's 0.683: its one win came against the winless C, while A also beat B, who has a win of its own. Laplace's rule alone would put D at $2/3$ and A at $3/4$; the opponent adjustment widens the gap.

## 5.4 Choosing the Benchmark

The benchmark rank $b = 12$ is a choice about what the resume is for, not a parameter that data could estimate: a resume is retrodictive, and there is no future outcome against which a backtest could score it. It was set to the size of the playoff field. Two facts complicate that. From 2014 to 2023 the playoff took four teams, so for those seasons the playoff boundary was the fourth-best team; and the twelve-team format reserves five places for conference champions, so the twelfth-*best* team is not the twelfth team *selected*, and the effective boundary for an at-large team is somewhat above twelve. Table 9 shows how much the choice matters, by recomputing WAB with $b = 4$ and $b = 25$ from each season's final posterior.

Over the whole FBS the order barely depends on the benchmark (median rank correlation 0.991 for $b = 4$ and 0.995 for $b = 25$), because a benchmark changes the price of each result but not which results are good. The top of the order does move: a strong benchmark makes a loss cheap and a win over a weak team nearly worthless, which favours difficult schedules, while a weak benchmark makes every win count and every loss costly. A typical season changes 0 of its top 12 under $b = 4$ and 1 under $b = 25$. The published $b = 12$ answers "what would a bubble team in the current format have done"; the benchmark is a setting (`benchmark_rank` in `model_config()`), so the four-team-era question can be asked of the earlier seasons directly.

Table 9: Sensitivity of the final resume to the benchmark rank, relative to $b = 12$, over complete seasons. Rank correlation of WAB among FBS teams; teams entering or leaving the top 4 and the top 12; the largest rank change within the $b = 12$ top 25. Median over seasons, with the range.

| Benchmark | $\rho$ with $b = 12$ | Top-4 changes | Top-12 changes | Max move, top 25 |
|---|---|---|---|---|
| $b = 4$ | 0.991 (0.985–0.995) | 0 (0–1) | 0 (0–2) | 8 (5–12) |
| $b = 25$ | 0.995 (0.987–0.998) | 0 (0–1) | 1 (0–1) | 6 (3–9) |

Table 10: Effect of a blowout margin on the resume, 2025 final posterior. For each game: the change in each team's WAB and power rating when the result is refitted as a one-point win for the same team, and the mean absolute change in WAB of every other FBS team.

| Game | Margin | ΔWAB winner | ΔWAB loser | Δ power winner | Δ power los |
|---|---|---|---|---|---|
| USC over Missouri State | 60 | -0.017 | -0.040 | -3.0 | |
| Oregon over Oklahoma State | 66 | -0.042 | 0.043 | -2.5 | |
| Nebraska over Akron | 68 | -0.031 | -0.028 | -2.5 | |
| Alabama over UL Monroe | 73 | -0.008 | -0.013 | -2.6 | |
| Michigan over Central Michigan | 60 | -0.031 | -0.030 | -2.8 | |
| Notre Dame over Syracuse | 63 | 0.012 | -0.027 | -3.7 | |

### 5.5 How Margin Reaches the Resume

WAB does not use margin directly, but margin informs the posterior strengths of the opponents and the benchmark, so it reaches the resume indirectly. A blowout win makes the loser look weaker, which lowers the winner's credit for beating it. Table 10 measures this on the 6 most lopsided FBS–FBS games of 2025: each is refitted with its regulation margin replaced by a one-point win for the same team (and its efficiency margin dropped), and the two teams' WAB recomputed.

Turning a blowout of 60–73 points into a one-point win moves the winner's power rating by 2.5–3.7 points, but its WAB by -0.042 to 0.012 wins and the loser's by -0.040 to 0.043; every other team's moves by at most 0.048 on average. For comparison, the Monte Carlo standard error of a published WAB is about 0.011, 0.008 (Section 6.1). The margin's influence on the resume runs through the opponents' strengths, as designed, and a team gains little by running up a score. The effect has no consistent sign: a close result makes the beaten opponent look stronger, which raises the winner's credit for that game, but it also changes the strengths that set the benchmark, and the two pull in different directions.

## 6 Computation and Pipeline

### 6.1 Sampling and Gates

Each posterior is sampled with the No-U-Turn sampler [32] in Stan [33] via CmdStan: 4 chains of 500 warm-up and 1,000 sampling iterations, target acceptance 0.9. A fit is accepted only if it passes four gates on every team strength and total tendency at the snapshot week and every structural

Table 11: Monte Carlo standard errors of the published resume quantities, FBS teams at the final posterior of every complete season: median and largest over teams and seasons. Thinned: the draws the pipeline keeps; all: every post-warm-up draw.

| Draws | WAB | $\Pr(\text{top } 12)$ | Rank, 10% quantile | Rank, 90% quantile |
|---|---|---|---|---|
| 1,000 (thinned) | 0.011 / 0.023 | 0.008 / 0.020 | 0.00 / 1.50 | 0.00 / 1.00 |
| 2,000 (thinned) | 0.008 / 0.013 | 0.005 / 0.012 | 0.00 / 0.50 | 0.00 / 0.50 |
| 4,000 (all) | 0.005 / 0.009 | 0.004 / 0.008 | 0.00 / 0.50 | 0.00 / 0.50 |
| 8,000 (all) | 0.004 / 0.006 | 0.002 / 0.006 | 0.00 / 0.50 | 0.00 / 0.50 |

parameter, including the random-walk steps and the family's shape parameters: rank-normalised $\widehat{R} \leq 1.01$ and bulk and tail effective sample sizes of at least 400 [34], no divergent transitions, and E-BFMI above 0.2 in every chain [28]. A fit that fails is rerun once with 1,000 warm-up iterations, target acceptance 0.95 and 2,000 sampling iterations (a much higher target acceptance makes the step size so small that chains stall in the neck of the random-walk funnel); the posterior for the current week halts the pipeline if it still fails, while a historical week is recorded with a warning. The resume and predictions are computed from 1,000 equally spaced draws, which keeps the per-game probability matrices of the resume (draws $\times$ team-games) small enough to store for every week. Thinning costs precision, and Table 11 measures how much, from the final-week refits of every complete season with all 4,000 draws kept: the Monte Carlo standard errors of the published quantities with the thinned draws, against all draws.

The largest Monte Carlo error of a published $\Pr(\text{top } 12)$ is 0.020, 0.012, and of a $\text{WAB}$ 0.023, 0.013 wins: small next to the posterior uncertainty they summarise, so the second decimal of a published probability is meaningful and the third is not. Thinning by four inflates the median error of $\text{WAB}$ by a factor of 2.2; for independent draws the factor would be two.

Of the 226 weekly posteriors, 226 pass the gates, 55 after the retry. Table 12 summarises them by season.

## 6.2 The Pipeline

The computation is a `targets` pipeline in R [35, 36], drawn in Figure 6. Its structure enforces the two properties the evaluation depends on. First, the inputs to the week-$w$ posterior contain only games of concluded weeks $\leq w$ and are hashed, so a cached posterior is reused exactly when its inputs are unchanged, and new results or betting lines for later weeks never invalidate it. Second, each season's preseason information $d_i$ is the final posterior of the previous season, a dependency the pipeline tracks, so refitting one season refits every later one. The margin family is fixed for the whole build by the `margin_comparison` target (Section 7.4), which fits every complete season without carry-over and therefore does not depend on the published fits it configures.

Table 12: Convergence gates by season, over all weekly posteriors. Worst values are taken over snapshots and over every monitored parameter.

| Season | Snapshots | Passed | Retried | Max $\widehat{R}$ | Min ESS | Divergent | Min E-BFMI | Minutes |
|---|---|---|---|---|---|---|---|---|
| 2014 | 17 | 17 | 4 | 1.010 | 431 | 0 | 0.79 | 21.9 |
| 2015 | 17 | 17 | 7 | 1.008 | 482 | 0 | 0.78 | 28.2 |
| 2016 | 17 | 17 | 5 | 1.009 | 408 | 0 | 0.82 | 23.5 |
| 2017 | 17 | 17 | 0 | 1.009 | 409 | 0 | 0.81 | 12.8 |
| 2018 | 17 | 17 | 5 | 1.009 | 406 | 0 | 0.81 | 21.5 |
| 2019 | 18 | 18 | 7 | 1.010 | 468 | 0 | 0.84 | 29.6 |
| 2020 | 30 | 30 | 3 | 1.010 | 431 | 0 | 0.80 | 19.5 |
| 2021 | 17 | 17 | 1 | 1.010 | 419 | 0 | 0.81 | 12.2 |
| 2022 | 17 | 17 | 5 | 1.009 | 471 | 0 | 0.78 | 25.6 |
| 2023 | 17 | 17 | 7 | 1.010 | 448 | 0 | 0.81 | 31.4 |
| 2024 | 18 | 18 | 5 | 1.009 | 402 | 0 | 0.78 | 29.3 |
| 2025 | 18 | 18 | 4 | 1.010 | 406 | 0 | 0.79 | 31.7 |
| 2026 | 6 | 6 | 2 | 1.010 | 594 | 0 | 0.80 | 4.6 |

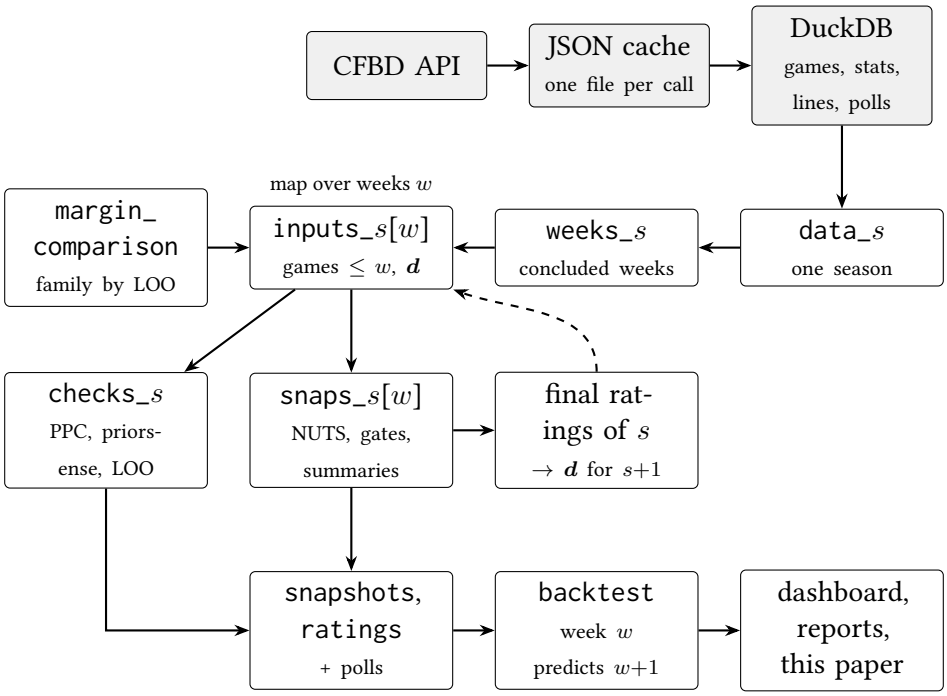



Figure 6: The computation as a `targets` pipeline. Grey boxes are external stores; white boxes are targets, with $[w]$ marking a dynamic branch per concluded week and $s$ a season. The dashed edge is the only cross-season dependency: season $s$'s final ratings become season $s+1$'s preseason information. The margin comparison, fitted once on every complete season, sets the family every fit uses. The checks feed the diagnostics report. A weekly update calls the API about six times and refits only the new week.

# 7 Model Checking

Each season's latest posterior is refitted with the pointwise log-likelihood stored, and three checks are run on it [24, 37]. This section summarises them across seasons; Section 11 examines two posteriors in full detail — sampler diagnostics, convergence, posterior geometry, prior-to-posterior learning, predictive distributions and influence.

## 7.1 Posterior Predictive Checks

Replicated regulation margins are drawn from Equation 2, each game at its expected margin in the week it was played, rounded to whole points so that they can land exactly on the thresholds of the statistics, and compared with five observed statistics: the standard deviation of the margin, the shares of games within seven and within three points, the share decided by 28 or more, and the home team's win rate in true home games. Replicated totals of the games without overtime are drawn from the marginal law of the total model and compared on their mean and standard deviation. Figure 7 shows the result for every season.

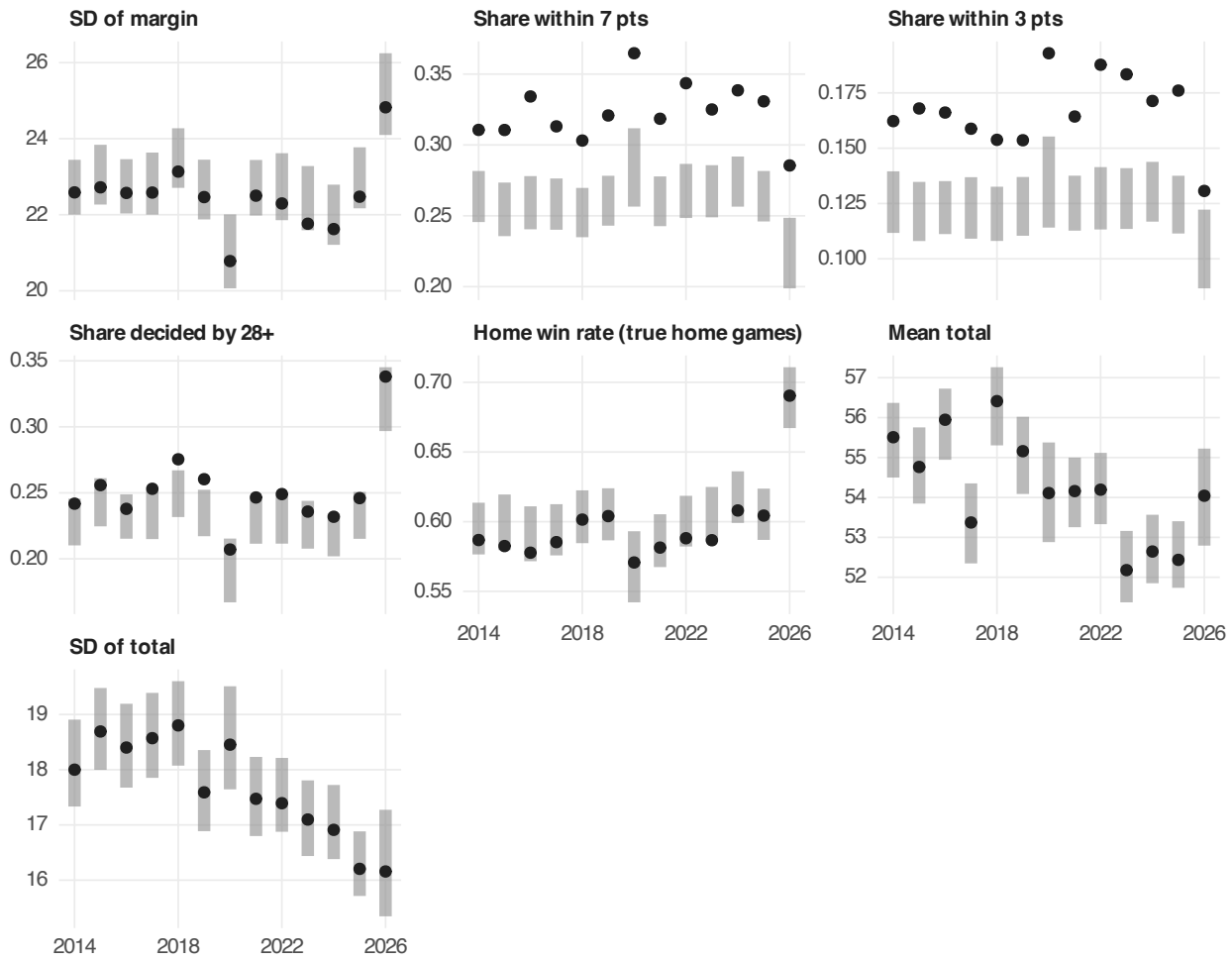



Figure 7: Posterior predictive checks at each season's latest posterior. Bars: central 90% interval of the statistic over 400 replicated seasons (as in Section 11.8); points: the observed value, filled when it falls outside the interval.

The model reproduces home win rate, mean total, sd of margin, sd of total (outside the interval in at most one season each). In 13 of 13 seasons the observed share of games decided by seven points or fewer (0.29–0.36) lies above the replicated interval, and in 13 the share within three points (0.13–0.19) does. Real margins bunch around one score more than the Student-$t$ around the rating difference allows: late-game behaviour (running out the clock when ahead, playing for a tying field goal when behind), key numbers, and the regulation ties that overtime games contribute all push mass towards zero. None of the families compared in Section 7.4 removes this misfit. It is in the *shape* of the margin distribution near zero; the win probabilities, which depend on the location and the tails, are well calibrated (Section 8).

## 7.2 Prior Sensitivity

Power-scaling sensitivity [38] perturbs the prior and the likelihood by raising each to a power near one, estimates the resulting change in each marginal posterior by importance sampling, and

flags a parameter whose posterior moves under both (prior–data conflict) or under neither (weak likelihood). Table 13 counts the flags across seasons.

5 of the 26 parameters are flagged for a weak likelihood in any season, in at most 1 season(s) each. Prior–data conflict flags are more common: $\phi_r$ in 13 of 13; $\phi_t$ in 13 of 13; $\nu$ in 10 of 13; $m_3$ in 8 of 13; $\tau_1$ in 8 of 13; $m_2$ in 6 of 13; $\tau_2$ in 6 of 13; $\omega$ in 5 of 13; $\sigma_e$ in 5 of 13; $\sigma_m$ in 4 of 13; $\tau^t$ in 4 of 13; $\beta_t$ in 3 of 13; $\theta$ in 3 of 13; $\eta$ in 3 of 13; $\omega_t$ in 3 of 13; $\alpha_s$ in 2 of 13; $\beta$ in 2 of 13; $\kappa$ in 2 of 13; $\lambda$ in 2 of 13; $\tau_3$ in 2 of 13; $\lambda_t$ in 1 of 13; $m_2^t$ in 1 of 13; $m_3^t$ in 1 of 13. Three groups stand out. The lower-division offset $m_3$ and spread $\tau_3$, and the FCS offset $m_2$, are identified each season by a few dozen lopsided cross-tier games, and a conflict in one season reflects that season's departure from the pooled mean on which the prior is centred (Section 3.4); none of the three enters an FBS rating except through those cross-tier games. Home field $\eta$ and $\sigma_m$ are flagged when a season sits at the edge of the historical range (for $\eta$: 2020, 2021, 2026). The degrees of freedom $\nu$ are the special case. Both of its sensitivities are large (median 0.10 to the prior and 0.23 to the likelihood, against the threshold of 0.05), and priorsense calls any parameter that is sensitive to both a prior–data conflict. For $\nu$ that label is misleading: the posterior means, 9–28, sit in the bulk of the $\mathrm{Gamma}(2, 0.1)$ prior, so prior and data do not disagree about where $\nu$ lies. Rather, the likelihood informs $\nu$ but only weakly once it exceeds about 15, where the $t$ and normal laws are hard to tell apart, so the shape of the prior's tail still moves the posterior. Section 11.11 refits with the prior bounded at $\nu > 3$. The sensitivity of $\nu$ matters for the tails of the predictive distribution, not for the ratings. The random-walk step $\phi_r$ is flagged in 13 of 13 seasons, as expected of a parameter the data identify only weakly and whose prior is informative on purpose (Section 3.2). Of the 12 parameters of the total model, 9 are flagged in some season; they affect the reported offence, defence and totals, not the power rating or the resume.

## 7.3 Leave-one-out Cross-validation

Pareto-smoothed importance-sampling leave-one-out cross-validation [39, 40] estimates the expected log predictive density of a new game, with the Pareto $\hat{k}$ diagnostic flagging games whose removal would change the posterior too much for the approximation.

The effective number of parameters is 52–87% of the number of teams, below it in every season, as expected of a hierarchical model that shrinks the least-informed teams towards their conference. Because `log_lik` is the joint density of both signals where efficiency is observed, its $\widehat{\mathrm{elpd}}$ cannot be compared with that of a model without efficiency; the program therefore also stores the marginal density of the margin alone, on which Section 4 compares the two models. A few games per season have high $\hat{k}$; they are almost always the only link between a lower-division team and the rest of the graph, whose strength is then determined by that single game.

## 7.4 Choosing the Margin Family

The family $\mathcal{D}$ of Equation 1 is chosen by PSIS-LOO. Each family is fitted to the final week of every complete season, 2014–2025, under the vague priors of Section 11.6, without carry-over and without the total model. The vague priors and the absent carry-over make the comparison independent of the published fits, which depend on its result; the total model is left out because the family enters only the margin likelihood. Within a season the pointwise $\widehat{\mathrm{elpd}}$ differences from

Table 13: Power-scaling sensitivity by parameter, at each season's latest posterior. Sensitivities are the cumulative Jensen–Shannon distance between the base and the power-scaled posteriors; the conventional threshold is 0.05. The parameters of the total model (Section 3.3) are listed last.

| Parameter | Prior (median) | Likelihood (median) | Seasons flagged: conflict | weak likelihood |
|---|---|---|---|---|
| $\beta$ | 0.041 | 0.081 | 2/13 | 1/13 |
| $\eta$ | 0.039 | 0.149 | 3/13 | 0/13 |
| $\kappa$ | 0.041 | 0.913 | 2/13 | 0/13 |
| $\lambda$ | 0.017 | 0.139 | 2/13 | 0/13 |
| $m_2$ | 0.050 | 0.077 | 6/13 | 0/13 |
| $m_3$ | 0.052 | 0.087 | 8/13 | 0/13 |
| $\nu$ | 0.105 | 0.234 | 10/13 | 0/13 |
| $\omega$ | 0.036 | 0.175 | 5/13 | 0/13 |
| $\sigma_e$ | 0.043 | 0.775 | 5/13 | 0/13 |
| $\sigma_m$ | 0.038 | 0.398 | 4/13 | 0/13 |
| $\tau_1$ | 0.064 | 0.125 | 8/13 | 0/13 |
| $\tau_2$ | 0.056 | 0.113 | 6/13 | 1/13 |
| $\tau_3$ | 0.026 | 0.494 | 2/13 | 0/13 |
| $\phi_r$ | 0.179 | 1.395 | 13/13 | 0/13 |
| $\alpha_s$ | 0.031 | 0.080 | 2/13 | 1/13 |
| $\beta_t$ | 0.024 | 0.077 | 3/13 | 0/13 |
| $\theta$ | 0.038 | 0.075 | 3/13 | 1/13 |
| $\kappa_t$ | 0.018 | 0.059 | 0/13 | 0/13 |
| $\lambda_t$ | 0.022 | 0.064 | 1/13 | 1/13 |
| $\omega_t$ | 0.047 | 0.119 | 3/13 | 0/13 |
| $\sigma_s$ | 0.012 | 0.070 | 0/13 | 0/13 |
| $\tau^t$ | 0.034 | 0.079 | 4/13 | 0/13 |
| $\sigma_x$ | 0.010 | 0.064 | 0/13 | 0/13 |
| $m_2^t$ | 0.030 | 0.077 | 1/13 | 0/13 |
| $m_3^t$ | 0.014 | 0.079 | 1/13 | 0/13 |
| $\phi_t$ | 0.093 | 0.085 | 13/13 | 0/13 |

Table 14: PSIS-LOO at each season's latest posterior. $p_{\text{loo}}$ is the effective number of parameters; the last column counts games with $\hat{k} > 0.7$.

| Season | Games | Teams | $\widehat{\text{elpd}}_{\text{loo}}$ | SE | $p_{\text{loo}}$ | $\hat{k} > 0.7$ |
|---|---|---|---|---|---|---|
| 2014 | 1,584 | 318 | -9,592 | 85 | 270 | 14 |
| 2015 | 1,536 | 300 | -9,477 | 80 | 241 | 3 |
| 2016 | 1,547 | 302 | -9,528 | 82 | 233 | 0 |
| 2017 | 1,549 | 296 | -9,533 | 80 | 238 | 5 |
| 2018 | 1,554 | 305 | -9,673 | 79 | 228 | 0 |
| 2019 | 1,621 | 310 | -9,868 | 81 | 260 | 8 |
| 2020 | 850 | 244 | -5,545 | 56 | 155 | 5 |
| 2021 | 1,570 | 298 | -9,707 | 79 | 259 | 24 |
| 2022 | 1,598 | 306 | -11,796 | 62 | 237 | 7 |
| 2023 | 1,603 | 302 | -11,896 | 56 | 232 | 2 |
| 2024 | 1,669 | 311 | -12,619 | 52 | 259 | 14 |
| 2025 | 1,687 | 315 | -12,922 | 49 | 239 | 4 |
| 2026 | 704 | 320 | -5,484 | 28 | 166 | 17 |

Table 15: Margin families by PSIS-LOO, final week of every complete season, 2014–2025. $\Delta\widehat{\text{elpd}}$: pooled difference from the Student-$t$ (positive is better) with its standard error; *Seasons better*: seasons in which the family beats the $t$, and by more than two of that season's standard errors; the last two columns count seasons whose observed share of games within 7 and within 3 points lies outside the 90% predictive interval.

| Family | $\widehat{\text{elpd}}$ | $\Delta\widehat{\text{elpd}}$ | Seasons better | Within 7 missed | Within 3 missed |
|---|---|---|---|---|---|
| Scale-$t$ | -122,333 | 25.5 (32.2) | 1, 0 of 12 | 12/12 | 12/12 |
| Student-$t$ (published) | -122,359 | – | – | 12/12 | 12/12 |
| Mixture | -122,441 | -82.0 (9.7) | 0, 0 of 12 | 12/12 | 12/12 |

the Student-$t$, the incumbent, give a difference and its standard error; seasons are independent, so the pooled difference is the sum and its standard error the root sum of squares. A family replaces the $t$ only if its pooled difference exceeds two pooled standard errors. The normal is not refitted here, since the $t$ contains it as the limit $\nu \to \infty$; it appears among the ablations of Section 8.4. Table 15 gives the result.

The Scale-$t$ family has the best pooled $\widehat{\text{elpd}}$, 25 above the $t$ (standard error 32), better in 1 of 12 seasons, and the published family is therefore the Student-$t$. The mixture also beats the $t$, by -82 (10), far less. The improvement does not extend to the one-score games. Figure 8 shows the shares of games within seven and within three points by season and family: for every family and every season the observed share within seven points lies above the replicated interval. The bunching of margins near zero is a property of the discrete score, with its key numbers and overtime ties, that a continuous location-scale law of any of these shapes does not reproduce (Section 10).

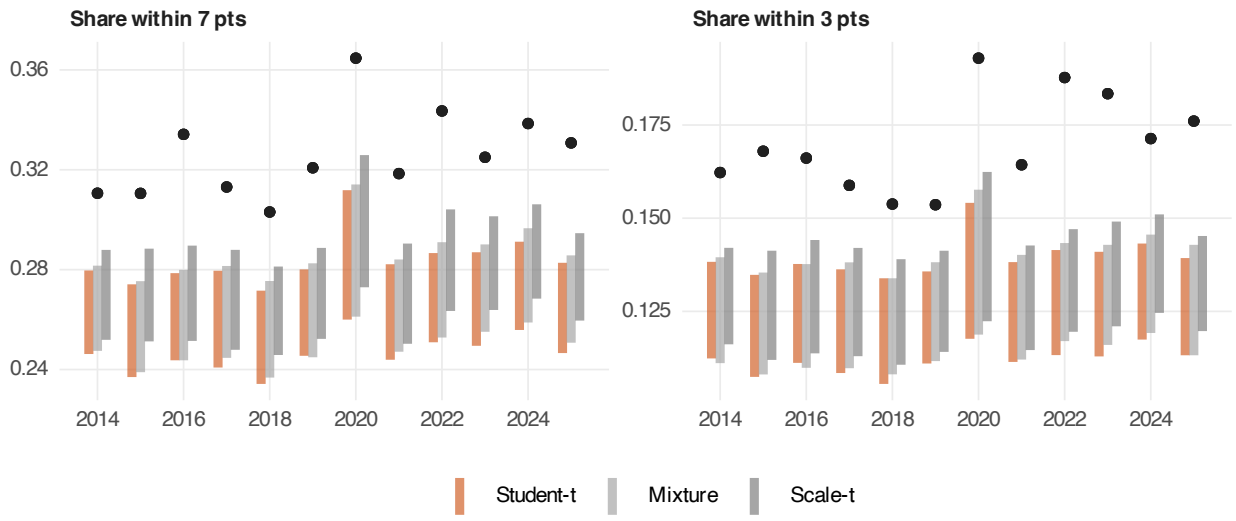



Figure 8: The share of games decided by seven or fewer points (left) and three or fewer (right) at each season's final week: observed (black points) and the 90% interval over replicates from each family's fit. The comparison fits use vague priors and no carry-over.

# 8 Out-of-sample Evaluation

## 8.1 Design

Every game of week $w + 1$ is predicted from the posterior fitted through week $w$, which by construction (Section 6.2) contains no information from week $w + 1$ or later. This is a rolling-origin evaluation [41] with one origin per week, and it covers every week of every season from 2015, when a preseason posterior first exists, including week 1 predicted from the preseason. Predictions are scored by the absolute error of the spread, the Brier score [42] and the log loss of the home win probability, both strictly proper [43], and against the closing betting line where one exists. A betting line is the sharpest public forecast available, since it aggregates information — injuries, weather, motivation — that no ratings model sees [44]. Beating it is not the aim; coming close to it with public results alone is evidence that the ratings are sound.

## 8.2 Accuracy

Table 16 scores every predicted game, and Table 17 compares the model with the closing line on the games that have one. The two tables have different denominators: every FBS–FBS game has a line, but only 70% of all games do, since lines are rarely posted for games against lower-division teams.

Over all 18,958 predicted games the mean absolute error is 12.9 points (95% interval 0.16–0.17), the Brier score 0.166 (9.081–9.311) and the log loss 0.497 (0.485–0.508), and the favourite wins 74.9% of the time. Games predicted by the same posterior share its errors, so these intervals come from a bootstrap that resamples whole season-weeks, and the comparisons below use standard

Table 16: Rolling out-of-sample accuracy, all predicted games in each scope. MAE and RMSE are errors of the predicted margin in points; CRPS scores the whole predictive distribution of the margin, in points; Brier score, log loss and accuracy score the home win probability. Week 5+ means the predicted week, so the posterior has at least four weeks of games.

| Games | N | MAE | RMSE | CRPS | Brier | Log loss | Accuracy |
|---|---|---|---|---|---|---|---|
| All games | 18,958 | 12.87 | 16.28 | 9.20 | 0.166 | 0.497 | 74.9% |
| FBS vs FBS | 9,325 | 12.86 | 16.27 | 9.20 | 0.180 | 0.533 | 72.5% |
| FBS vs FBS, week 5+ | 6,916 | 12.81 | 16.17 | 9.15 | 0.186 | 0.548 | 71.2% |

Table 17: The model against the closing betting line, games with a line only. The line's win probability applies the model's own predictive distribution to the line's spread. ATS is the share of games in which the model's side of the line covered, pushes excluded; the break-even rate against a standard $-110$ price is 52.4%.

| Games | N with line | MAE, model | MAE, line | Brier, model | Brier, line | ATS |
|---|---|---|---|---|---|---|
| All games | 13,236 | 12.77 | 12.24 | 0.166 | 0.159 | 49.9% |
| FBS vs FBS | 9,325 | 12.86 | 12.29 | 0.180 | 0.172 | 49.4% |
| FBS vs FBS, week 5+ | 6,916 | 12.81 | 12.36 | 0.186 | 0.178 | 49.0% |

errors clustered the same way. On the 9,325 FBS–FBS games the model's error is 12.86 points against the line's 12.29. The difference, 0.57 points per game with standard error 0.05 ($z = 11.5$ in a Diebold–Mariano comparison of absolute errors [45]), is clearly distinguishable from zero: the market is better. Its win probabilities are better too (Brier difference 0.0082, $z = 9.6$). Against the line the model picks the covering side in 49.4% of 9,156 games (95% interval 48.4%–50.4%), consistent with the 50% of a forecast with no edge on the market and well short of the 52.4% needed to profit at standard prices.

Figure 9 shows how accuracy evolves through the season. The error is largest in week 1 (13.8 points), predicted from the preseason alone, and from week $\infty$ on it stays within 0.6 points of its week-5-and-later mean of 12.9. The gap to the line ranges from -0.0 to 1.4 points across regular-season weeks. The Brier score *rises* through the season, and the right panel shows why: the mean distance of the predicted probability from one half falls from 0.25 in the early weeks to 0.14 later on, as conference play pairs more evenly matched teams, and the line's own Brier score rises in step.

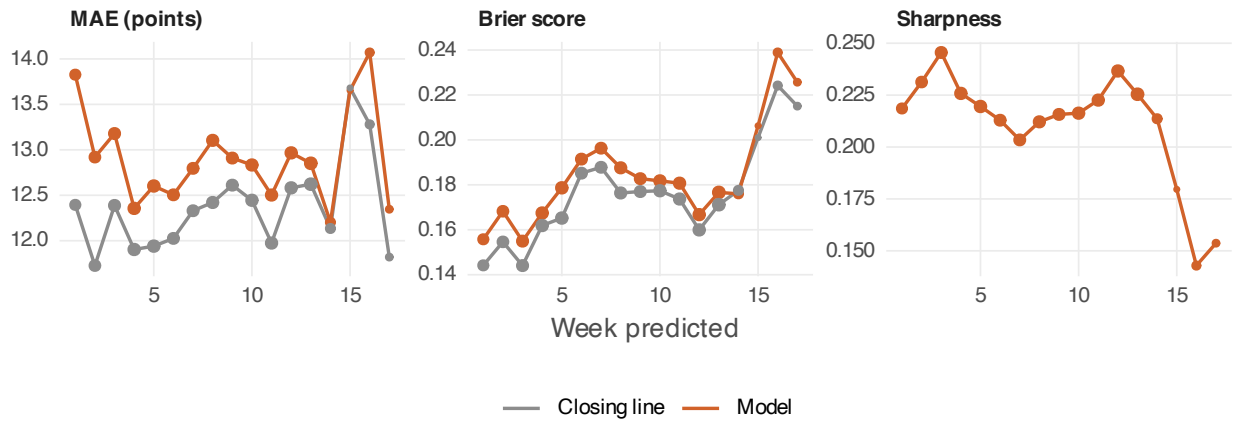



Figure 9: Rolling out-of-sample accuracy on FBS–FBS games by the week predicted. Left: mean absolute error of the model's spread and of the closing line on the same games. Centre: Brier score of the model's win probability and of the line's (the line's spread through the model's predictive distribution). Right: sharpness, the mean of $|p - 1/2|$ over the model's predicted probabilities. Weeks after the last regular-season week are the postseason and have few games.

**The predictive distribution of the margin.** A spread is a point forecast; the model also implies a full predictive distribution of the margin. For each game of the week ahead it is represented by quantiles of posterior predictive draws: for each posterior draw, the expected margin plus a week of random-walk drift for both teams, and then a margin from the Student-$t$ law around it, so that the uncertainty in the ratings, the drift and the game noise all enter. Its central 50%, 80% and 95% intervals contain 50.7%, 80.3% and 95.6% of observed margins (FBS–FBS: 49.9%, 79.2%, 95.1%), and its continuous ranked probability score [43, 46], computed from the quantiles, is 9.20 points (12.70–13.04). The intervals are close to nominal in the aggregate, but they are the right width for the wrong reason near zero: Section 7.1 shows that real margins pile up within one score more than the model allows, so central intervals are slightly too wide for close games and slightly too narrow in the shoulders.

**Totals.** The total model of Section 3.3 predicts the final combined score of each game, and the closing over/under is the market's forecast of the same quantity. Both are scored against the final total, overtime included, since that is how the over/under is settled; the model's total, fitted on regulation-decided games only, therefore runs slightly low for games that go to overtime. On the 9,325 FBS–FBS games with an over/under the model's mean absolute error is 13.26 points against the over/under's 12.92 (all games: 13.18 against 12.85 on 12,885). The gap, 0.34 points, is smaller than the gap on the spread, 0.57: the market knows about weather, tempo changes and injuries to quarterbacks, which matter for totals as well as for margins.

## 8.3 Calibration

A win probability is useful to the resume only if it is calibrated, because WAB subtracts $\pi_g$ from an observed result. Figure 10 groups all predicted games into ten fixed-width bins of their win

probability and compares the mean prediction in each bin with the observed frequency, overall and split by kind of game and part of the season.

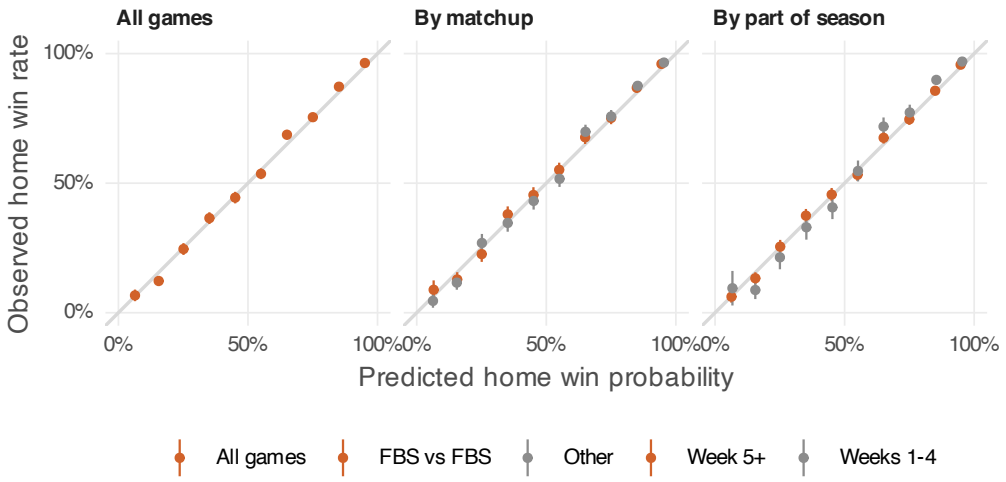



Figure 10: Calibration of out-of-sample home win probabilities. Games are grouped into fixed-width bins of predicted probability, $[0, 0.1], (0.1, 0.2], \ldots$; each point is a bin's mean prediction and observed home win rate, with a 95% binomial interval. The diagonal is perfect calibration. Left: all games. Centre: FBS–FBS games and the rest. Right: predicted weeks 1–4 and week 5 on.

Every bin lies within 3.6 percentage points of the diagonal, but with 18,958 games a gap of a few points can be real. 4 of the 10 bins are more than two standard errors from it: (0.1,0.2], predicted 0.155 against observed 0.122 with $n = 1{,}078$ ($z = -3.3$); (0.6,0.7], predicted 0.650 against observed 0.687 with $n = 2{,}341$ ($z = 3.8$); (0.8,0.9], predicted 0.851 against observed 0.872 with $n = 2{,}522$ ($z = 3.2$); (0.9,1], predicted 0.951 against observed 0.964 with $n = 2{,}932$ ($z = 3.7$). The expected calibration error, the game-weighted mean gap between prediction and outcome within bins, is 0.016, and Spiegelhalter's test [47] gives $z = -4.8$ ($p = 1.5e-06$). The splits locate the misfit. In weeks 1–4, $z = -6.6$ (expected calibration error 0.032); from week 5, $z = -1.3$ (0.013). For FBS–FBS games $z = -2.6$, and for the rest $z = -4.2$. The early-season and cross-tier probabilities are thus less well calibrated than the rest: they lean on the preseason carry-over and on the tier offsets, both of which are least certain before any game.

The calibration of predicted games bears on the resume only indirectly. The benchmark probabilities $\pi_g$ of Equation 7 use the same predictive distribution, but they describe a hypothetical fixture — the benchmark against the opponent — and are computed in sample, from the posterior that includes the game itself; no outcome can check them directly. That the model's out-of-sample probabilities for real fixtures are close to calibrated, especially in the second half of the season when resumes matter, is supporting evidence that the scale of $\pi_g$ is right, not proof.

Figure 11 repeats the comparison with the line season by season. The gap is stable, which suggests it reflects information the market has and the model does not rather than a drifting model.

Table 18: Rolling out-of-sample accuracy of model variants, every game of 2022–2025, each predicted from the posterior through the previous week. The published model has the Student-$t$ margin and the weekly random walk; each variant changes one thing: no efficiency likelihood, no carry-over, no conference level, ratings frozen within the season, or another margin family. $\Delta$ columns: difference from the published model on the same games, with a standard error clustered by season-week; positive is worse.

| Variant | MAE | Brier | CRPS | $\Delta$ MAE | $\Delta$ Brier $\times 10^3$ | $\Delta$ CRPS |
|---|---|---|---|---|---|---|
| Published model | 12.53 | 0.1657 | 8.96 | | | |
| Margin only | 12.52 | 0.1655 | 8.95 | -0.01 (0.01) | -0.3 (0.2) | -0.01 (0.01) |
| No carry-over | 13.35 | 0.1758 | 9.56 | +0.82 (0.17) | +10.0 (1.7) | +0.60 (0.11) |
| No conference level | 12.53 | 0.1658 | 8.96 | +0.01 (0.01) | +0.1 (0.2) | +0.01 (0.01) |
| Static ratings | 12.54 | 0.1660 | 8.97 | +0.01 (0.01) | +0.2 (0.1) | +0.01 (0.00) |
| Normal margin | 12.53 | 0.1656 | 8.96 | -0.00 (0.01) | -0.1 (0.2) | +0.00 (0.01) |
| Mixture margin | 12.57 | 0.1664 | 8.99 | +0.04 (0.02) | +0.6 (0.3) | +0.03 (0.01) |
| Scale-t margin | 12.56 | 0.1665 | 9.00 | +0.03 (0.01) | +0.7 (0.3) | +0.04 (0.01) |

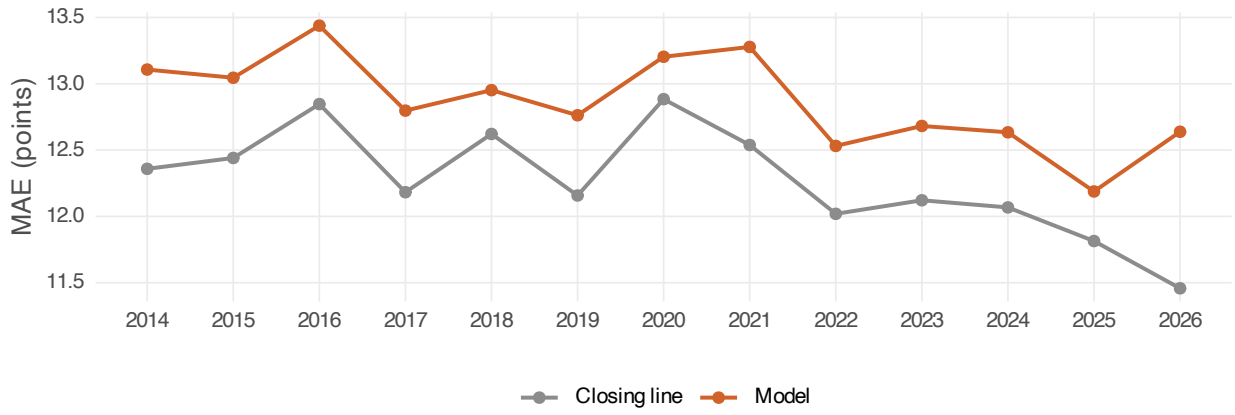



Figure 11: Out-of-sample mean absolute error by season on FBS–FBS games, model and closing line on the same games. The last season is in progress.

## 8.4 Ablations and Baselines

Each component of the model is tested by removing it and repeating the rolling backtest on the same games. Refitting every week of every season for each variant is expensive, so the ablations cover the 4 most recent complete seasons, 2022–2025 (6,557 games), with shorter chains than the published fits (two chains of 400 draws; the predictions are posterior means, for which that is ample). The published configuration is refitted the same way, so every row of Table 18 is comparable.

Removing a component makes the spreads measurably worse (by more than two standard errors) for: no carry-over, static ratings, scale-t margin. No variant beats it by more than two standard errors. The margin-only model settles the question of Section 4 empirically: the efficiency

Table 19: Baselines against the published model on the same games (predicted weeks 2 and later, from the second season). Least squares covers every game; the CFBD Elo ratings exist only for games between rated teams.

| Baseline | Games | N | MAE, baseline | MAE, model | Brier, baseline | Brier, model |
|---|---|---|---|---|---|---|
| CFBD Elo | All games | 8,104 | 13.11 | 12.77 | 0.185 | 0.180 |
| CFBD Elo | FBS vs FBS | 8,091 | 13.11 | 12.78 | 0.185 | 0.180 |
| Least squares (Massey) | All games | 15,919 | 13.64 | 12.72 | 0.182 | 0.170 |
| Least squares (Massey) | FBS vs FBS | 8,084 | 13.60 | 12.78 | 0.192 | 0.180 |

likelihood changes the mean absolute error by +0.01 points and the Brier score by +0.3 $\times 10^{-3}$ (published minus margin-only), in line with the few-percent information gain the proposition predicts. The carry-over matters most in the first weeks, where without it every team starts at its conference mean. Freezing ratings within the season (*static ratings*, $\phi_r = 0$) changes the mean absolute error by +0.01 points (se 0.01) and the CRPS by +0.01 (se 0.00) relative to the random walk: letting ratings drift helps measurably. The other margin families, fitted with the same published priors and carry-over, change the CRPS and the Brier score by +0.00 and -0.1 $\times 10^{-3}$ (normal); +0.03 and +0.6 $\times 10^{-3}$ (mixture); +0.04 and +0.7 $\times 10^{-3}$ (scale-t); the family shapes the distribution of the margin, which the CRPS scores, more than its centre or the win probability.

The model can also be compared with rating systems built differently. Table 19 scores two baselines on the same games from 2015 (their win-probability scales are fitted on earlier seasons only): least-squares ratings on the margin, refitted weekly with a weak ridge penalty (the Massey method, unpooled), and the pregame Elo ratings that CFBD publishes, mapped to a spread and a win probability by regressions on earlier seasons. Published SP+ and FPI ratings are available only at the end of each season, after the games they would predict, so they cannot be backtested; at the end of each complete season the final power rating's rank correlation with SP+ is 0.91–0.97 and with FPI 0.98–1.00.

## 8.5 Priors Set Without the Season

The published priors pool every complete season, so the early weeks of, say, 2016 are predicted with priors informed by 2017–2025. To measure what that borrows, weeks 0–3 of every season from 2015 are refitted with priors pooled from the *other* seasons only, and their predictions of weeks 1–4 scored against the published ones on the same 5,046 games. The leave-one-season-out predictions have a mean absolute error of 13.40 points against 13.38 for the published ones, and a Brier score of 0.1352 against 0.1350: the leakage from pooling is negligible in the weeks where the priors matter most, and none at all later, when the data dominate them.

## 8.6 Preseason Ratings

Week-1 games are predicted from the preseason posterior, which rests on last season's final ratings through the carry-over $\beta$. The preseason power ranks agree with the preseason AP poll on the teams it ranks (rank correlation 0.57–0.82 by season), although the poll uses information the model

lacks — recruiting, coaching changes, returning starters and transfers. Against the season's final ratings, the preseason ratings have a rank correlation of 0.76–0.90. A comparison with preseason SP+ would be the natural next check, but CFBD publishes SP+ only as an end-of-season rating.

The transfer portal and the one-time transfer exception, from 2021, might be expected to weaken the link between seasons. Figure 12 shows $\beta$ by season. The mean estimate is 0.76 for 2015–2020 and 0.77 for 2021–2025; with posterior standard deviations of about 0.05 per season, the data show no clear change.

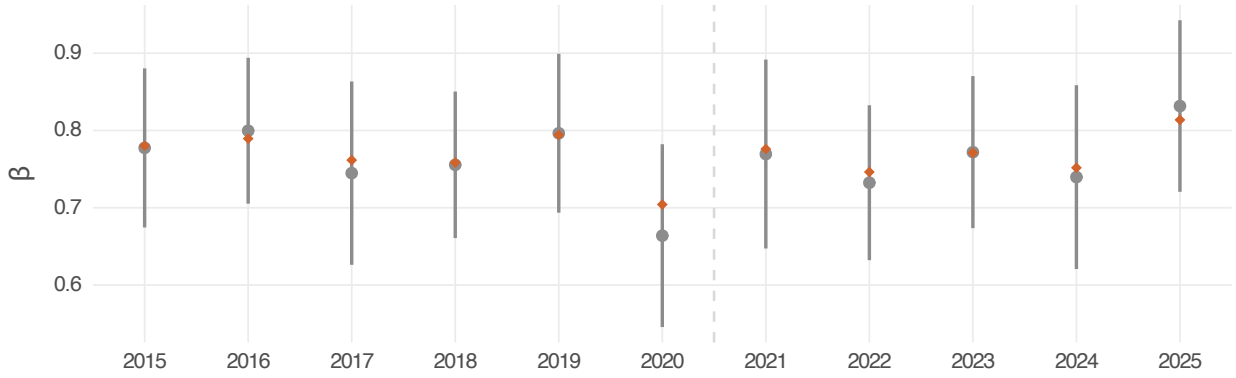



Figure 12: The carry-over $\beta$ by season: full-season posterior mean $\pm 2$ sd under vague priors (grey) and the published final posterior mean (orange). The first season has no previous ratings, so $\beta$ is not identified there.

## 8.7 Tuned Choices

Every choice that was tuned or set from data is listed in Table 20, with the data it used. Several used the same seasons that the backtest reports, which makes the reported backtest slightly optimistic for those choices; Section 8.5 bounds the effect for the pooled priors. The margin family was chosen on in-sample LOO at each season's final week, not on the backtest, but on the same seasons.

# 9 The 2025 Season

This section uses the 2025 season to show what the two products look like and how they relate. Most of it uses the final posterior, after the postseason.

*Example.* The 2025 national championship game, Indiana against Miami at a neutral site, was predicted by the posterior through the week before it. There the posterior mean ratings were $r = 31.6$ and $20.1$ points, so the predicted spread is 11.4 points for Indiana (there is no home field, $\delta_g = 0$). With the posterior means $\bar{\sigma} = 12.94$ and $\nu = 10.8$, the plug-in win probability is $F_\nu(11.4/12.94) = 0.802$. The published probability averages $1 - F^{(s)}(0 \mid \cdot)$ over the posterior draws instead, 0.784, which is closer to one half because the uncertainty in the two ratings and a week of random-walk drift (together a posterior sd of the spread of 5.8 points) widen the predictive

Table 20: Every tuned or data-driven setting, how it was set, and whether the data overlap the reported backtest.

| Setting | Value | How set |
|---|---|---|
| Structural priors (12 parameters) | Table 5 | Pooled full-season refits under vague priors, all co |
| Margin family | Student-$t$ | Pooled PSIS-LOO, final week of every complete sea |
| Random-walk step priors $\phi_r, \phi_t$ | Table 6 | Preliminary full-season fit, 2025 |
| Total-model priors | Table 6 | FBS games, 2022–2025 |
| Seasons left out of the pooling | 2014; 2020 for $\eta$ | Structural reasons, not tuned |
| Benchmark rank $b$ | 12 | Playoff size (Section 5.4) |
| Family shape priors | Gamma(2, 0.1), $\nu > 1$ | Literature default, weakly informative |
| Minimum plays for $e_g$ | 20 | Judgement |
| Garbage-time filter | provider's | CFBD definition |
| Sampler and gates | 4 chains, 500/1000 | Conventional thresholds |

distribution. The total model projected a score of 31–20, a total of 50.9 against a closing over/under of 47.2. The closing line had Indiana favoured by 7.2, and the result was 27–21.

Table 21 lists the top of the final resume.

The rank intervals are the point of reporting a posterior. Figure 13 shows them for the top 25 in both products. Among the resume top 12 the median width of the 80% rank interval is 4 places for the resume and 12 for power. Resume ranks are the sharper, because once the results are fixed only the valuation of the opponents is uncertain, while power ranks describe strength itself.

Table 21: Final 2025 FBS resume. WAB with its 80% interval; the resume rank with its 80% rank interval; the power rating in points above an average FBS team with its rank; and the Colley and final AP ranks for reference.

| | Team | Record | WAB | Rank 80% | $\Pr(\leq 12)$ | Power | Colley | AP |
|---|---|---|---|---|---|---|---|---|
| 1 | Indiana | 16–0 | 5.29 (4.8, 5.8) | 1–1 | 1.00 | 35.0 (1) | 1 | 1 |
| 2 | Oregon | 13–2 | 3.31 (2.8, 3.8) | 2–2 | 1.00 | 26.2 (4) | 2 | 4 |
| 3 | Miami | 13–3 | 2.31 (1.7, 2.9) | 3–5 | 1.00 | 22.8 (6) | 4 | 2 |
| 4 | Ohio State | 12–2 | 2.20 (1.7, 2.7) | 3–7 | 1.00 | 29.0 (2) | 3 | 5 |
| 5 | Georgia | 12–2 | 1.85 (1.3, 2.4) | 4–8 | 1.00 | 20.4 (8) | 6 | 6 |
| 6 | Ole Miss | 13–2 | 1.76 (1.2, 2.3) | 4–8 | 1.00 | 19.8 (9) | 5 | 3 |
| 7 | Texas A&M | 11–2 | 1.66 (1.2, 2.1) | 5–9 | 1.00 | 16.7 (15) | 12 | 8 |
| 8 | BYU | 12–2 | 1.57 (1.0, 2.2) | 5–10 | 1.00 | 14.5 (19) | 7 | 11 |
| 9 | Alabama | 11–4 | 1.25 (0.7, 1.8) | 7–11 | 0.98 | 17.8 (10) | 13 | 9 |
| 10 | Texas Tech | 12–2 | 1.13 (0.6, 1.7) | 7–12 | 0.93 | 23.6 (5) | 8 | 7 |
| 11 | Texas | 10–3 | 0.81 (0.4, 1.2) | 10–13 | 0.81 | 16.4 (17) | 19 | 12 |
| 12 | Oklahoma | 10–3 | 0.66 (0.1, 1.1) | 10–15 | 0.58 | 15.6 (18) | 9 | 13 |
| 13 | Notre Dame | 10–2 | 0.57 (0.2, 1.0) | 11–15 | 0.45 | 27.5 (3) | 10 | 10 |
| 14 | Utah | 11–2 | 0.29 (-0.1, 0.8) | 12–17 | 0.14 | 21.0 (7) | 14 | 14 |
| 15 | Vanderbilt | 10–3 | 0.27 (-0.2, 0.7) | 13–17 | 0.07 | 16.5 (16) | 22 | 15 |
| 16 | Iowa | 9–4 | -0.03 (-0.5, 0.4) | 14–19 | 0.04 | 17.5 (11) | 26 | 17 |
| 17 | Navy | 11–2 | -0.11 (-0.4, 0.3) | 15–20 | 0.00 | 1.6 (66) | 18 | 23 |
| 18 | Illinois | 9–4 | -0.21 (-0.6, 0.2) | 15–20 | 0.00 | 11.4 (23) | 21 | – |
| 19 | James Madison | 12–2 | -0.35 (-0.6, -0.0) | 17–21 | 0.00 | 9.6 (30) | 16 | 19 |
| 20 | USC | 9–4 | -0.51 (-0.9, -0.1) | 18–23 | 0.00 | 17.3 (13) | 20 | 20 |
| 21 | Michigan | 9–4 | -0.51 (-0.9, -0.1) | 18–24 | 0.00 | 13.4 (20) | 23 | 21 |
| 22 | Tulane | 11–3 | -0.64 (-1.0, -0.2) | 19–25 | 0.00 | 4.3 (54) | 15 | 18 |
| 23 | North Texas | 12–2 | -0.82 (-1.1, -0.5) | 21–25 | 0.00 | 8.9 (33) | 11 | 24 |
| 24 | Houston | 10–3 | -0.99 (-1.4, -0.5) | 22–27 | 0.00 | 5.4 (47) | 24 | 22 |
| 25 | Virginia | 11–3 | -1.00 (-1.5, -0.5) | 21–27 | 0.00 | 8.4 (36) | 17 | 16 |

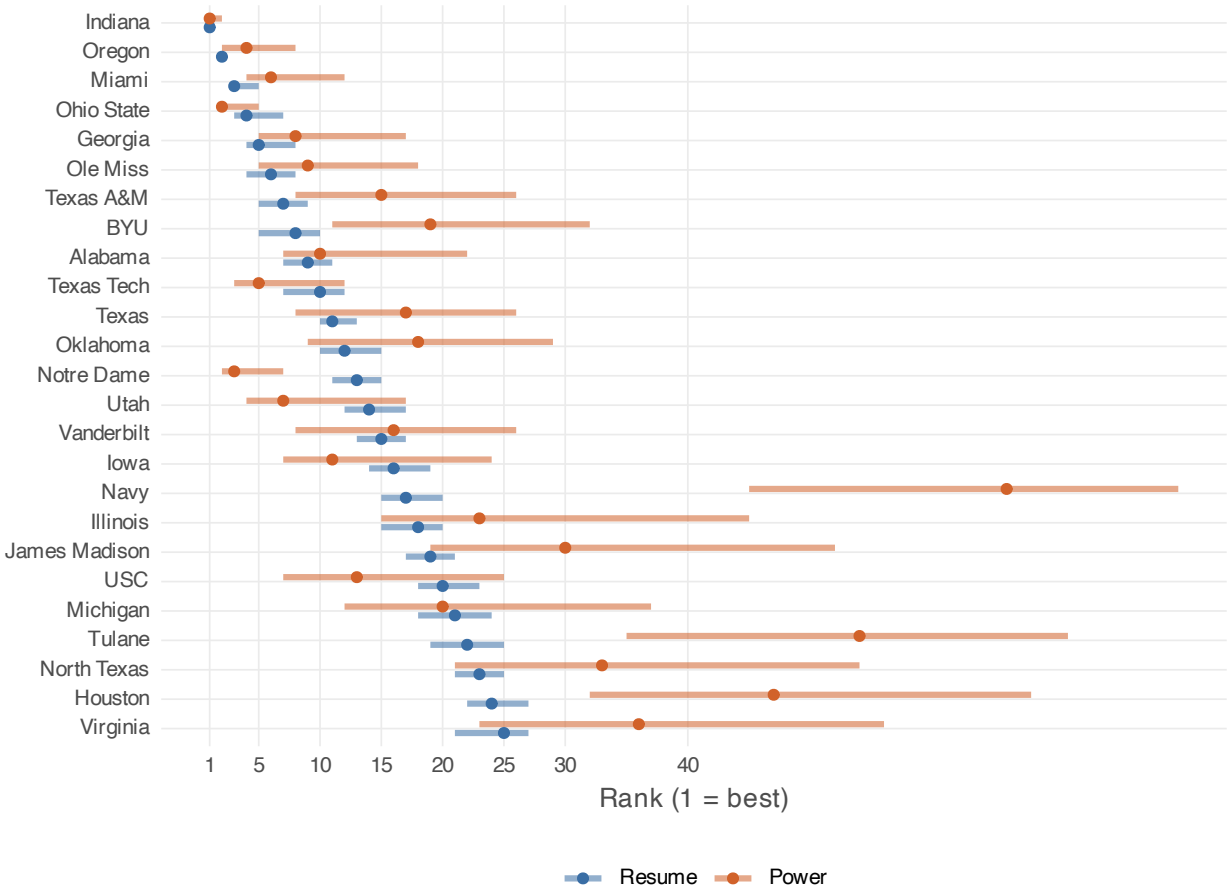



Figure 13: Posterior rank intervals at the end of 2025 for the top 25 of the resume. Points are the published ranks; bars the central 80% of the rank across posterior draws. Teams are ordered by resume rank.

Figure 14 plots the two products against each other for every FBS team. The relation is monotone but loose. The right panel shows what the scatter around the curve is: the vertical distance of each team from the curve against its *win luck*, actual wins minus the wins its own rating would predict against its schedule (the sum of its win probabilities in each game under the final posterior). The two are correlated at 0.90: teams above the curve have won more games than their strength predicts, often close ones, and teams below it have lost them.

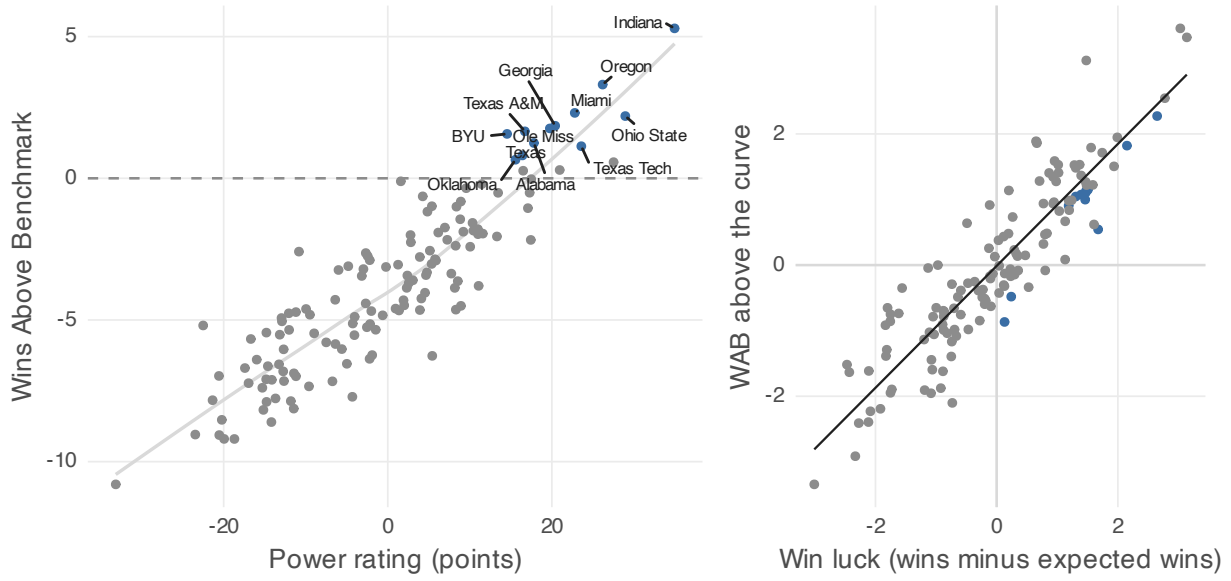



Figure 14: Resume against power at the end of 2025, all FBS teams. Left: the dashed line is zero Wins Above Benchmark, teams above it did better than the twelfth-best team would have with their schedule; the grey curve is a LOESS fit; labelled, the resume top 12. Right: each team's distance above or below that curve against its win luck, actual minus expected wins.

Figure 15 follows the power rating of the resume top six through the season. Preseason ratings come from the carry-over alone and are wide: the median 80% interval is 22.0 points in week 0, 11.4 in week 8, and 9.3 at the end of the season.

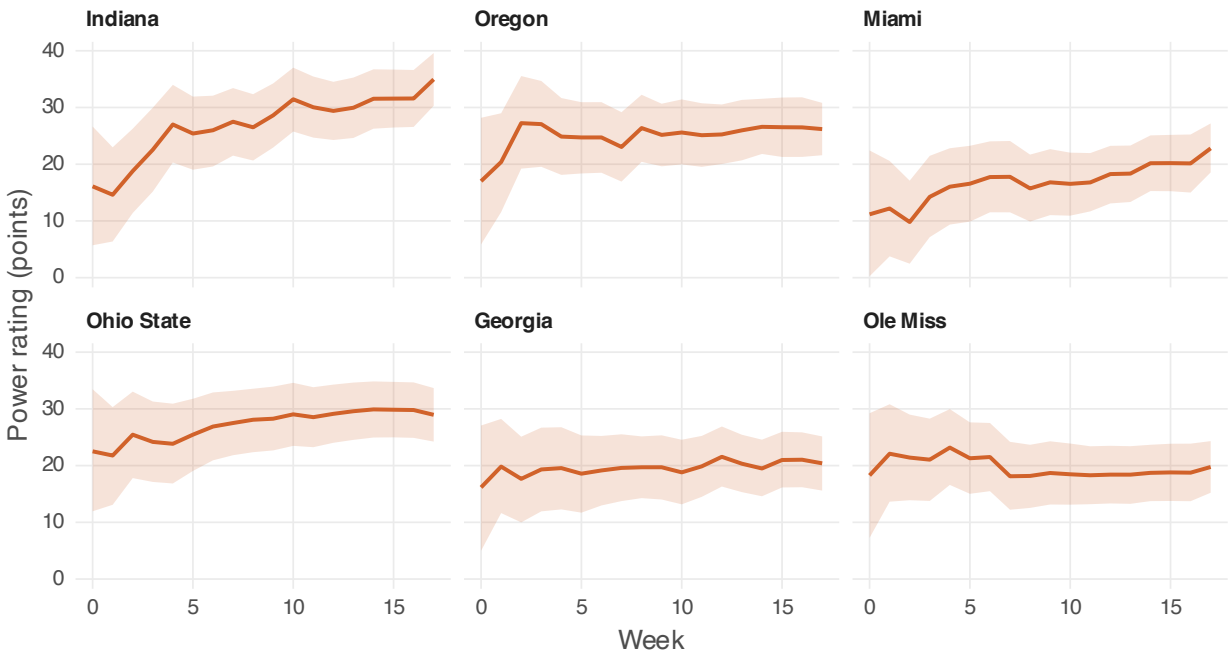



Figure 15: Weekly power rating in 2025 for the final resume top six, with 80% posterior intervals. Week 0 is the preseason posterior; the last point is the final posterior after the postseason.

With the random walk a rating can fall as well as rise late in the season, and the bands do not narrow to zero: the walk keeps adding uncertainty of $\phi_r$ per week while each game removes some.

Figure 16 splits the final 2025 power ratings into offence and defence (Section 3.3). The best offence by this measure is Indiana's, 44.0 points against an average FBS team, and the best defence Indiana's, 9.1 points allowed. Across FBS teams good offences and good defences tend to go together (correlation of Off with $-$Def 0.77), and teams with the same power rating can get there in very different ways: every point on one diagonal of the figure has the same Off $-$ Def.

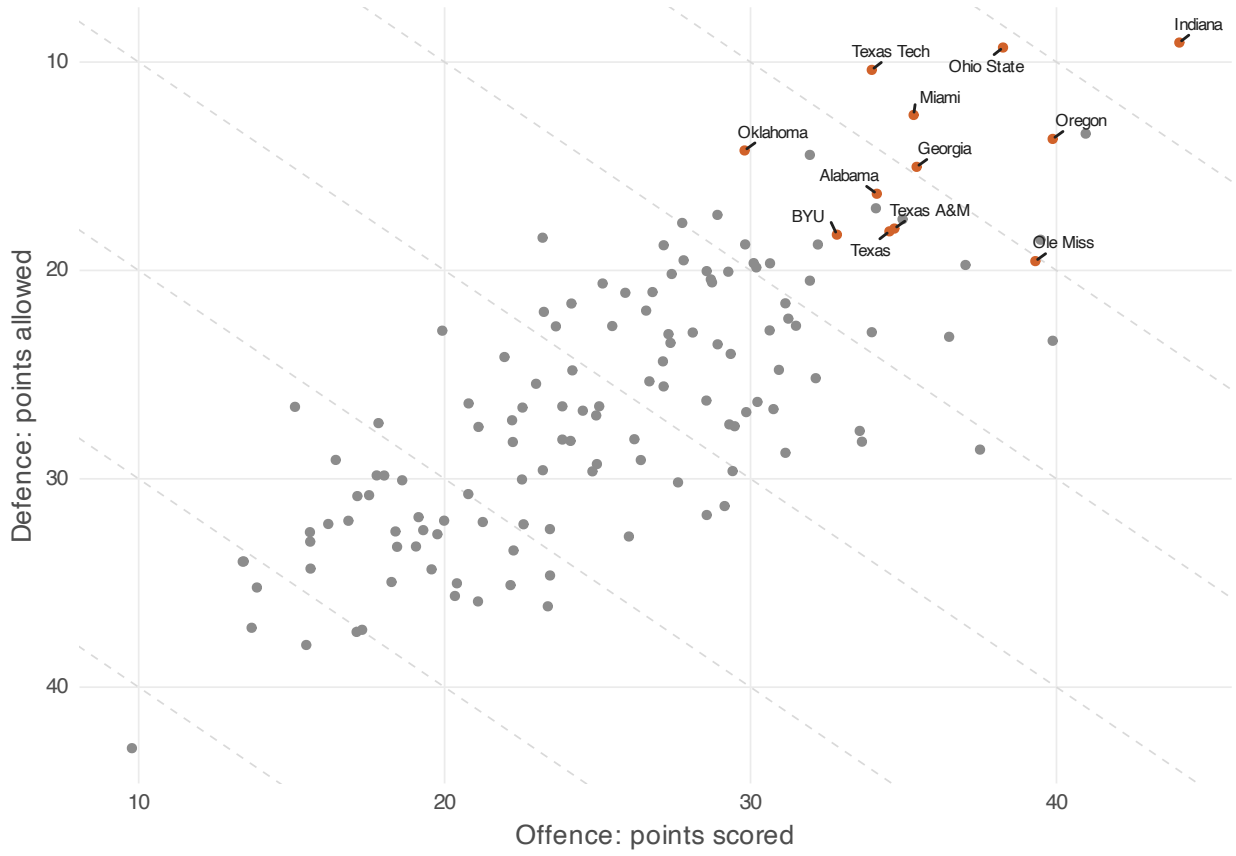



Figure 16: Offence and defence at the end of 2025, all FBS teams: points scored and allowed against an average FBS team on a neutral field. The defence axis is reversed, so better teams are up and to the right; dashed diagonals are lines of equal power rating, 10 points apart. Labelled, the resume top 12.

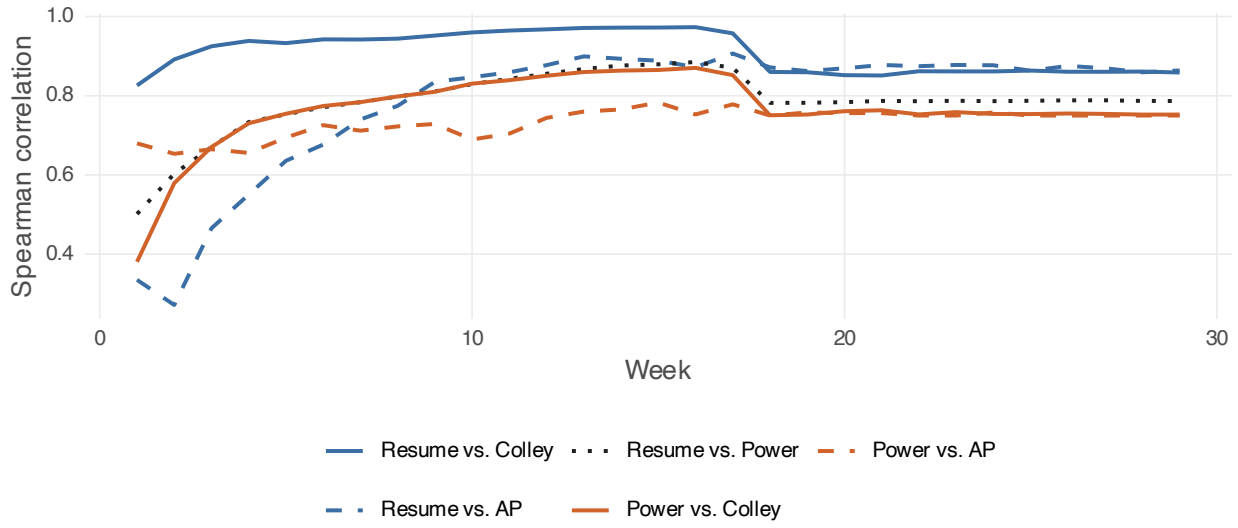



Figure 17: Spearman rank correlation between rankings of FBS teams, by week, averaged over complete seasons. Correlations with the AP poll use the teams it ranks.

Figure 17 measures agreement between the products and the references by Spearman rank correlation among FBS teams, week by week, averaged over the complete seasons. The resume tracks the record-only Colley ranking closely from the first weeks (mean correlation 0.88 in weeks 1–3 and 0.97 in weeks 10–14) — both are functions of the record — and diverges from it only in how opponents are valued. The power rating is less correlated with Colley (0.54 early, 0.85 late), as it should be, because it uses margin and efficiency. Agreement with the AP poll, which ranks only 25 teams, is computed on those 25.

*Example.* The products disagree most for teams that are strong but lost games. In 2025 the highest-rated team outside the resume top 25 is Penn State: power rank 12 (17.4 points) but resume rank 40, at 7–6. Table 22 shows why, game by game. Its own posterior gave it an expected 8.9 wins against this schedule, so its win luck is -1.9; 6 losses, 5 of them by one score, cost it 2.91 wins of WAB, while its 7 wins earned 0.73. The power rating reads those losses as close games that a strong team loses sometimes; the resume counts them as losses.

## 9.1 The 2025 Playoff Field

The first question a reader will ask of a resume is whether it picks the teams the committee picked. Table 23 compares the 2025 playoff field with the model at the committee's final ranking (the posterior through the conference championships, week 15). The resume's top 12 contains 9 of the 12 teams in the field. In the field but outside the resume top 12: Miami (resume 13, CFP 10); Tulane (resume 20, CFP 20); James Madison (resume 18, CFP 24). In the resume top 12 but not in the field: BYU (resume 7, CFP 12); Texas (resume 12, CFP 13); Vanderbilt (resume 11, CFP 14). The field is not simply the committee's top 12: the five highest-ranked conference champions qualify automatically, which is how a team outside the top 12 of any ranking can be selected.

Table 22: Resume against power, game by game, for Penn State in 2025. $p_g$: the team's own win probability under the final posterior; $\bar{\pi}_g$: the benchmark's.

| Week | Opponent | Score | $p_g$ | $\bar{\pi}_g$ | $W_g - \bar{\pi}_g$ |
|---|---|---|---|---|---|
| Week 1 | Nevada | 46–11 | 0.98 | 0.99 | 0.01 |
| Week 2 | Florida International | 34–0 | 0.98 | 0.98 | 0.02 |
| Week 3 | Villanova | 52–6 | 0.99 | 0.99 | 0.01 |
| Week 5 | Oregon | 24–30 | 0.33 | 0.36 | -0.36 |
| Week 6 | UCLA | 37–42 | 0.87 | 0.90 | -0.90 |
| Week 7 | Northwestern | 21–22 | 0.83 | 0.86 | -0.86 |
| Week 8 | Iowa | 24–25 | 0.42 | 0.46 | -0.46 |
| Week 10 | Ohio State | 14–38 | 0.17 | 0.18 | -0.18 |
| Week 11 | Indiana | 24–27 | 0.15 | 0.16 | -0.16 |
| Week 12 | Michigan State | 28–10 | 0.82 | 0.85 | 0.15 |
| Week 13 | Nebraska | 37–10 | 0.84 | 0.86 | 0.14 |
| Week 14 | Rutgers | 40–36 | 0.80 | 0.83 | 0.17 |
| Postseason | Clemson | 22–10 | 0.73 | 0.77 | 0.23 |

Table 23: The 2025 playoff field and the model at the committee's final ranking. Teams in the field, in the resume top 12, or in the committee's top 12. Field: selected for the playoff.

| Team | Record | Field | CFP | Resume | WAB | SOR | Power | AP |
|---|---|---|---|---|---|---|---|---|
| Indiana | 13–0 | yes | 1 | 1 | 3.39 | 0.0048 | 1 | 1 |
| Ohio State | 12–1 | yes | 2 | 2 | 2.42 | 0.0690 | 2 | 3 |
| Georgia | 12–1 | yes | 3 | 4 | 2.36 | 0.0940 | 6 | 2 |
| Texas Tech | 12–1 | yes | 4 | 8 | 1.46 | 0.2300 | 5 | 4 |
| Oregon | 11–1 | yes | 5 | 3 | 2.41 | 0.0670 | 4 | 5 |
| Ole Miss | 11–1 | yes | 6 | 6 | 1.55 | 0.2100 | 10 | 6 |
| Texas A&M | 11–1 | yes | 7 | 5 | 2.15 | 0.1000 | 13 | 7 |
| Oklahoma | 10–2 | yes | 8 | 9 | 1.29 | 0.3000 | 15 | 8 |
| Alabama | 10–3 | yes | 9 | 10 | 1.01 | 0.3800 | 9 | 11 |
| Miami | 10–2 | yes | 10 | 13 | 0.56 | 0.5000 | 8 | 10 |
| Notre Dame | 10–2 |  | 11 | 14 | 0.49 | 0.5200 | 3 | 9 |
| BYU | 11–2 |  | 12 | 7 | 1.48 | 0.2400 | 19 | 12 |
| Texas | 9–3 |  | 13 | 12 | 0.58 | 0.4900 | 17 | 14 |
| Vanderbilt | 10–2 |  | 14 | 11 | 0.86 | 0.4100 | 12 | 13 |
| Tulane | 11–2 | yes | 20 | 20 | -0.33 | 0.7900 | 49 | 17 |
| James Madison | 12–1 | yes | 24 | 18 | -0.18 | 0.8200 | 32 | 19 |

Table 24: Rankings against the committee, 2014–2025 (mean over seasons), at the snapshot before the committee's final ranking, and retrodictive accuracy at the end of each season. SOR ranks teams by strength of record.

| Ranking | Field overlap | Top 4 (of 4) | Top 12 (of 12) | $\rho$, CFP top 25 | Retrodictive accuracy |
|---|---|---|---|---|---|
| Resume | 85% | 3.42 | 10.2 | 0.90 | 80.9% |
| SOR | 83% | 3.33 | 10.2 | 0.90 | 81.7% |
| Colley | 83% | 3.25 | 10.2 | 0.83 | 81.6% |
| Power | 68% | 2.50 | 9.2 | 0.78 | 78.1% |
| AP | 94% | 3.75 | 11.1 | 0.96 | – |

## 9.2 The Resume Against the Committee

A retrodictive ranking has no future outcome to score it against, so the evidence for the resume has to be external. Table 24 compares each ranking with the committee's final ranking in every season since the playoff began: overlap with the playoff field (the top four through 2023, the twelve selected teams since 2024), with the committee's top 4 and top 12, and the rank correlation over the committee's 25. It also reports retrodictive accuracy: the share of the season's FBS–FBS games whose winner the final ranking places higher. The legacy composite that this model replaced is not included, because its code was retired with the defects listed in the project plan and cannot be rerun on the same data.

The AP poll agrees with the committee most closely, which is unsurprising: both are made by people watching the same games, and the poll is known to the committee. Among the computer rankings the resume and SOR match the committee's field in 85% and 83% of places, Colley in 83% and the power rating in 68%. The committee selects on results, as the resume does, more than on strength. Retrodictive accuracy is 80.9% for the resume, 81.7% for SOR, 81.6% for Colley and 78.1% for power: a ranking built from the record explains the record better than one built to predict.

# 10 Discussion

The design principle of this paper is that a ranking should say which question it answers, and that different questions can share a model without sharing a formula. The power rating and the resume come from one posterior and are never mixed: the resume borrows the model's estimate of how hard each game was, and nothing else. That separation has a cost — two numbers where the public expects one — but each number can then be checked against something observable: the power rating against future games (Section 8), and the resume, less directly, against the committee's selections and the season's own results (Section 9.2).

**Limitations.** Several choices deserve more work.

1. *The margin distribution near zero.* The model underpredicts one-score games in 13 of 13 seasons (Section 7.1), and neither a two-normal scale mixture nor a scale that grows with the mismatch removes the misfit, although the latter improves the LOO density considerably (Section 7.4). The remaining reason is that margins are integers with spikes at key numbers

(3 and 7) and at zero (overtime games, whose regulation margin is zero by construction), while the likelihood is continuous. This affects spread and cover probabilities rather than win probabilities. Alternatives that respect the discreteness are models for the two scores themselves, such as the bivariate Poisson of Karlis and Ntzoufras [48] or the hierarchical score model of Baio and Blangiardo [49], or a rounded (interval-censored) version of the present likelihood. They should be compared on the margin's LOO density.

2. *One step size for every team and week.* The random walk of Equation 4 moves every team by Gaussian steps of the same size, so a team that loses its quarterback in October is pulled down only gradually, at the rate the whole sport's drift allows, and a slow decline and a sudden one look alike. Heavier-tailed steps, or a step size that varies by team or by part of the season, would let a rating jump; the data identify even one step size only weakly (Section 3.2), so such extensions would need informative structure, such as injury reports, rather than more parameters alone.
3. *Asymmetric garbage time.* The efficiency margin excludes garbage time (the provider's definition), but the scoreboard margin cannot: a team that leads by 30 and rests its starters concedes late points that enter $y_g$ but not $e_g$. That shrinks the scoreboard margin of lopsided games relative to their efficiency margin, which the model can only absorb in $\lambda$ (the efficiency scale per point of margin) and $\kappa$; both are therefore averages over close and lopsided games. A margin-dependent $\lambda$, or a garbage-time-filtered scoreboard margin, would separate the two.
4. *Priors from the whole history.* The structural priors use every complete season (Section 3.4). The leave-one-season-out comparison of Section 8.5 bounds what that borrows; a strictly prospective version would set each season's priors from earlier seasons only.
5. *A simple total model.* Offence and defence come from a total model that is Gaussian, independent of the margin given the ratings, and blind to the score of overtime games (Section 3.3). It trails the closing over/under (Section 8.2), and the priors of the total model and of the random-walk steps are provisional: they were set from preliminary fits rather than pooled from the vague-prior refits as the other structural priors were, and the twelve pooled priors themselves date from the static Student-$t$ model (Section 3.4). Re-pooling all of them under the published model is the next step.
6. *Games are exchangeable.* Motivation, rest, travel, injuries and weather are all absent, and home field is one number per season for every venue (Figure 3); the gap to the closing line in Section 8 is a measure of how much they matter.

# 11 Appendix: Supplementary Posterior Checks

The weekly posteriors of Section 6 are checked by gates and summaries only, because keeping every draw of 226 fits is not needed for publication. This appendix examines two posteriors in full, following the checklist of Kahle et al. [50] and Miyakawa and Kahle [51] for applied Bayesian computation — sampler behaviour; convergence (smooth, chain-consistent marginal densities, well-mixed traces, quickly damping autocorrelation, a Gelman–Rubin statistic approaching one); posterior geometry; the justification of informative priors and the priors they induce on derived quantities; sensitivity of the posterior to changes in the prior; predictive adequacy; and influence — together with the visual workflow of Gabry et al. [37]. The two posteriors bracket the range the

pipeline fits:

- **2025, week 1**: 141 games, about one per team. The data barely constrain a team's strength, the hierarchy and the carry-over do most of the work, and this is where the hierarchical geometry is hardest for the sampler.
- **2025, final**: all 1,687 games of the season, including the postseason.

Both are refitted with the production settings of Section 6.1 (4 chains, 1,000 post-warm-up draws each, 4,000 in all) by the `supplement` target in `R/06_supplement.R`.

## 11.1 Sampler Behaviour Across All Snapshots

Before the two detailed fits, Figure 18 shows the gate statistics of every weekly posterior. The retry of Section 6.1 is visible as the fits with doubled sampling.

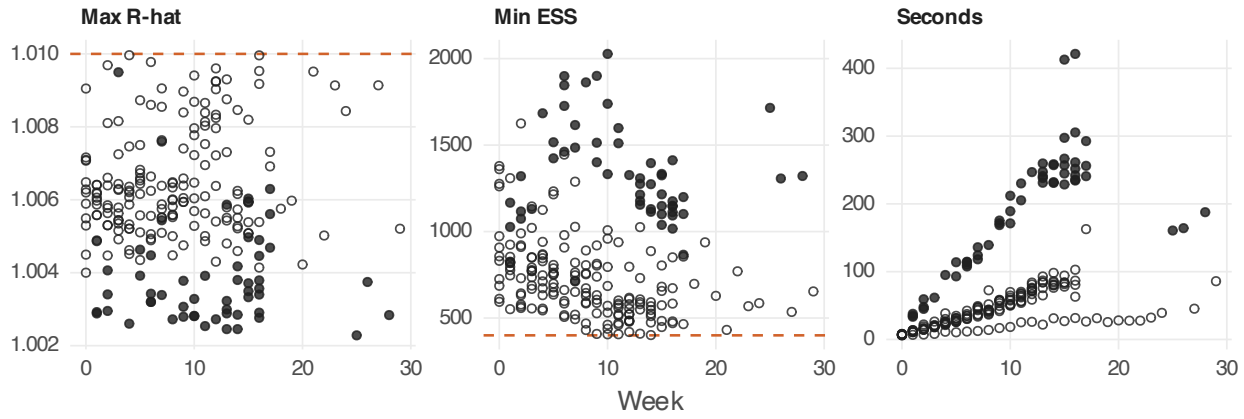



Figure 18: Gate statistics of all weekly posteriors by week, every season overlaid. Left: the largest rank-normalised $\widehat{R}$ over all monitored parameters, with the gate at 1.01. Centre: the smallest bulk or tail effective sample size, with the gate at 400. Right: wall-clock time of the fit, four chains in parallel. Filled points are fits that needed the retry.

## 11.2 Sampler Diagnostics

Table 25 reports the Hamiltonian Monte Carlo diagnostics of each chain [28]: the adapted step size, the mean number of leapfrog steps per iteration, how often the trajectory hit the maximum tree depth (which would indicate a truncated, inefficient exploration), divergent transitions (which indicate curvature the integrator cannot follow and a potentially biased posterior), and the energy Bayesian fraction of missing information, E-BFMI [52].

Neither fit has a single divergent transition, no iteration reaches the maximum tree depth, and the smallest E-BFMI is 0.85, far above 0.2. The week-1 fit, with few data, needs no longer trajectories than the final one: the non-centred hierarchy keeps its geometry close to a product of normals.

Figure 19 compares the marginal energy distribution with the distribution of energy changes between iterations. When the second is much narrower than the first, the sampler explores the

Table 25: Sampler diagnostics by chain. E-BFMI below 0.2 would indicate that the momentum resampling cannot move the chain across energy levels fast enough; the maximum tree depth is 10.

| Fit | Chain | Step size | Leapfrog steps | Max depth | At max depth | Divergent | E-BFMI | Warm |
|---|---|---|---|---|---|---|---|---|
| 2025, week 1 | 1 | 0.163 | 31.0 | 5 | 0 | 0 | 0.90 | |
| | 2 | 0.153 | 31.0 | 5 | 0 | 0 | 0.85 | |
| | 3 | 0.165 | 31.0 | 5 | 0 | 0 | 0.97 | |
| | 4 | 0.155 | 31.0 | 5 | 0 | 0 | 0.89 | |
| 2025, final | 1 | 0.107 | 31.0 | 5 | 0 | 0 | 0.90 | |
| | 2 | 0.111 | 31.0 | 5 | 0 | 0 | 0.96 | |
| | 3 | 0.089 | 63.0 | 6 | 0 | 0 | 0.89 | |
| | 4 | 0.108 | 31.0 | 5 | 0 | 0 | 0.89 | |

energy levels of the posterior slowly and tails may be under-sampled; overlapping histograms are what a healthy fit looks like.

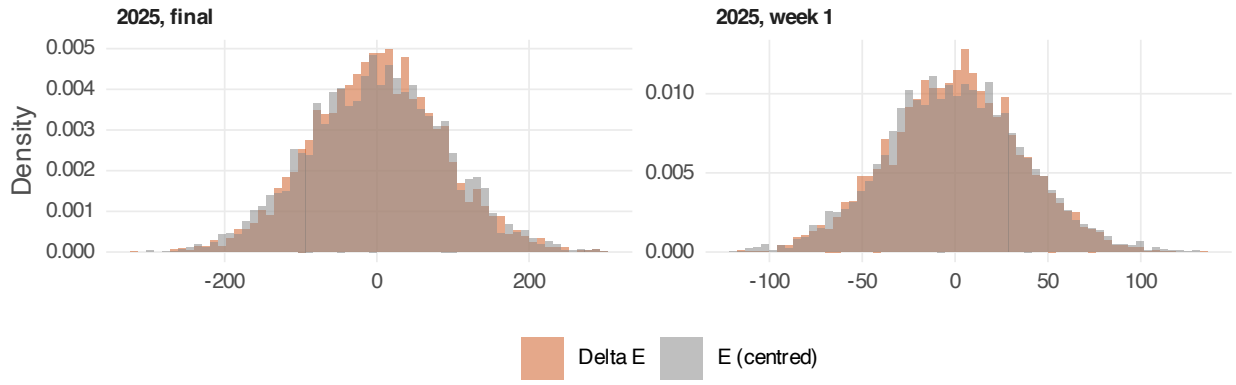



Figure 19: Energy diagnostic. Grey: marginal energy $E$, centred; orange: the change in energy between successive iterations, $\Delta E$. Pooled over chains.

## 11.3 Convergence

Table 26 gives posterior summaries and convergence diagnostics of the structural parameters: the rank-normalised split-$\widehat{R}$, which compares between- and within-chain variation after rank-normalising and folding the draws [34, 53]; bulk and tail effective sample sizes, which account for autocorrelation [54]; and the Monte Carlo standard error of the posterior mean.

Every structural $\widehat{R}$, including those of the 12 total-model parameters, is at most 1.007, and the smallest bulk effective sample size is 465, for $\phi_r$ in the final fit (smallest among the total model's: 629). The random-walk step $\phi_r$ mixes most slowly: the data inform it only weakly, and its posterior is correlated with every step of every team (Section 3.2). Monte Carlo standard errors are at most 4.6% of the posterior standard deviations.

Table 26: Posterior summaries and convergence diagnostics for the structural parameters of the margin model, the random-walk step and the family's shape parameters. *MCSE* is the Monte Carlo standard error of the posterior mean. The total model's parameters are summarised in the text.

| Fit | Parameter | Mean | SD | 5% | 95% | $\widehat{R}$ | ESS bulk | ESS tail | MCSE |
|---|---|---|---|---|---|---|---|---|---|
| 2025, week 1 | $\eta$ (home field) | 2.57 | 0.74 | 1.35 | 3.80 | 1.000 | 8,239 | 3,395 | 0.008 |
| | $\beta$ (carry-over) | 0.75 | 0.07 | 0.63 | 0.86 | 1.000 | 5,288 | 2,376 | 0.001 |
| | $\lambda$ | 0.70 | 0.03 | 0.65 | 0.74 | 1.001 | 5,604 | 3,343 | 0.000 |
| | $\kappa$ | 0.76 | 0.05 | 0.68 | 0.84 | 1.000 | 4,949 | 3,322 | 0.001 |
| | $\sigma_e$ | 13.97 | 0.59 | 13.03 | 14.93 | 1.000 | 6,693 | 3,176 | 0.007 |
| | $\sigma_m$ | 8.38 | 0.61 | 7.36 | 9.37 | 1.001 | 4,865 | 3,330 | 0.009 |
| | $\nu$ | 14.77 | 11.08 | 4.38 | 35.66 | 1.001 | 4,653 | 3,621 | 0.162 |
| | $\omega$ | 2.11 | 1.27 | 0.25 | 4.35 | 1.000 | 1,490 | 1,714 | 0.032 |
| | $\tau_1$ (FBS) | 6.14 | 1.22 | 4.09 | 8.10 | 1.000 | 3,107 | 2,955 | 0.022 |
| | $\tau_2$ (FCS) | 6.46 | 1.34 | 4.24 | 8.60 | 1.001 | 2,680 | 2,395 | 0.026 |
| | $\tau_3$ (lower) | 11.37 | 5.83 | 2.13 | 21.49 | 1.003 | 1,737 | 1,909 | 0.135 |
| | $m_2$ (FCS) | -29.37 | 2.18 | -32.96 | -25.75 | 1.002 | 6,310 | 3,358 | 0.027 |
| | $m_3$ (lower) | -64.46 | 5.30 | -73.15 | -55.77 | 1.001 | 6,172 | 3,091 | 0.068 |
| | $\phi_r$ | 0.60 | 0.24 | 0.22 | 1.00 | 1.001 | 3,207 | 1,456 | 0.004 |
| 2025, final | $\eta$ (home field) | 2.53 | 0.33 | 1.98 | 3.07 | 1.000 | 3,894 | 2,817 | 0.005 |
| | $\beta$ (carry-over) | 0.81 | 0.05 | 0.74 | 0.89 | 1.002 | 1,187 | 2,111 | 0.001 |
| | $\lambda$ | 0.76 | 0.01 | 0.74 | 0.78 | 1.002 | 4,542 | 3,155 | 0.000 |
| | $\kappa$ | 0.72 | 0.02 | 0.70 | 0.75 | 1.001 | 3,683 | 2,869 | 0.000 |
| | $\sigma_e$ | 14.06 | 0.25 | 13.65 | 14.47 | 1.000 | 4,490 | 2,749 | 0.004 |
| | $\sigma_m$ | 8.00 | 0.25 | 7.61 | 8.41 | 1.001 | 2,956 | 2,963 | 0.005 |
| | $\nu$ | 10.56 | 2.99 | 7.05 | 16.03 | 1.000 | 2,788 | 2,808 | 0.058 |
| | $\omega$ | 1.99 | 1.00 | 0.37 | 3.66 | 1.004 | 502 | 714 | 0.043 |
| | $\tau_1$ (FBS) | 6.53 | 0.65 | 5.51 | 7.60 | 1.000 | 1,304 | 2,407 | 0.018 |
| | $\tau_2$ (FCS) | 6.46 | 0.66 | 5.43 | 7.59 | 1.000 | 1,410 | 2,112 | 0.018 |
| | $\tau_3$ (lower) | 8.63 | 3.46 | 2.58 | 14.29 | 1.003 | 1,098 | 1,130 | 0.105 |
| | $m_2$ (FCS) | -31.17 | 1.64 | -33.86 | -28.46 | 1.000 | 2,120 | 2,528 | 0.036 |
| | $m_3$ (lower) | -64.69 | 3.32 | -70.07 | -59.19 | 1.001 | 2,482 | 2,642 | 0.067 |
| | $\phi_r$ | 0.61 | 0.22 | 0.24 | 0.97 | 1.007 | 465 | 990 | 0.010 |

The team strengths are too many to tabulate. Figure 20 shows the distribution of their $\widehat{R}$ and of their bulk effective sample size as a fraction of the number of draws.

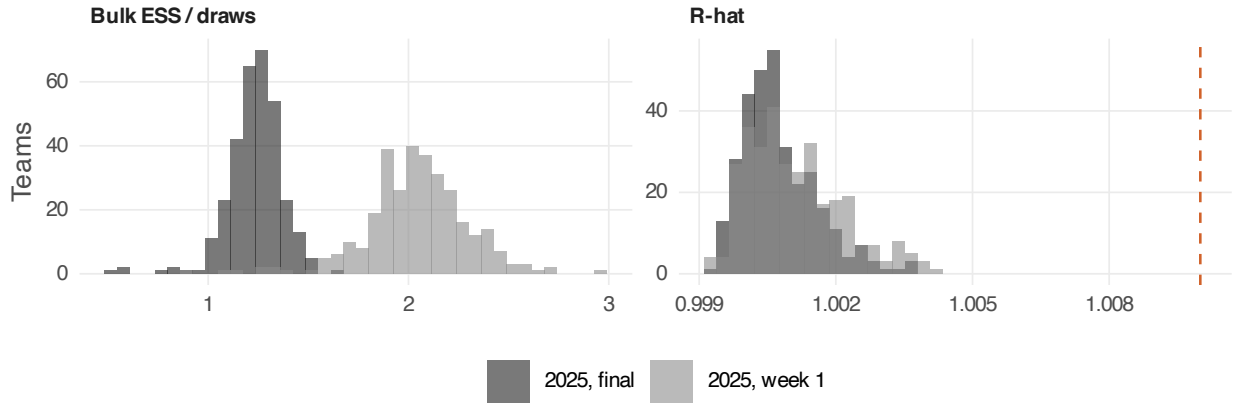



Figure 20: Convergence of all team strengths $r_i$. Left: rank-normalised $\widehat{R}$, with the gate at 1.01. Right: bulk effective sample size per draw; values above one are possible for anti-correlated chains, which NUTS often produces for well-identified parameters.

Trace plots (Figure 21, Figure 22) show the chains iteration by iteration; well-mixed chains overlap and look like stationary noise, with no trends, no stuck stretches and no chain wandering off on its own. Rank plots (Figure 23) are more sensitive: the draws of all chains are ranked together, and if the chains sample the same distribution each chain's ranks are uniform [34]. Kernel density estimates of each chain's marginal (Figure 24) should be smooth, unimodal where the model implies it, and indistinguishable between chains. Autocorrelation (Figure 25) should damp quickly with the lag, since it determines the effective sample size. Finally, the Gelman–Rubin shrink factor (Figure 26) should fall to 1 as the chains lengthen and stay there [53].

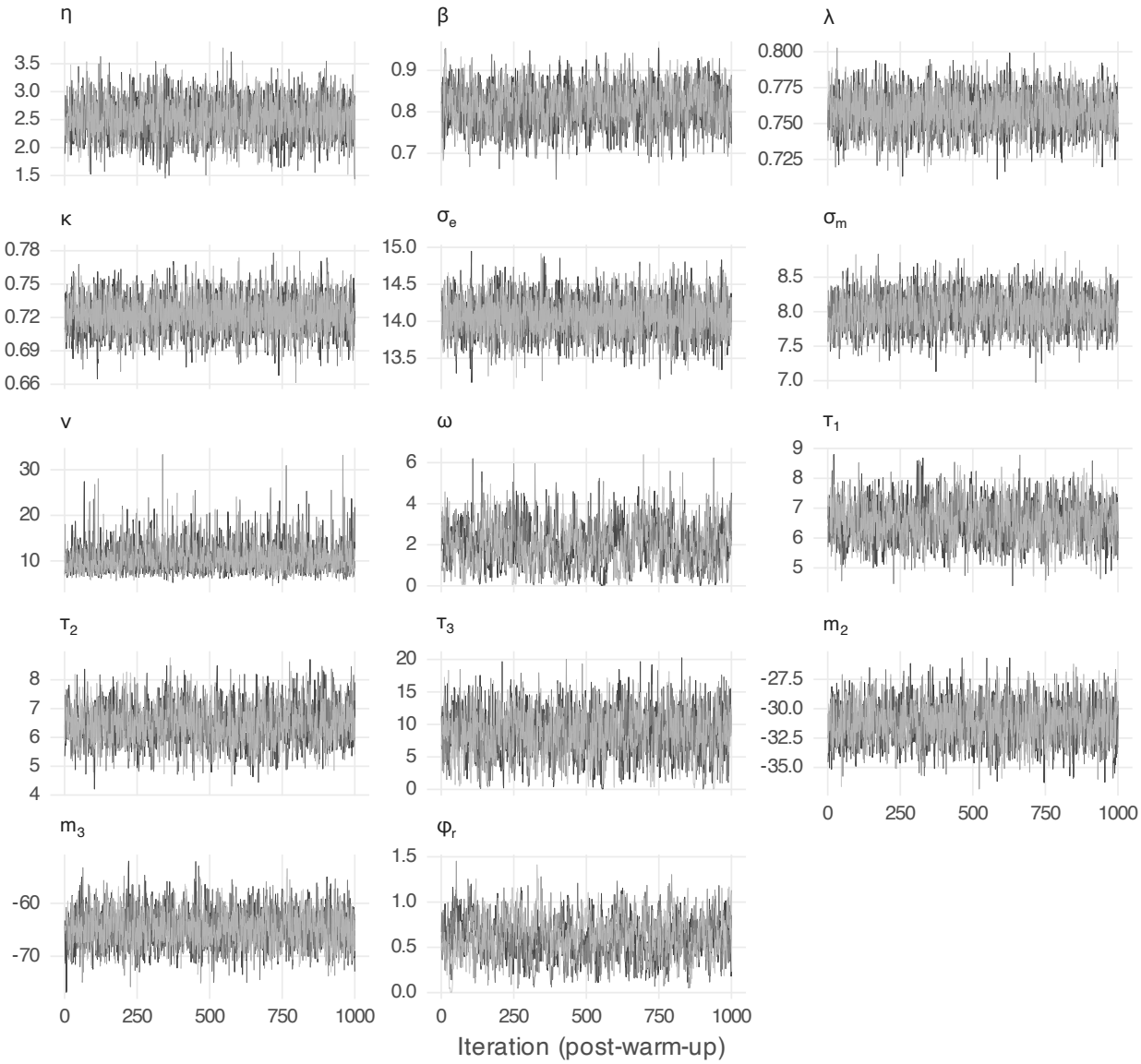



Figure 21: Trace plots of the structural parameters, 2025, final. One shade of grey per chain.

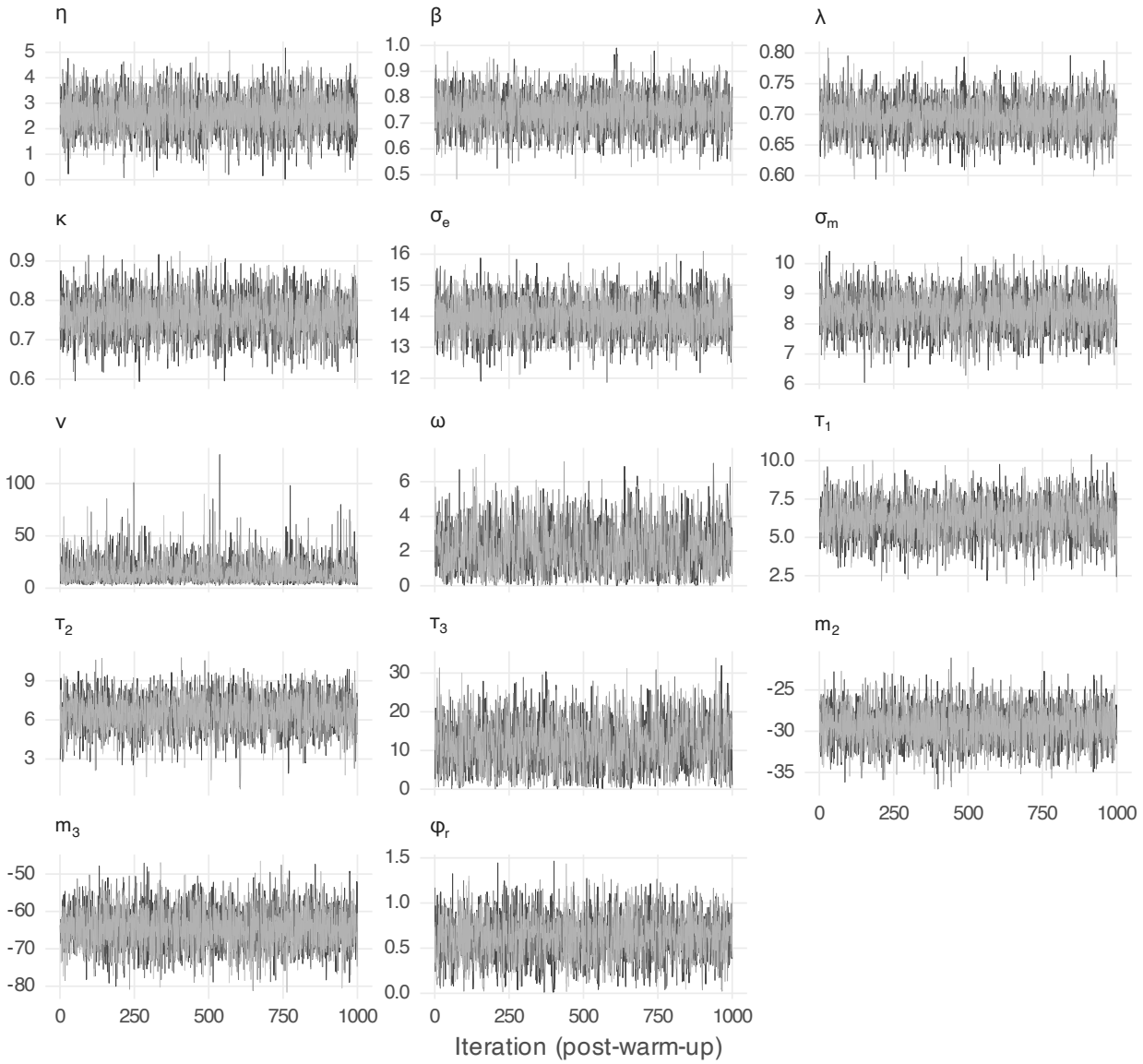



Figure 22: Trace plots of the structural parameters, 2025, week 1. With about one game per team several parameters are close to their priors, and the chains explore those priors freely.

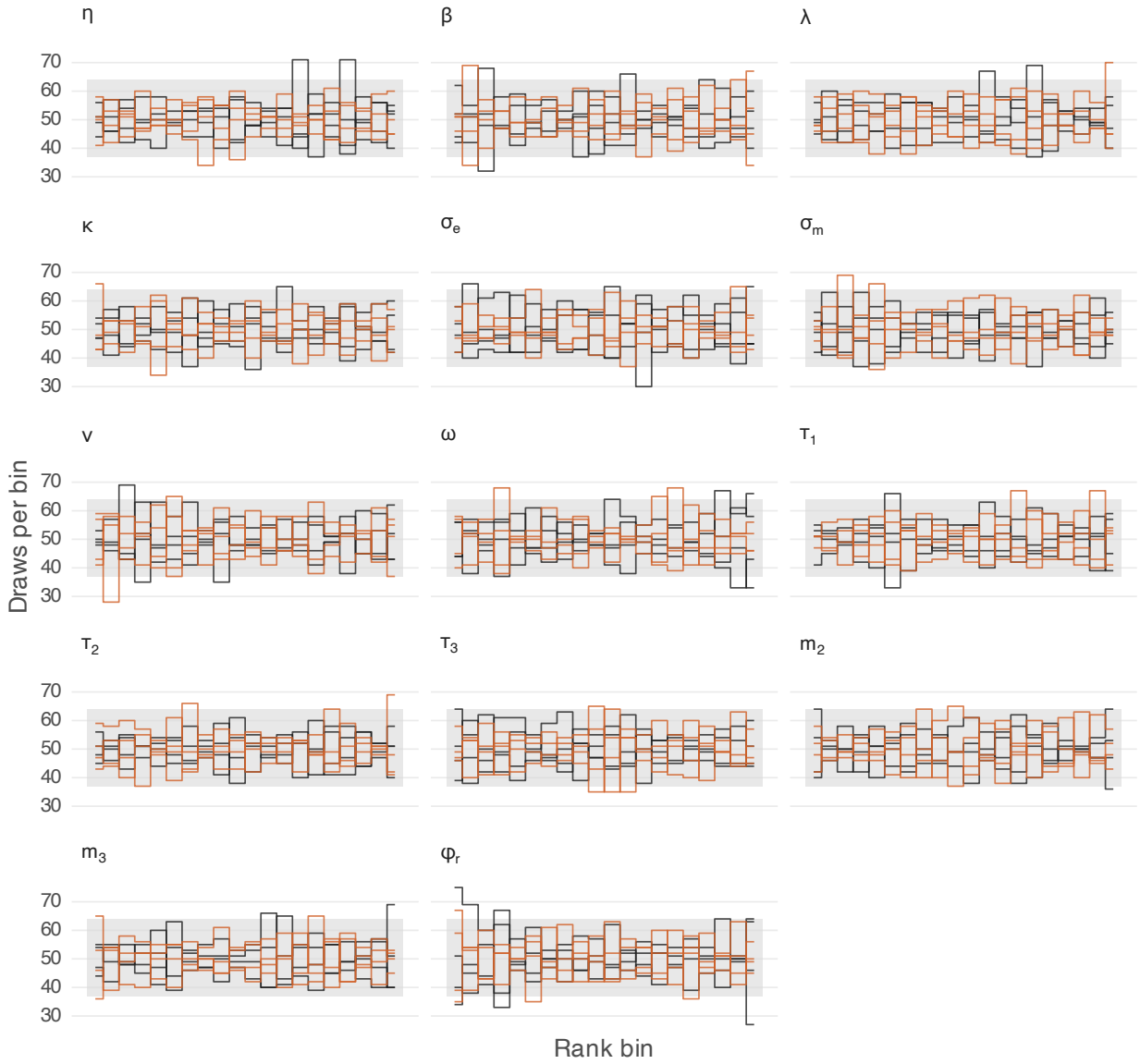



Figure 23: Rank plots of the structural parameters for both fits. For each parameter the draws of all chains are ranked jointly and each chain's ranks are binned into 20 bins; the band is the central 95% range of a bin count under uniformity. Lines are chains; 2025, week 1 in orange, 2025, final in grey.

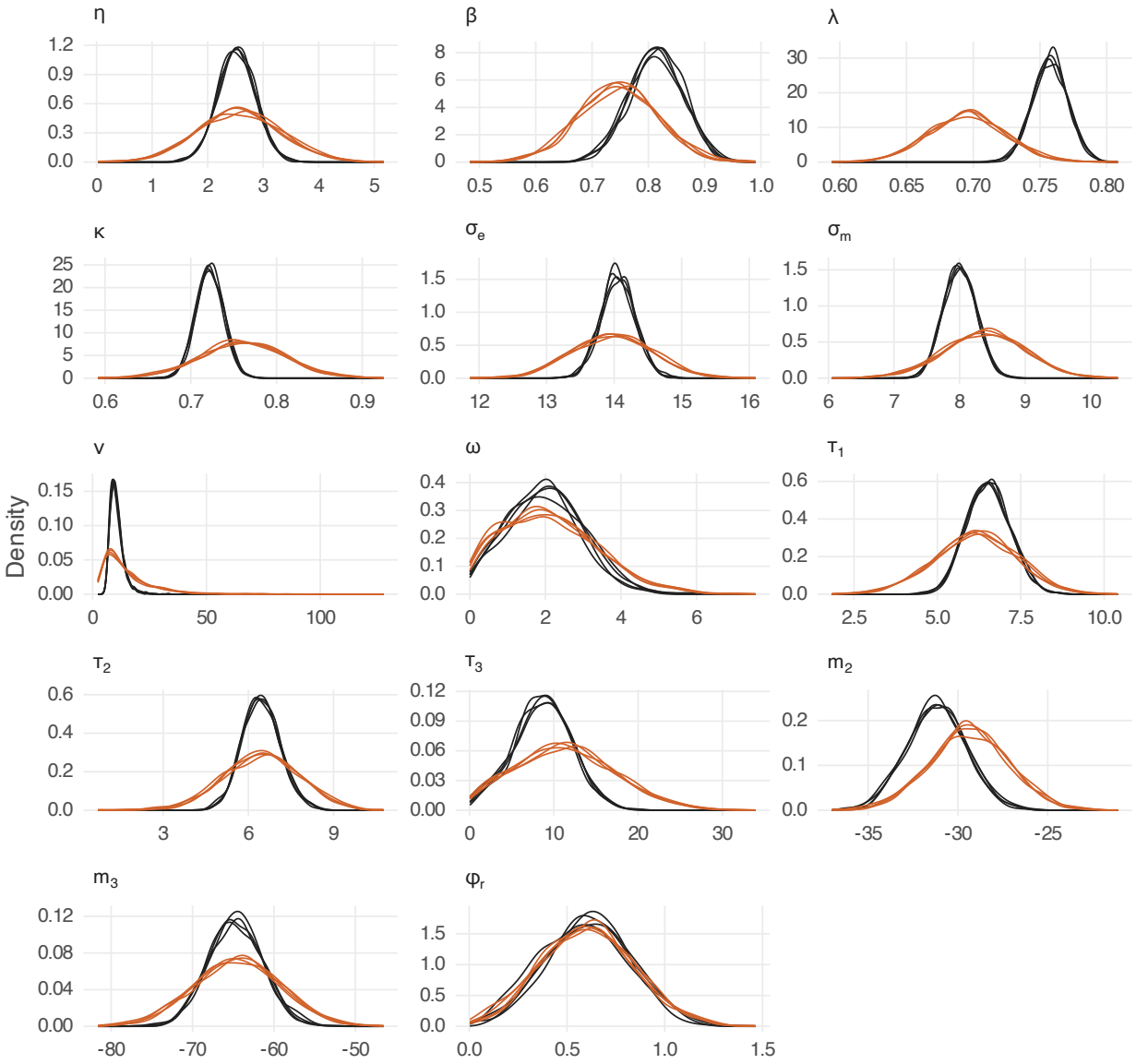



Figure 24: Kernel density estimates of each structural parameter's marginal posterior, one line per chain. Orange: week 1; grey: final. Chains that agree give overlapping curves; a bumpy or multimodal curve, or one chain apart from the others, would indicate poor exploration.

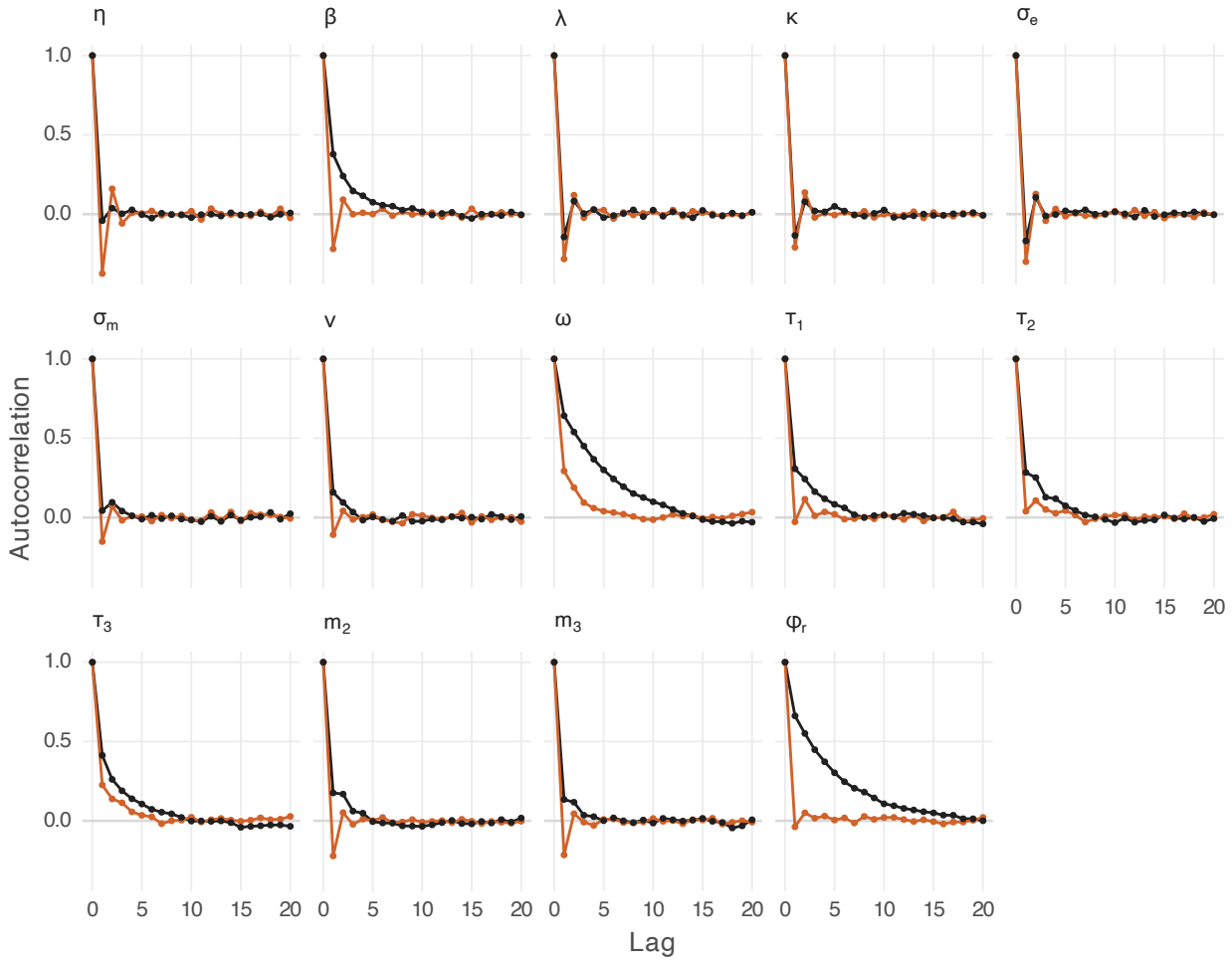



Figure 25: Autocorrelation of the structural parameters up to lag 20, averaged over chains. Orange: week 1; grey: final.

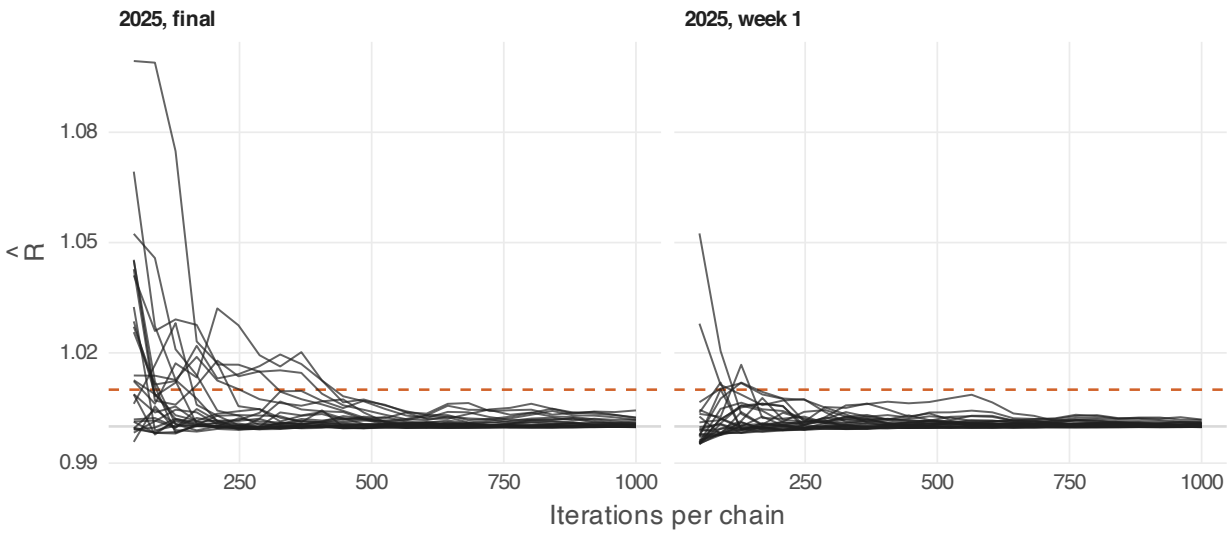



Figure 26: Gelman–Rubin shrink factor (rank-normalised split-$\widehat{R}$) of each structural parameter computed on the first $n$ post-warm-up iterations of every chain, as $n$ grows. The dashed line is the gate at 1.01.

The autocorrelation of 21 of the 28 parameter–fit pairs drops below 0.1 within three lags; the slowest is $\phi_r$ in the final fit, which needs 11 lags, and even there the effective sample size is well above the gate. The shrink factors of every parameter are below the gate after 446 iterations per chain in both fits and stay there, so the 1,000 iterations used are more than enough.

## 11.4 Posterior Geometry

Hierarchical models have funnels: when a scale parameter such as $\omega$ or $\tau_1$ is small, the group effects it governs must be close to their mean, and the posterior narrows into a region the sampler can struggle to enter. The non-centred parameterisation of Equation 3 removes the funnel from the sampled coordinates when the data are weak [27]. Figure 27 plots each scale against a standardised effect it governs, in the sampled coordinates; divergent transitions, if any, would be drawn in orange and would typically concentrate at the neck of a funnel.

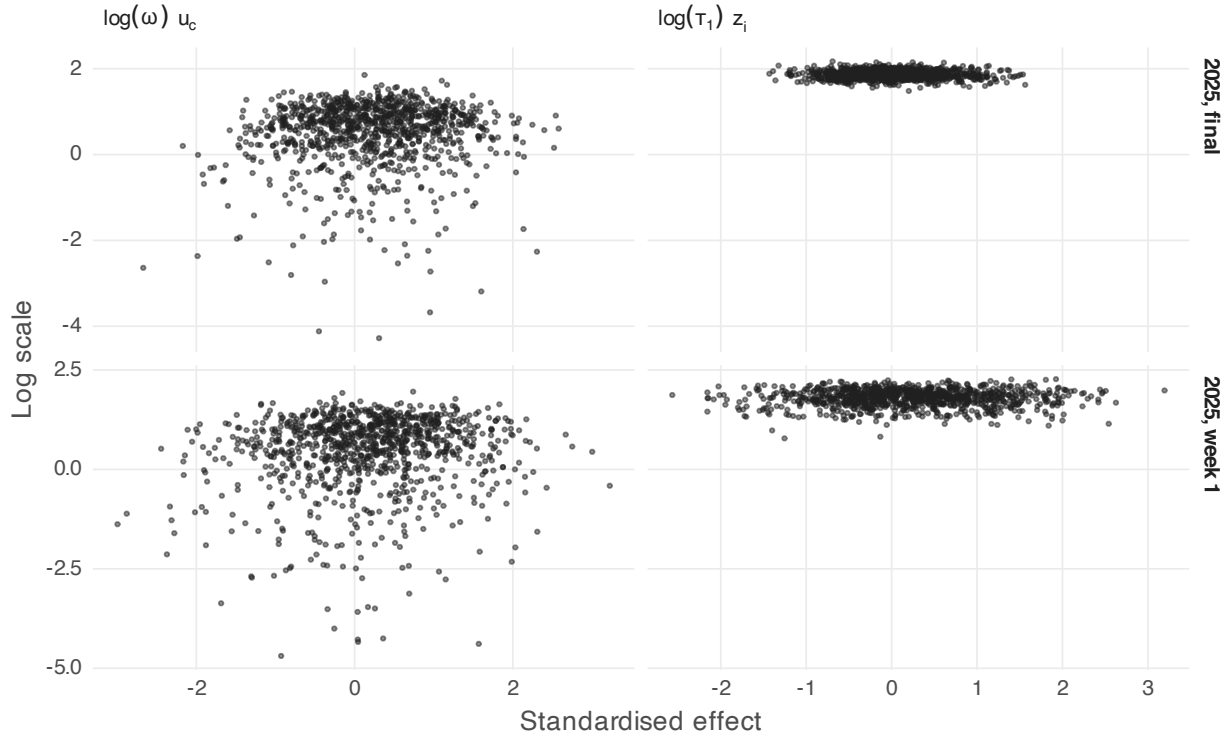



Figure 27: Posterior geometry in the sampled coordinates, 1,000 draws per panel. Each panel plots the log of a scale parameter against a standardised effect it governs: the conference effect $u_c$ of the first FBS conference against $\log \omega$, and the team effect $z_i$ of the FBS team with the fewest games against $\log \tau_1$. Divergent transitions would be orange.

Neither fit shows a funnel in the sampled coordinates: the standardised effects have the same spread at every value of the scale, which is the purpose of the non-centred parameterisation. In the final fit the team effect is nearly uncorrelated with $\tau_1$ and concentrated, because a full season pins the team down; in week 1 it spreads over its $\mathcal{N}(0, 1)$ prior, as it should with one game. The random walk has a funnel of the same kind, between the step $\phi_r$ and the standardised steps $\varepsilon_{i,k}$ of Equation 4, which is not drawn because the steps are not kept; the evidence that the sampler enters its neck is the absence of divergent transitions, which the gates require of every published fit (Section 6.1), together with the informative prior that keeps $\phi_r$ away from zero.

## 11.5 Prior and Posterior

Figure 28 overlays the prior density of each structural parameter on its posterior in both fits. The comparison shows what the data teach: in week 1 many posteriors stay close to the prior, which is why the priors of Section 3.4 were set empirically, while by the end of the season most have contracted well inside it.

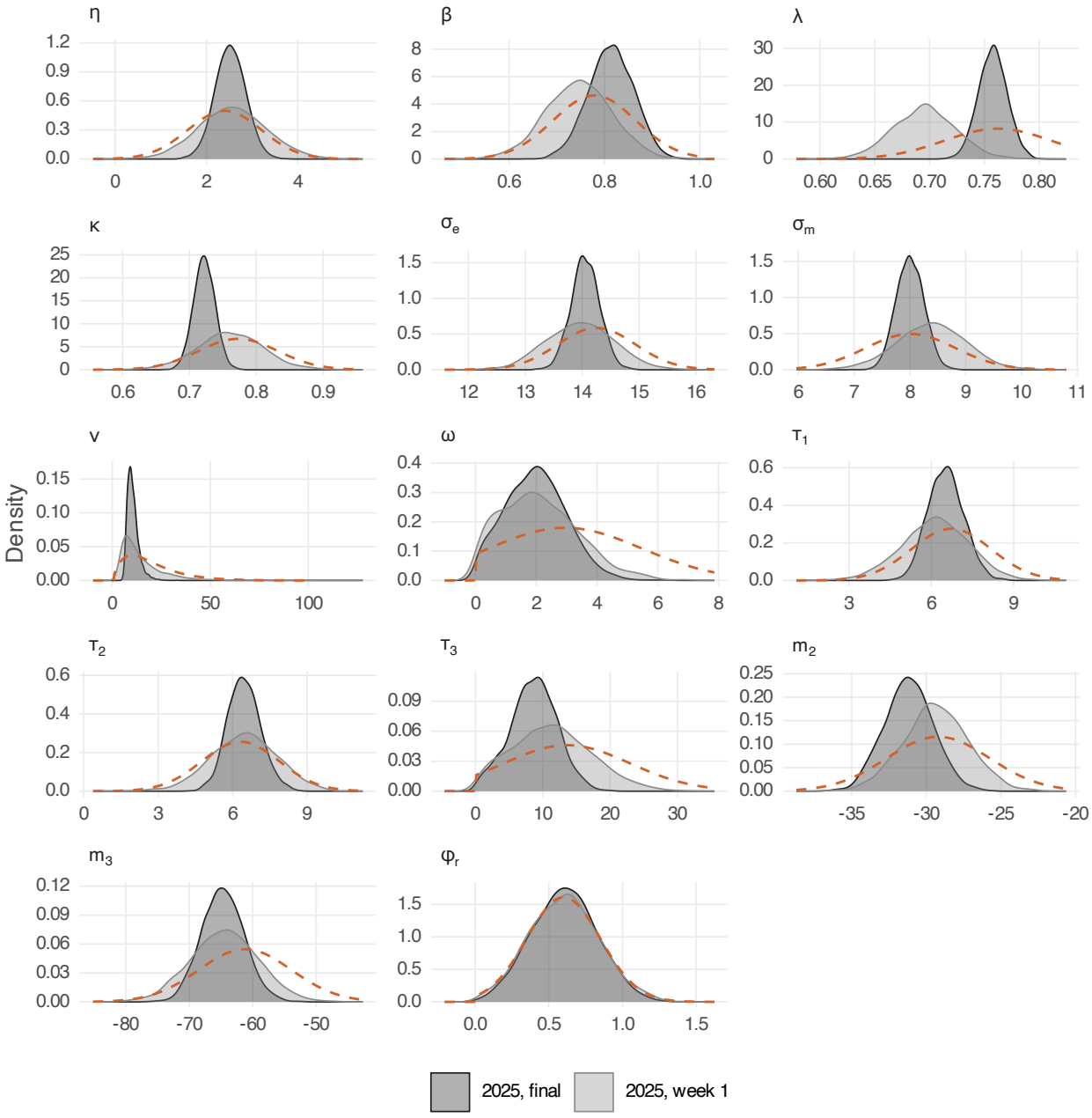



Figure 28: Prior (dashed) and posterior densities of the structural parameters. Priors are normal, truncated to the parameter's support, except $\nu \sim \mathrm{Gamma}(2, 0.1)$ on $\nu > 1$.

Figure 29 condenses the same comparison into two numbers per parameter [55]: the posterior contraction $1 - \mathrm{Var}_{\mathrm{post}} / \mathrm{Var}_{\mathrm{prior}}$, near one when the data dominate the prior, and the posterior $z$-score $(\mathbb{E}_{\mathrm{post}} - \mathbb{E}_{\mathrm{prior}})/\mathrm{sd}_{\mathrm{post}}$, large in magnitude when the posterior sits far from where the prior expected it. Contraction near one with small $|z|$ is the ideal; high contraction with large $|z|$ indicates a prior–data conflict; low contraction indicates that the prior, not the data, is determining the answer.

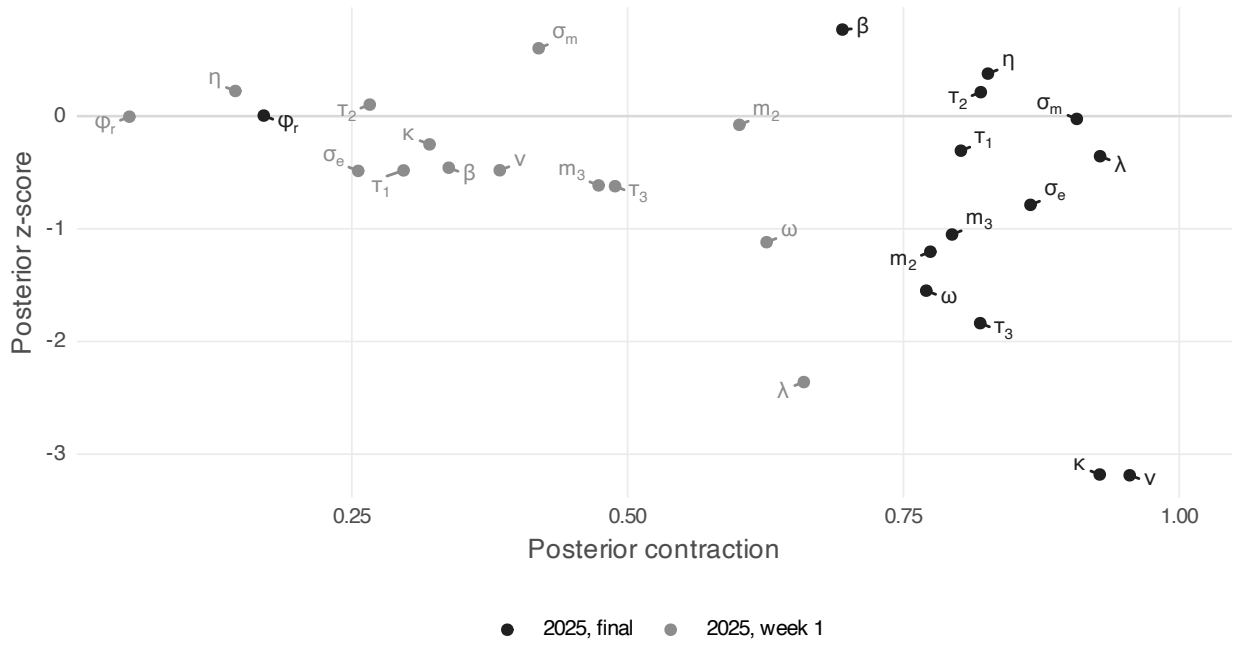



Figure 29: Posterior contraction against posterior $z$-score for the structural parameters, for both fits. Prior moments are those of the truncated priors.

In week 1, 11 of 14 structural parameters contract by less than half — $\eta$, $\sigma_m$, $\sigma_e$, $\kappa$, $\beta$, $m_3$, $\tau_1$, $\tau_2$, $\tau_3$, $\nu$ and $\phi_r$ — so their priors carry the fit, as intended in a week with one game per team. By the end of the season every parameter except $\phi_r$ contracts by more than half. Posteriors more than two of their own standard deviations from the prior mean are $\lambda$ (week 1, $z = -2.4$); $\kappa$ (final, $z = -3.2$); $\nu$ (final, $z = -3.2$). These sit in the tail of their priors: not a failure of fit, but a sign that for that fit the prior is tighter than the data would choose. Section 11.11 checks whether it matters for the published ratings.

## 11.6 Justification of the Informative Priors

Informative priors need a justification that does not come from the data they are applied to. Ours come from the history of the sport. The structural parameters describe the game rather than any team — how much a home field is worth, how noisy a margin is, how far FCS teams sit below FBS teams, how much of a team's strength carries over a summer — and they should change little from year to year. The `prior_refits` target checks this directly: it refits the full-season posterior of every complete season, each with about 1,550 games, under deliberately vague priors — $\eta \sim \mathcal{N}(0, 10^2)$, tier offsets $\mathcal{N}(0, 100^2)$, $\beta$ nearly uniform on $(0, 1)$, half-normal priors with scale 25 on $\tau_\ell$ and $\omega$, scale 2 on $\lambda$ and $\kappa$, and scale 50 on $\sigma_e$ and $\sigma_m$, and comparably wide priors on the total model and the family's shape parameters — so that a season's estimate owes nothing to the published priors. The exception is the random-walk step, which keeps its published prior, because under a vague one the sampler stalls in the funnel at small $\phi_r$ (Section 3.2). Figure 30 plots the resulting per-season posteriors against the priors now in use, and Table 27 gives the numbers.

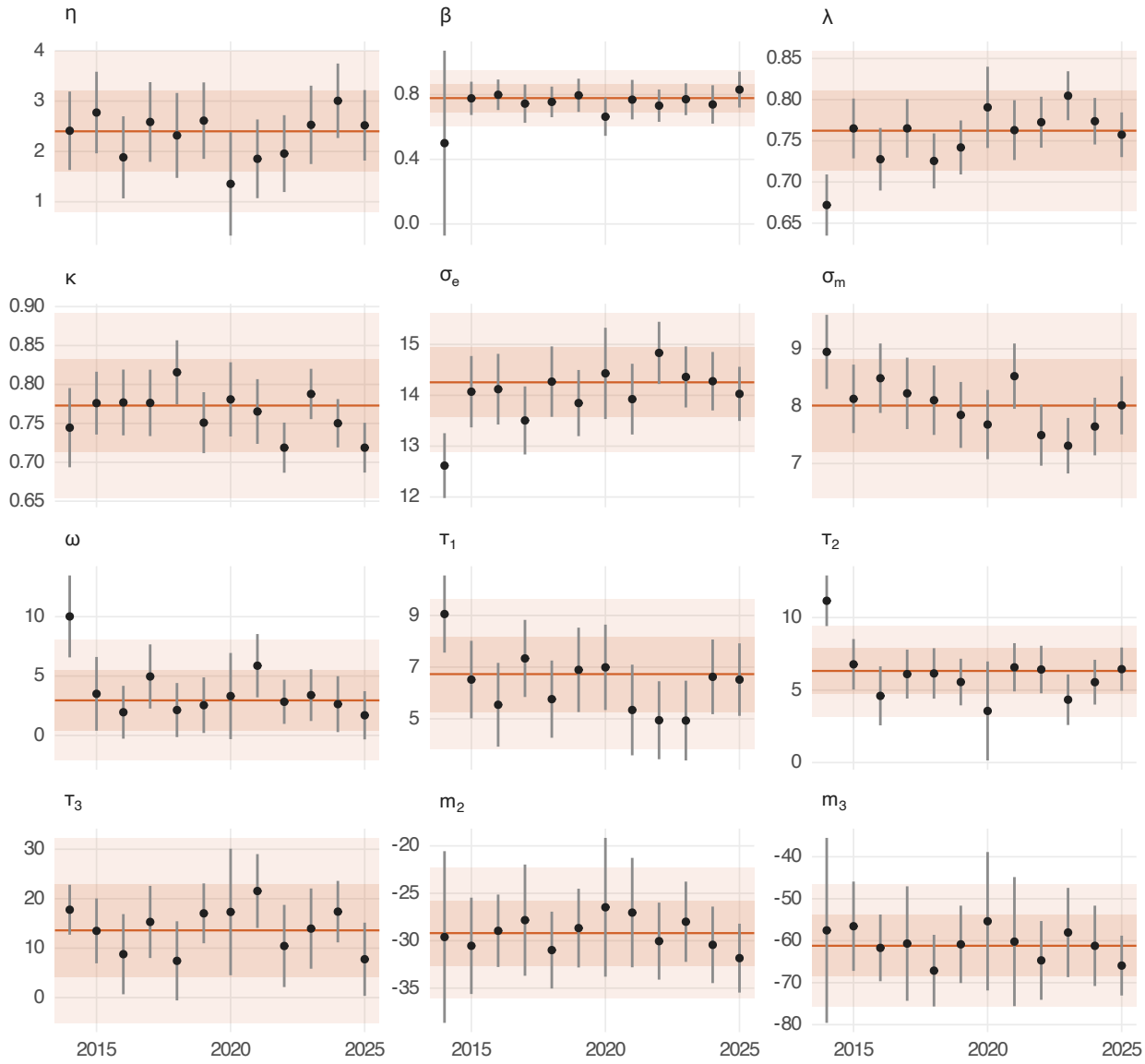



Figure 30: Per-season full-season posteriors under vague priors (points: mean; bars: $\pm 2$ posterior sd) against the prior now in use (line: prior mean; bands: $\pm 1$ and $\pm 2$ prior sd). The priors are centred on the pooled mean of the seasons after the first and are twice as wide as the between-season spread.

Four points qualify this justification. First, pooling every complete season is empirical Bayes at the level of the sport, and saying that a season's estimate lies inside a prior partly built from it would be circular. The last column of Table 27 avoids that: for each season it pools the *other* seasons only and asks whether the season's own estimate lies within two prior standard deviations of that leave-one-season-out prior. For every parameter it does in 100% or more of seasons, so a new season should be expected to fall inside its prior. Second, 2014 is left out of the pooling for the structural reason given in Section 3.4, and 2020 for $\eta$. Third, the prior standard deviation is twice the between-season spread of the posterior means, and that spread is itself inflated by estimation

Table 27: Empirical basis of the structural priors: the pooled mean of the per-season full-season posterior means under vague priors, their between-season standard deviation, the typical (median) within-season posterior sd, and the prior now in use. *Ratio* is the prior sd over the between-season sd; *LOSO* is the share of seasons whose own estimate lies within two prior sds of the prior pooled from the other seasons.

| Parameter | Pooled mean | Between-season sd | Posterior sd | Prior mean | Prior sd | Ratio | LOSO |
|---|---|---|---|---|---|---|---|
| $\eta$ (home field) | 2.40 | 0.39 | 0.39 | 2.40 | 0.80 | 2.0 | 100% |
| $\beta$ (carry-over) | 0.76 | 0.04 | 0.05 | 0.78 | 0.09 | 2.0 | 100% |
| $\lambda$ | 0.76 | 0.02 | 0.02 | 0.76 | 0.05 | 2.0 | 100% |
| $\kappa$ | 0.77 | 0.03 | 0.02 | 0.77 | 0.06 | 2.0 | 100% |
| $\sigma_e$ | 14.15 | 0.35 | 0.33 | 14.26 | 0.68 | 2.0 | 100% |
| $\sigma_m$ | 7.95 | 0.39 | 0.29 | 8.01 | 0.81 | 2.1 | 100% |
| $\omega$ | 3.17 | 1.27 | 1.17 | 2.95 | 2.53 | 2.0 | 100% |
| $\tau_1$ (FBS) | 6.13 | 0.85 | 0.75 | 6.73 | 1.45 | 1.7 | 100% |
| $\tau_2$ (FCS) | 5.64 | 1.05 | 0.84 | 6.32 | 1.56 | 1.5 | 100% |
| $\tau_3$ (lower) | 13.68 | 4.60 | 3.73 | 13.60 | 9.35 | 2.0 | 100% |
| $m_2$ (FCS) | -29.16 | 1.73 | 2.08 | -29.20 | 3.45 | 2.0 | 100% |
| $m_3$ (lower) | -61.12 | 3.70 | 4.79 | -61.20 | 7.31 | 2.0 | 100% |

noise: for $\beta$, $\eta$, $m_2$, $m_3$, $\omega$, $\sigma_e$, $\tau_1$, $\tau_2$ and $\tau_3$ it is about the size of the within-season posterior sd. The priors are therefore somewhat wider than the true season-to-season variation of the sport, which errs on the side of letting each season's data speak. Fourth, the parameters of Table 6 are not pooled at all: the degrees of freedom $\nu$ keep the weakly informative $\mathrm{Gamma}(2, 0.1)$ prior of Juárez and Steel [30], which puts most of its mass between about 3 and 50 and so lets the data choose between heavy and nearly normal tails, and the random-walk and total-model priors are provisional (Section 3.4). The twelve pooled priors are also a version behind: Figure 30 shows the refits under the published model against priors pooled under the static Student-$t$ model, and 0 of the 12 prior means are more than half a prior sd from the current pooled mean.

## 11.7 Induced Priors

A prior on parameters induces a prior on everything computed from them, and a set of individually reasonable priors can combine into an unreasonable one. The quantities that matter here are not $\omega$ or $\tau_1$ but the spread of team strengths, the probability that the home team wins an even game, and the distribution of margins the model expects before seeing any game. Table 28 simulates them from the priors alone, with the 2025 schedule and carry-over, each game played at the random-walk states of its week and its margin drawn from the Student-$t$ law, and Figure 31 compares the prior predictive distribution of three margin statistics with the values observed that season.

Table 28: Induced priors, by simulation from the priors alone (4,000 draws) on the 2025 schedule, with the carry-over $d_i$ from 2024. The last column gives the value observed in 2025 where one exists.

| Quantity | 5% | Median | 95% | Observed |
|---|---|---|---|---|
| Marginal margin scale $\bar{\sigma}$ (pts) | 12.09 | 13.63 | 15.18 | |
| Share of margin variance from efficiency, $R^2$ | 0.55 | 0.65 | 0.75 | |
| Degrees of freedom $\nu$ | 3.62 | 16.82 | 45.34 | |
| Weekly random-walk step $\phi_r$ (pts) | 0.21 | 0.60 | 1.01 | |
| SD of FBS strengths (pts) | 10.35 | 12.73 | 15.58 | |
| Best minus 12th-best FBS team (pts) | 7.76 | 13.73 | 23.20 | |
| P(home win), evenly matched teams | 0.53 | 0.57 | 0.61 | |
| P(best FBS team beats an average one), neutral | 0.93 | 0.98 | 1.00 | |
| SD of regulation margin, all games (pts) | 21.16 | 23.49 | 27.92 | 22.47 |
| Share of games within 7 points | 0.23 | 0.26 | 0.29 | 0.33 |
| Home win rate, true home games | 0.57 | 0.60 | 0.63 | 0.60 |

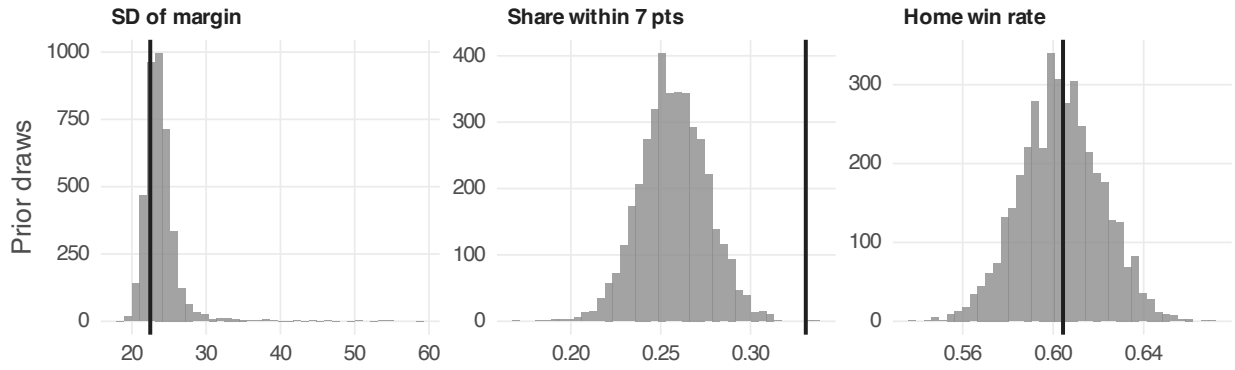



Figure 31: Prior predictive distributions of three statistics of the 2025 regulation margins, simulated from the priors alone; the vertical line is the observed value. The SD panel omits the 1.1% of draws above 60 points, all with $\nu < 3$ (see text).

The induced priors are on the scale of the sport: before any game, the model expects the best FBS team to be about 14 points better than the twelfth (90% prior interval 8–23), an even game to be won at home 57% of the time, and margins with a standard deviation of about 23 points. 2 of the three observed statistics lie inside the central 95% of their prior predictive distribution; the exception is the share within 7 pts, the statistic the fitted model also misses (Section 7.1). Three induced quantities deserve care. The first is a warning. The $\mathrm{Gamma}(2, 0.1)$ prior on $\nu$ puts 3.4% of its mass below 3, where the $t$ law has little or no variance, and those draws produce prior-predictive seasons with a margin standard deviation above 50 points (1.2% of draws; 99th percentile 62). This is harmless here, because $\nu$ is estimated between 9 and 28 in every season and its posterior contracts well (Figure 29), but it is precisely the kind of implausible implication a prior on one parameter can have for an observable, and a prior bounded away from Cauchy tails

(say $\nu > 3$) would be a defensible tightening. The spread of FBS strengths combines $\omega$, $\tau_1$ and $\beta d_i$, so a prior that looks modest for each piece can imply a wide or narrow league; the table shows it lands at a plausible 13 points. And the prior on $\bar{\sigma} = (\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2}$, which with the family's shape parameters sets every win probability and therefore every $\pi_g$ in the resume, is induced by three separate priors; its 90% prior interval, 12.1–15.2 points, is narrow, which is appropriate only because each of the three is well estimated in every season (Figure 30).

## 11.8 Posterior Predictive Checks

Section 7.1 compared four summary statistics with their replicates. Here the whole distributions are compared (Figure 32), and then the location and spread of the margin within each kind of matchup (Figure 33), since a model that fits the pooled distribution can still misfit a subgroup, in particular the games across tiers.

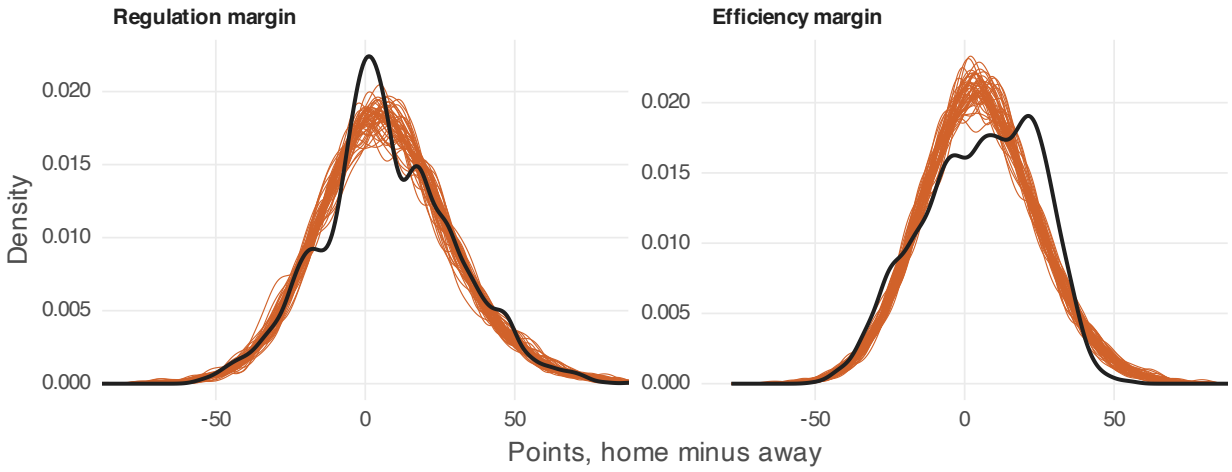



Figure 32: Observed distribution (black) against 50 replicated data sets (orange) for the 2025, final fit. Left: regulation margin, home minus away, replicated from the marginal law of Equation 2 and rounded. Right: efficiency margin, for games that have one.

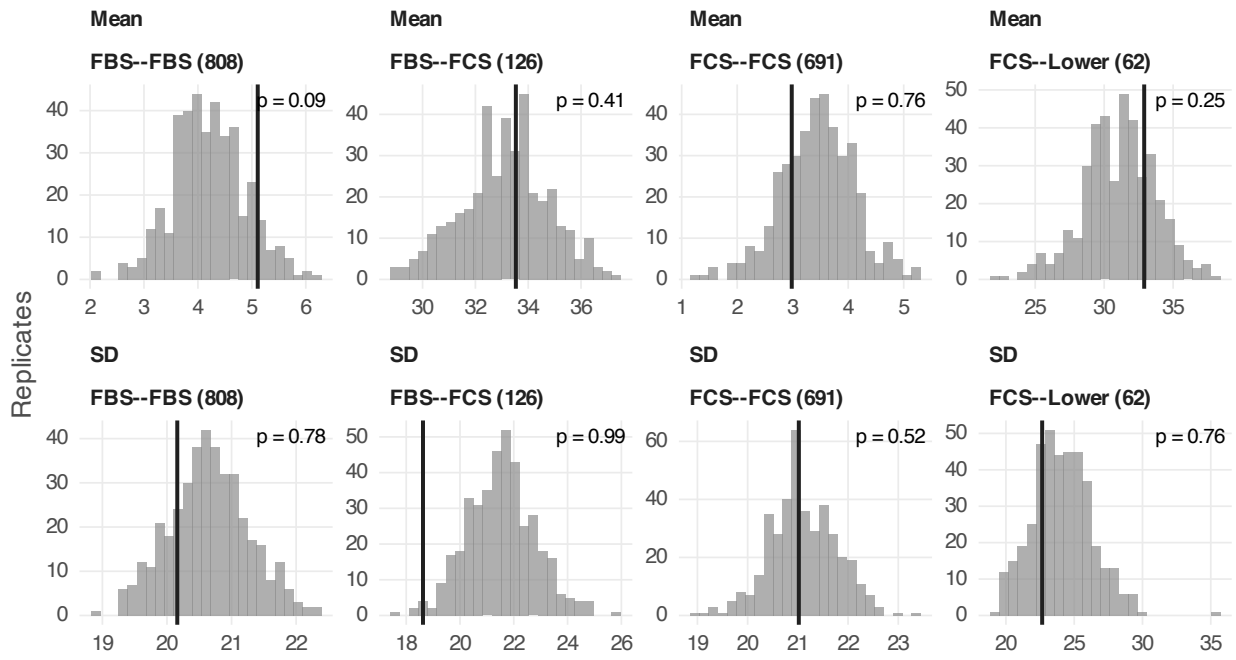



Figure 33: Grouped test statistics, 2025, final. Histograms: the mean and standard deviation of the regulation margin over 400 replicated data sets (as in Figure 7), within each kind of matchup; vertical line: the observed value. The label gives the posterior predictive $p$-value, the share of replicates at least as large as the observed statistic.

Across the 8 grouped statistics the posterior predictive $p$-values run from 0.09 to 0.99, and 1 falls outside $(0.025, 0.975)$: the sd in FBS–FCS games, where the observed value is smaller than almost every replicate. The tier offsets therefore place cross-tier games about right in location, and the misfit of Section 7.1 is mainly one of shape. Games across tiers are, if anything, less variable than a single margin scale for all games allows: a tier-specific scale is a natural refinement. The density overlay shows the same shape misfit directly: the observed margin has a sharper peak near zero than any replicate.

The leave-one-out probability integral transform (LOO-PIT) checks calibration game by game without using a game to predict itself: for each game, $\Pr(y_g^{\text{rep}} \le y_g \mid \text{all other games})$, estimated with the Pareto-smoothed importance weights of the LOO computation. If the model is calibrated these values are uniform; a U shape indicates predictive distributions that are too narrow, a hump that they are too wide [37, 56]. Figure 34 shows them for both signals and both fits.

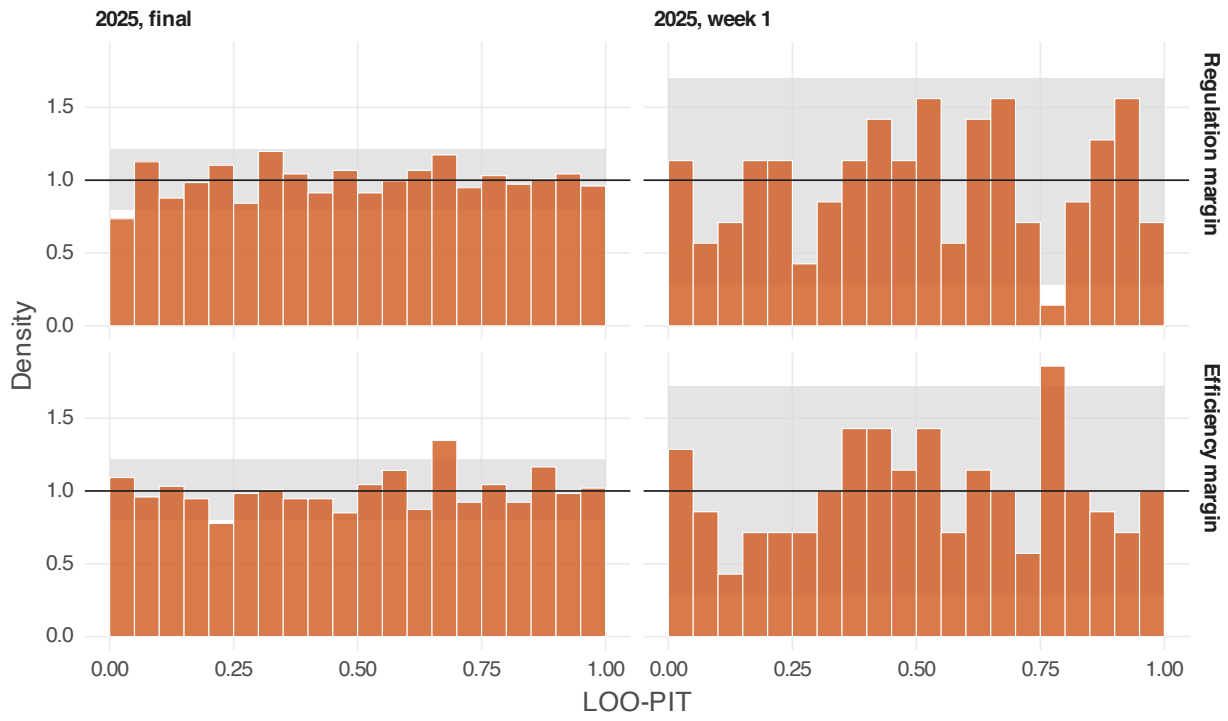



Figure 34: LOO-PIT histograms, 20 bins; the band is the central 95% range of a bin count under uniformity. Margins are whole numbers while the model is continuous, which spreads each PIT value over an interval of width about $1/\bar{\sigma} \approx 0.08$ and does not by itself cause non-uniformity at this bin width.

Of the 80 histogram bins across the four panels, 5 fall outside their 95% band, against about 4 expected by chance. There is no U shape or hump, so the predictive distributions are neither systematically too narrow nor too wide, in week 1 or at the end of the season.

Figure 35 plots the observed margin minus its posterior mean against that mean. A systematic trend would mean that the model's spreads are biased for lopsided or for close matchups — for example, if strong favourites won by less than predicted.

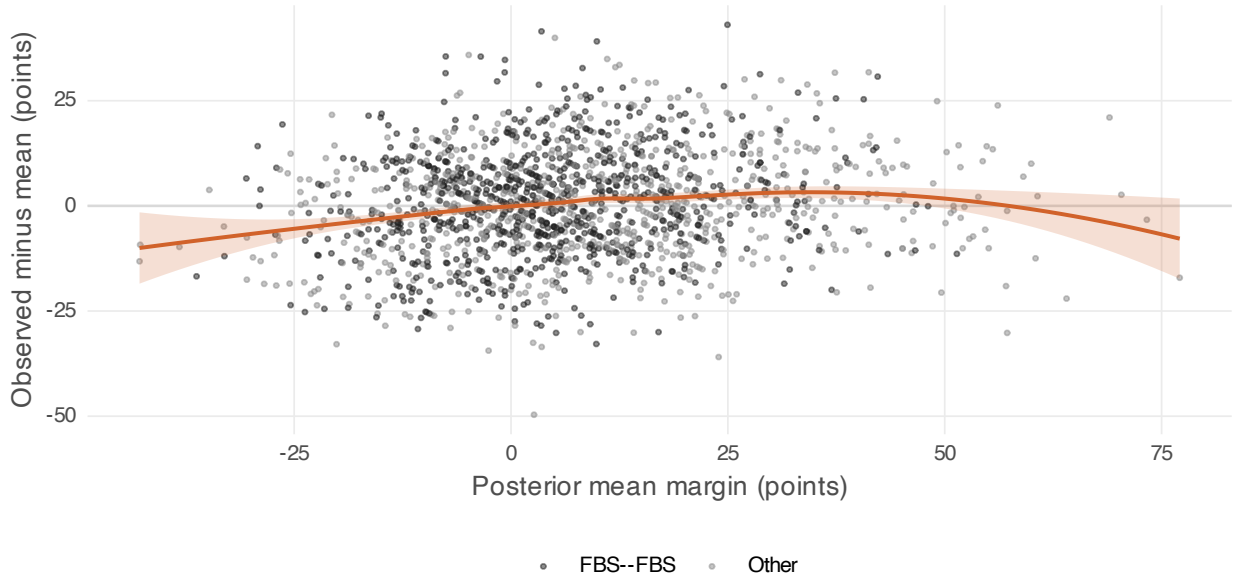



Figure 35: Residual regulation margin against the posterior mean margin, 2025, final. The line is a LOESS smooth with a 95% band; points are games, by kind of matchup.

## 11.9 Influence and PSIS Diagnostics

The Pareto $\hat{k}$ of each game estimates the tail shape of its importance ratios in PSIS-LOO; values above 0.7 mean the leave-one-out approximation is unreliable for that game, and in practice that the game is highly influential on the posterior [40]. Figure 36 plots every game's $\hat{k}$ in both fits, and Table 29 lists the games above 0.7 in the final fit.

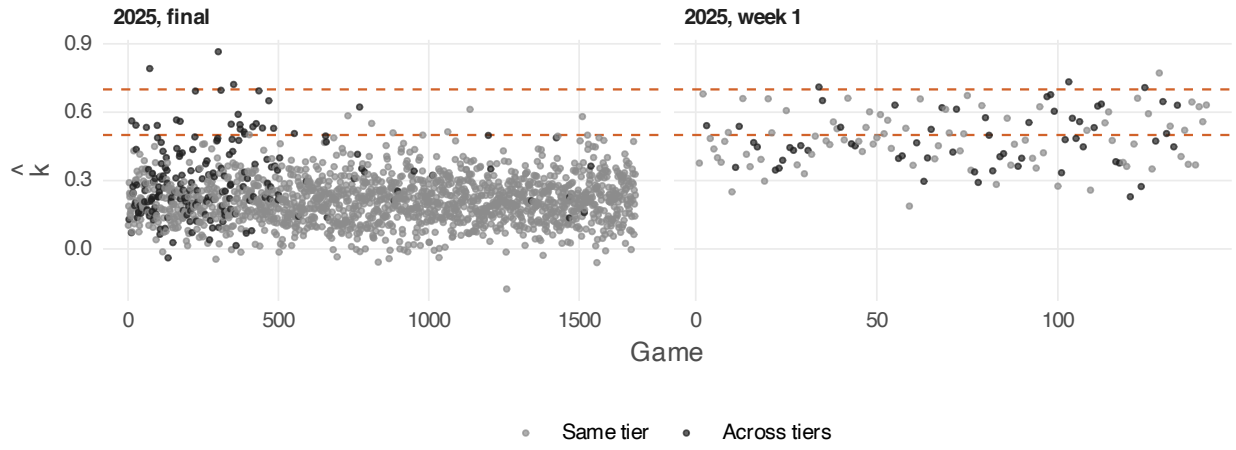



Figure 36: Pareto $\hat{k}$ for each game, in schedule order. Dashed lines at 0.5 and 0.7.

3 of 1,687 games exceed 0.7 in the final fit and 4 of 141 in week 1. In the final fit 3 of them involve

Table 29: Games with Pareto $\hat{k} > 0.7$ in the 2025, final fit, largest first. *Games* counts each team's games in the fit.

| Week | Home (tier, games) | Away (tier, games) | Score | $\hat{k}$ |
|---|---|---|---|---|
| 3 | Hanover (Lower, 1) | Butler (FCS, 12) | 7–16 | 0.87 |
| 1 | Northern Colorado (FCS, 12) | Chadron State (Lower, 1) | 17–3 | 0.79 |
| 3 | Robert Morris (FCS, 12) | West Liberty (Lower, 1) | 28–14 | 0.72 |

a lower-division team and 3 are the only game of one of the two teams: such a game is the only evidence about that team, so removing it changes the team's posterior completely. None involves an FBS team, and the estimated expected log predictive density is not materially affected. A $K$-fold refit would remove the approximation for these games if it were needed.

## 11.10 Power-scaling Sensitivity

Table 13 summarised power-scaling sensitivity by a single distance per parameter. Figure 37 shows what lies behind it for the parameters flagged most often: the posterior density as the prior (top row) or the likelihood (bottom row) is raised to powers between 0.8 and 1.25 [38]. A density that shifts with the prior power is prior-sensitive; one that shifts with the likelihood power is informed by the data.

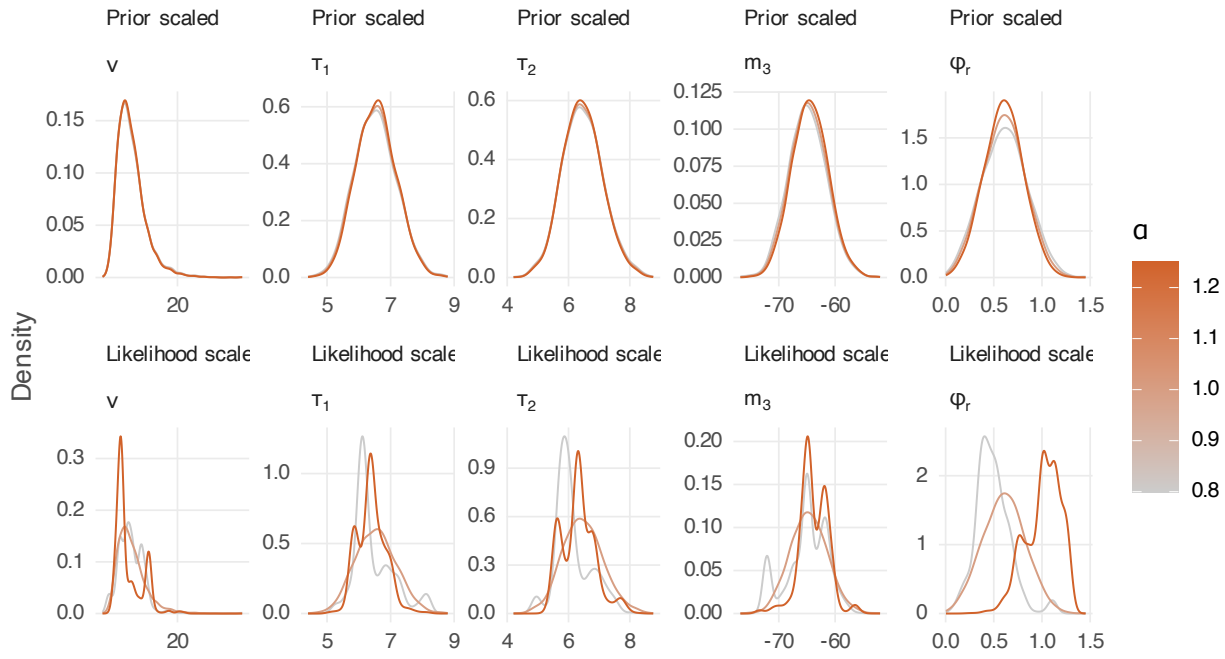



Figure 37: Power-scaled posterior densities, 2025, final. Darker lines are larger powers $\alpha$; the base posterior is $\alpha = 1$.

Table 30: Posterior means (posterior sd) of selected structural parameters under each prior. *Gates* reports whether the refit passed the convergence gates of Section 6.1.

| Fit | Prior | $\eta$ | $\beta$ | $\sigma_m$ | $\kappa$ | $\omega$ | $m_2$ | Gates |
|---|---|---|---|---|---|---|---|---|
| week 1 | Baseline | 2.57 (0.74) | 0.75 (0.07) | 8.38 (0.61) | 0.76 (0.05) | 2.11 (1.27) | -29.37 (2.18) | pass |
| | SD $\times$ 2 | 2.78 (1.24) | 0.72 (0.09) | 8.61 (0.84) | 0.76 (0.07) | 2.22 (1.45) | -29.34 (2.79) | fail |
| | SD $\times$ 5 | 3.15 (1.81) | 0.70 (0.11) | 8.76 (1.05) | 0.75 (0.10) | 2.29 (1.57) | -29.05 (3.23) | fail |
| | Means $+1$ SD | 3.36 (0.74) | 0.79 (0.07) | 8.88 (0.61) | 0.80 (0.05) | 2.58 (1.44) | -27.29 (2.25) | pass |
| | Means $-1$ SD | 1.76 (0.72) | 0.71 (0.07) | 7.94 (0.63) | 0.73 (0.05) | 1.79 (1.17) | -31.32 (2.08) | pass |
| | $\nu > 3$ | 2.56 (0.74) | 0.75 (0.07) | 8.39 (0.60) | 0.76 (0.05) | 2.10 (1.31) | -29.38 (2.12) | pass |
| final | Baseline | 2.53 (0.33) | 0.81 (0.05) | 8.00 (0.25) | 0.72 (0.02) | 1.99 (1.00) | -31.17 (1.64) | pass |
| | SD $\times$ 2 | 2.52 (0.36) | 0.82 (0.05) | 8.01 (0.25) | 0.72 (0.02) | 1.82 (1.04) | -31.65 (1.80) | pass |
| | SD $\times$ 5 | 2.52 (0.37) | 0.83 (0.05) | 8.01 (0.26) | 0.72 (0.02) | 1.79 (1.05) | -31.83 (1.81) | pass |
| | Means $+1$ SD | 2.71 (0.33) | 0.83 (0.05) | 8.09 (0.25) | 0.72 (0.02) | 2.26 (1.07) | -29.99 (1.68) | pass |
| | Means $-1$ SD | 2.36 (0.33) | 0.80 (0.05) | 7.93 (0.24) | 0.72 (0.02) | 1.80 (0.95) | -32.22 (1.56) | pass |
| | $\nu > 3$ | 2.52 (0.33) | 0.81 (0.05) | 8.01 (0.25) | 0.72 (0.02) | 1.98 (1.01) | -31.15 (1.60) | pass |

## 11.11 Sensitivity to the Prior: Perturbation Refits

Power-scaling is a local check: it asks what happens under small multiplicative changes to the prior. The requirement that matters in practice is global — that a reasonable analyst with a different prior would publish essentially the same ratings. We therefore refit both posteriors under four joint alternatives for every structural prior: all prior standard deviations doubled, all multiplied by five (close to vague on the scale of these parameters), and all prior means moved up, or down, by one prior standard deviation. The last two move every mean at once in the same direction, a deliberately harsh test. A fifth alternative bounds the degrees of freedom at $\nu > 3$, the tightening suggested by the induced priors (Section 11.7). Joint changes can cancel, so each prior group is also widened fivefold on its own. Table 30 gives the structural posteriors and Table 31 compares what is published: the FBS power ratings and the resume. Table 32 gives the one-at-a-time results, and Table 33 repeats the joint alternatives for 2020, the most atypical season.

Table 31: Effect of the prior on the published FBS products, relative to the baseline prior. Power: Spearman correlation of the ratings, and the mean and largest absolute change in a team's rating (points). Resume: Spearman correlation of WAB, the largest change in resume rank among the baseline top 25, and the number of teams whose membership of the top 12 changes.

| Fit | Prior | $\rho$ power | Mean $\lvert\Delta\rvert$ | Max $\lvert\Delta\rvert$ | $\rho$ WAB | Max rank $\Delta$ (top 25) | Top-12 changes |
|---|---|---|---|---|---|---|---|
| week 1 | SD $\times$ 2 | 0.999 | 0.31 | 1.04 | 1.000 | 3 | 0 |
| | SD $\times$ 5 | 0.999 | 0.51 | 1.69 | 0.999 | 4 | 0 |
| | Means $+1$ SD | 0.999 | 0.42 | 1.44 | 0.999 | 2 | 0 |
| | Means $-1$ SD | 1.000 | 0.40 | 1.32 | 0.999 | 5 | 1 |
| | $\nu > 3$ | 1.000 | 0.08 | 0.27 | 1.000 | 2 | 0 |
| final | SD $\times$ 2 | 1.000 | 0.07 | 0.24 | 1.000 | 0 | 0 |
| | SD $\times$ 5 | 1.000 | 0.11 | 0.41 | 1.000 | 0 | 0 |
| | Means $+1$ SD | 1.000 | 0.19 | 0.67 | 1.000 | 0 | 0 |
| | Means $-1$ SD | 1.000 | 0.16 | 0.60 | 1.000 | 0 | 0 |
| | $\nu > 3$ | 1.000 | 0.05 | 0.20 | 1.000 | 0 | 0 |

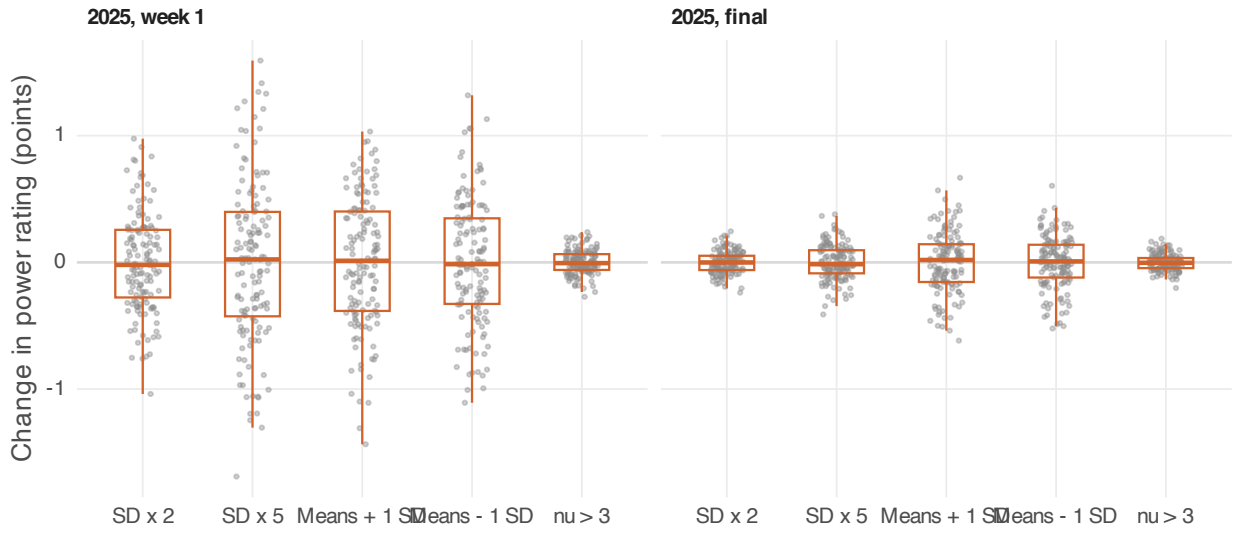



Figure 38: Change in each FBS team's power rating relative to the baseline prior, by alternative prior. Each point is a team; the box spans the middle half. The week-1 ratings depend on the prior by design, the final ratings hardly at all.

At the end of the season the prior is immaterial to what is published: under every alternative the FBS power ratings correlate with the baseline at $\rho > 0.999$, no team's rating moves by more than 0.7 points (mean 0.19), and the resume top 12 is unchanged. More strikingly, the week-1 ratings are also robust, although several structural posteriors there stay close to their priors (Section 11.5): widening every prior fivefold moves no FBS rating by more than 1.7 points, shifting every prior mean by one prior standard deviation moves none by more than 1.4, and the ratings keep their order ($\rho \geq 0.999$). The reason is that a week-1 rating rests mainly on the carry-over $\beta d_i$ from last

Table 32: One prior group at a time: each group's prior standard deviation multiplied by five, all others at baseline. Columns as in Table 31.

| Fit | Prior | $\rho$ power | Mean $\lvert\Delta\rvert$ | Max $\lvert\Delta\rvert$ | $\rho$ WAB | Max rank $\Delta$ (top 25) | Top-12 chang |
|---|---|---|---|---|---|---|---|
| week 1 | $\eta$ sd $\times$ 5 | 1.000 | 0.14 | 0.47 | 1.000 | 3 | |
| | $m_2, m_3$ sd $\times$ 5 | 1.000 | 0.09 | 0.39 | 1.000 | 4 | |
| | $\beta$ sd $\times$ 5 | 0.999 | 0.37 | 1.28 | 1.000 | 2 | |
| | $\tau_\ell$ sd $\times$ 5 | 0.997 | 0.63 | 1.72 | 1.000 | 3 | |
| | $\omega$ sd $\times$ 5 | 1.000 | 0.08 | 0.32 | 1.000 | 2 | |
| | $\lambda$ sd $\times$ 5 | 1.000 | 0.16 | 0.66 | 1.000 | 2 | |
| | $\kappa$ sd $\times$ 5 | 1.000 | 0.12 | 0.40 | 1.000 | 2 | |
| | $\sigma_e$ sd $\times$ 5 | 1.000 | 0.15 | 0.51 | 1.000 | 2 | |
| | $\sigma_m$ sd $\times$ 5 | 1.000 | 0.14 | 0.40 | 1.000 | 3 | |
| | $\phi_r, \phi_t$ sd $\times$ 5 | 1.000 | 0.08 | 0.28 | 1.000 | 2 | |
| | total-model sd $\times$ 5 | 1.000 | 0.11 | 0.41 | 1.000 | 1 | |
| final | $\eta$ sd $\times$ 5 | 1.000 | 0.06 | 0.18 | 1.000 | 1 | |
| | $m_2, m_3$ sd $\times$ 5 | 1.000 | 0.05 | 0.18 | 1.000 | 0 | |
| | $\beta$ sd $\times$ 5 | 1.000 | 0.06 | 0.18 | 1.000 | 0 | |
| | $\tau_\ell$ sd $\times$ 5 | 1.000 | 0.07 | 0.21 | 1.000 | 0 | |
| | $\omega$ sd $\times$ 5 | 1.000 | 0.06 | 0.24 | 1.000 | 1 | |
| | $\lambda$ sd $\times$ 5 | 1.000 | 0.06 | 0.17 | 1.000 | 0 | |
| | $\kappa$ sd $\times$ 5 | 1.000 | 0.05 | 0.13 | 1.000 | 1 | |
| | $\sigma_e$ sd $\times$ 5 | 1.000 | 0.06 | 0.22 | 1.000 | 0 | |
| | $\sigma_m$ sd $\times$ 5 | 1.000 | 0.05 | 0.17 | 1.000 | 1 | |
| | $\phi_r, \phi_t$ sd $\times$ 5 | 1.000 | 0.10 | 0.35 | 1.000 | 0 | |
| | total-model sd $\times$ 5 | 1.000 | 0.05 | 0.19 | 1.000 | 1 | |

Table 33: The joint prior alternatives for the 2020 season (week 1 and the final posterior). Columns as in Table 31.

| Fit | Prior | $\rho$ power | Mean $\lvert\Delta\rvert$ | Max $\lvert\Delta\rvert$ | $\rho$ WAB | Max rank $\Delta$ (top 25) | Top-12 chan |
|---|---|---|---|---|---|---|---|
| 2020, week 1 | SD $\times$ 2 | 0.994 | 0.75 | 4.37 | 1.000 | 0 | |
| | SD $\times$ 5 | 0.972 | 2.12 | 10.09 | 1.000 | 2 | |
| | Means $+1$ SD | 0.999 | 0.83 | 3.10 | 1.000 | 0 | |
| | Means $-1$ SD | 0.999 | 0.80 | 3.27 | 1.000 | 0 | |
| | $\nu > 3$ | 0.999 | 0.14 | 0.54 | 1.000 | 0 | |
| 2020, final | SD $\times$ 2 | 1.000 | 0.21 | 0.66 | 1.000 | 3 | |
| | SD $\times$ 5 | 0.999 | 0.33 | 1.01 | 0.999 | 3 | |
| | Means $+1$ SD | 1.000 | 0.21 | 0.92 | 1.000 | 2 | |
| | Means $-1$ SD | 1.000 | 0.19 | 0.74 | 1.000 | 1 | |
| | $\nu > 3$ | 1.000 | 0.09 | 0.38 | 1.000 | 1 | |

season's final ratings, and $\beta$ is estimated from the week-1 games together with last season's ratings, whatever its prior; the priors on the scales and offsets change how much each team is shrunk towards its conference, which moves ratings by a point or two, not their order. The structural priors therefore stabilise the sampler in data-poor weeks (Section 3.4) without determining what is published. The priors on lower-division offsets and spreads, which the diagnostics flag most, affect FBS ratings only through the handful of cross-tier games.

One prior group at a time (Table 32), the largest effect in the final fit comes from widening the prior on $\phi_r, \phi_t$, which moves no FBS rating by more than 0.3 points; in week 1 it is the prior on $\tau_\ell$, at 1.7 points. Bounding $\nu > 3$ moves no final rating by more than 0.20 points, as expected when every season's $\nu$ is estimated well above 3. For 2020 (Table 33), a season with atypical home field, few early games and a spring FCS season, the final ratings under the joint alternatives keep a rank correlation of at least 0.999 and no rating moved by more than 1.0 points.

## 11.12 Simulation-based Calibration

The checks so far ask whether the model fits the data. Simulation-based calibration asks whether the *computation* recovers the model: if parameters are drawn from the prior, data simulated from them, and the posterior computed, then the rank of each true value among the posterior draws is uniform when the sampler and the code are correct [57, 58]. We simulate 100 seasons on the real 2025 schedule through week 8 (952 games, with the same pattern of missing efficiency margins and the same carry-over $d_i$ and $s_i$), each with its own random walk and margins from the Student-$t$ law, refit each with four chains of 250 draws, and rank the true values among 99 thinned draws. The simulation covers the margin model only; the total model, which shares no parameter with it, is switched off. Besides structural parameters the test quantities include what is published: every FBS team's centred power rating, its WAB (the true value computed from the true strengths and the simulated results), and both ranks.

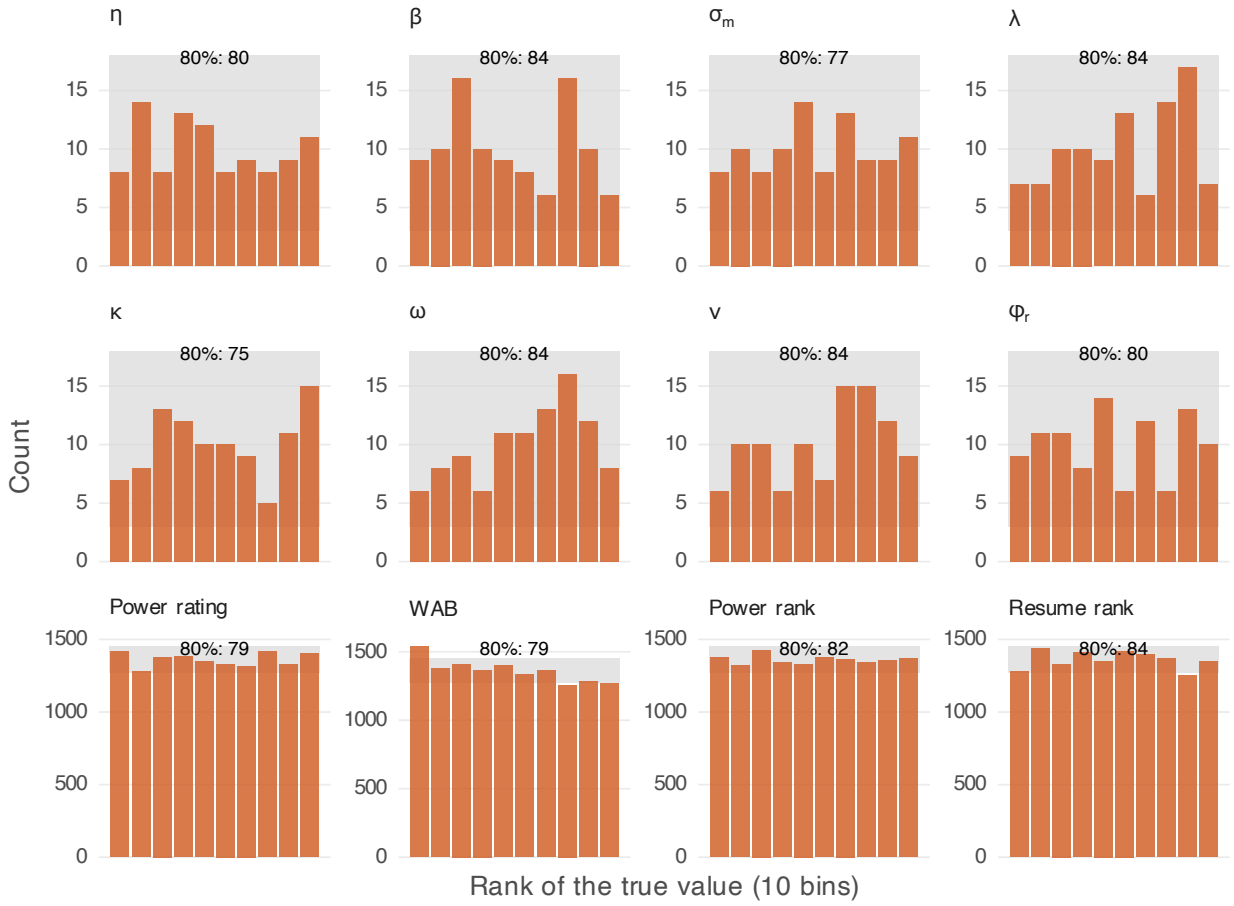



Figure 39: Simulation-based calibration, 100 simulated seasons. Histograms of the rank of the true value among 99 posterior draws, in 10 bins; the band is the central 99% range of a bin count under uniformity (for the team-level quantities, pooled over all FBS teams, so the band is narrow). The label gives the coverage of the central 80% posterior interval.

The structural parameters are recovered with coverage of the 80% interval between 75% and 84%, and for the published quantities it is power rank 82%, power rating 79%, resume rank 84%, wab 79%. A chi-square test of uniformity of the rank histograms rejects at the 1% level for Resume rank, WAB; with 544 team-level values per simulated season the team-level tests are very sensitive, and the rank statistics of different teams in one simulation are not independent, so the size of the departure, visible in the figure, matters more than its significance. Ranks are discrete, so ties between a true rank and the posterior draws are broken at random.

# 12 Appendix: The Stan Program

The program below is `stan/power.stan`, reproduced verbatim at render time.

```
// ============================================================================
// Power model: each team has a net strength r (points, neutral field) and a total tendency
```

```
// t (how many points its games produce), both moving week to week as random walks.
// Offence and defence follow: off = (r + t) / 2, def = (r - t) / 2, so a game's expected
// home score is base + (mu + T) / 2 and away score base + (T - mu) / 2, with
//   mu = r[home] - r[away] + hfa * home_adv      (expected margin)
//   T  = t[home] + t[away]                       (expected total, relative to 2 * base)
//
// Margin, observed through two per-game signals that share r:
//   eff    : net garbage-time-filtered PPA (home offence total - away offence total)
//            eff ~ Normal(lambda * mu, sigma_eff)
//   margin : regulation margin (OT games -> 0), conditioned on the eff residual so the
//            two signals are not double-counted as independent evidence
//            margin ~ F(mu + kappa * (eff - lambda * mu); sigma_m, ...)
// Games without eff use the margin's marginal law: the same family with the scales widened by
// kappa^2 sigma_eff^2 (exact for the normal and the mixture, an approximation for the t).
// F is one of four families (margin_family):
//   1 normal(loc, sigma)
//   2 Student-t(nu, loc, sigma)
//   3 two-normal scale mixture: (1 - p_wide) normal(loc, sigma) + p_wide normal(loc, sigma_w),
//     sigma_w = sigma + sigma_gap > sigma (ordered, so the labels are identified)
//  4 Student-t whose scale grows with the expected margin: sigma_g = sqrt(sigma^2 + gamma_m^2 mu^2)
//
// Total (final score, games without overtime), observed the same way through the summed
// offensive PPA of both teams:
//   eff_sum ~ Normal(alpha_es + lambda_t * T, sigma_es)
//   total   ~ Normal(2 base + T + kappa_t * (eff_sum - alpha_es - lambda_t * T), sigma_tot)
//
// Preseason level (week 1): r0 = mu_conf[conf] + b_prev * prev_dev + s_team * z,
//         mu_conf ~ Normal(mu_tier[tier], sigma_conf), tiers FBS (anchored at 0) / FCS / non-D1;
//         t0 has the same structure with its own conference spread, tier offsets and carry-over.
// prev_dev is last season's final rating relative to its tier's mean (0 when unknown); prev_sd
// is its posterior sd, integrated out: s_team = sqrt(sigma_team^2 + (b_prev prev_sd)^2).
//
// Dynamics: a team's state at each week it plays ("states", in week order per team) is its
// previous state plus tau * sqrt(weeks elapsed) * N(0, 1); the first state steps from r0 (the
// week-1 level). r in generated quantities is the rating at the snapshot week: the last state
// plus the drift since. dynamic = 0 freezes ratings within the season (tau = 0).
//
// Switches for the ablations: conf_level = 0 drops the conference effects; use_total = 0 drops
// the total model (its parameters are then absent).
// ============================================================================

functions {
  // Margin density for family fam; m is the expected margin from the ratings (family 4 only)
 real margin_lpdf(real y, int fam, real loc, real s, real s_w, real p_w, real nu, real gam, real m) {
    if (fam == 1) return normal_lpdf(y | loc, s);
    if (fam == 2) return student_t_lpdf(y | nu, loc, s);
    if (fam == 3) return log_mix(p_w, normal_lpdf(y | loc, s_w), normal_lpdf(y | loc, s));
    return student_t_lpdf(y | nu, loc, sqrt(square(s) + square(gam * m)));
  }
  // The same, summed over a vector of games (vectorised except for the mixture)
 real margins_lpdf(vector y, int fam, vector loc, real s, real s_w, real p_w, real nu, real gam, vector m) {
    if (fam == 1) return normal_lpdf(y | loc, s);
    if (fam == 2) return student_t_lpdf(y | nu, loc, s);
    if (fam == 3) {
```

```
      real lp = 0;
    for (i in 1:rows(y)) lp += log_mix(p_w, normal_lpdf(y[i] | loc[i], s_w), normal_lpdf(y[i] | loc[i], s));
      return lp;
    }
    return student_t_lpdf(y | nu, loc, sqrt(square(s) + square(gam * m)));
  }
}

data {
  int<lower=1> N_teams;
  int<lower=1> N_conf;
  array[N_teams] int<lower=1, upper=N_conf> team_conf;
  array[N_conf] int<lower=1, upper=3> conf_tier;
  array[N_teams] int<lower=1, upper=3> team_tier;
  vector[N_teams] prev_dev;
  vector<lower=0>[N_teams] prev_sd;
  vector[N_teams] prev_dev_t;                 // last season's total tendency vs. its tier mean
  vector<lower=0>[N_teams] prev_sd_t;

  // Weekly states: one per (team, week with a game), grouped by team and in week order within
  // a team, so a team's walk is a cumulative sum over its block
  int<lower=0> N_states;
  array[N_states] int<lower=1, upper=N_teams> state_team;
  array[N_states] int<lower=1, upper=N_states> state_first;  // first state of the same team
  vector<lower=0>[N_states] state_gap;               // weeks since the previous state (r0 for the first)
  array[N_teams] int<lower=0, upper=N_states> now_state;     // latest state (0 = no game yet)
  vector<lower=0>[N_teams] now_gap;                          // weeks from it to the snapshot

  int<lower=0> N_games;
  array[N_games] int<lower=1, upper=N_states> home_state;
  array[N_games] int<lower=1, upper=N_states> away_state;
  vector<lower=0, upper=1>[N_games] home_adv;   // 1 = true home game, 0 = neutral site
  vector[N_games] margin;                       // regulation margin, home - away
  array[N_games] int<lower=0, upper=1> has_eff;
  vector[N_games] eff;                          // 0 where has_eff == 0
  vector[N_games] eff_sum;               // home + away offensive PPA totals; 0 where has_eff == 0
  array[N_games] int<lower=0, upper=1> has_total;
  vector[N_games] total;                        // final total points; 0 where has_total == 0

  real hfa_mean;
  real<lower=0> hfa_sd;
  vector[2] tier_mean;                          // prior means for FCS, non-D1 tier offsets
  vector<lower=0>[2] tier_sd;
  real b_prev_mean;
  real<lower=0> b_prev_sd;
  // Informative priors (mean, sd) for the structural noise/scale parameters: they barely move
  // between seasons, and without them week-1 fits (~1 game per team) funnel
  vector[3] sigma_team_mean;
  vector<lower=0>[3] sigma_team_sd;
  vector[2] sigma_conf_prior;
  vector[2] lambda_prior;
  vector[2] kappa_prior;
  vector[2] sigma_eff_prior;
  vector[2] sigma_m_prior;
```

```
  vector<lower=0>[2] nu_prior;                // Gamma(shape, rate), truncated to nu > nu_lower
  real<lower=0> nu_lower;
  vector[2] tau_r_prior;                       // weekly step sd of r (normal, truncated at 0)
  vector[2] sigma_gap_prior;                   // mixture: sigma_w - sigma_m
  vector<lower=0>[2] p_wide_prior;            // mixture: Beta(a, b) weight of the wide component
  vector[2] gamma_prior;                       // scale-t: growth of the scale with |mu|
  // Total model
  vector[2] base_prior;                        // points per team, average FBS matchup
  vector[2] t_tier_mean;
  vector<lower=0>[2] t_tier_sd;
  vector[2] sigma_t_team_prior;
  vector[2] sigma_conf_t_prior;
  vector[2] b_prev_t_prior;
  vector[2] tau_t_prior;
  vector[2] lambda_t_prior;
  vector[2] alpha_es_prior;
  vector[2] kappa_t_prior;
  vector[2] sigma_es_prior;
  vector[2] sigma_tot_prior;

  int<lower=1, upper=4> margin_family;
  int<lower=0, upper=1> dynamic;
  int<lower=0, upper=1> use_total;
  int<lower=0, upper=1> conf_level;
  int<lower=0, upper=1> compute_ll;
}

transformed data {
  int N_eff = sum(has_eff);
  array[N_eff] int idx_eff;
  array[N_games - N_eff] int idx_noeff;
  int is_mix = margin_family == 3;
  int is_sct = margin_family == 4;
  int dyn_t = dynamic * use_total;
  vector[N_states] sqrt_gap = sqrt(state_gap);
  // Totals: games with eff_sum; with eff_sum and a total; with a total but no eff_sum
  int N_es_tot = 0;
  int N_tot_only = 0;
  for (k in 1:N_games) {
    if (has_eff[k] == 1 && has_total[k] == 1) N_es_tot += 1;
    if (has_eff[k] == 0 && has_total[k] == 1) N_tot_only += 1;
  }
  array[N_es_tot] int idx_es_tot;
  array[N_tot_only] int idx_tot_only;
  {
    int i = 1;
    int j = 1;
    for (k in 1:N_games) {
      if (has_eff[k] == 1 && has_total[k] == 1) { idx_es_tot[i] = k; i += 1; }
      if (has_eff[k] == 0 && has_total[k] == 1) { idx_tot_only[j] = k; j += 1; }
    }
  }
  {
    int i = 1;
```

```
    int j = 1;
    for (k in 1:N_games) {
      if (has_eff[k] == 1) { idx_eff[i] = k; i += 1; }
      else { idx_noeff[j] = k; j += 1; }
    }
  }
}

parameters {
  vector[2] mu_tier_raw;
  real<lower=0> sigma_conf;
  vector[N_conf] conf_raw;
  vector<lower=0>[3] sigma_team;
  real<lower=0, upper=1> b_prev;
  vector[N_teams] z;
  vector<lower=0>[dynamic] tau_r_raw;
  vector[dynamic ? N_states : 0] eps_r;

  real hfa;
  real<lower=0> lambda;
  real<lower=0> sigma_eff;
  real kappa;
  real<lower=0> sigma_m;
  real<lower=nu_lower> nu;                        // used by families 2 and 4 only
  vector<lower=0>[is_mix] sigma_gap_raw;
  vector<lower=0, upper=1>[is_mix] p_wide_raw;
  vector<lower=0>[is_sct] gamma_raw;

  // Total model
  vector[use_total] base_raw;
  vector[use_total ? 2 : 0] t_tier_raw;
  vector<lower=0>[use_total] sigma_conf_t_raw;
  vector[use_total ? N_conf : 0] conf_t_raw;
  vector<lower=0>[use_total] sigma_t_team_raw;
  vector<lower=0, upper=1>[use_total] b_prev_t_raw;
  vector[use_total ? N_teams : 0] z_t;
  vector<lower=0>[dyn_t] tau_t_raw;
  vector[dyn_t ? N_states : 0] eps_t;
  vector<lower=0>[use_total] lambda_t_raw;
  vector[use_total] alpha_es_raw;
  vector<lower=0>[use_total] sigma_es_raw;
  vector[use_total] kappa_t_raw;
  vector<lower=0>[use_total] sigma_tot_raw;
}

transformed parameters {
  vector[3] mu_tier = append_row(0, mu_tier_raw);
  vector[N_conf] mu_conf = mu_tier[conf_tier] + conf_level * sigma_conf * conf_raw;
  vector[N_teams] r0 = mu_conf[team_conf] + b_prev * prev_dev
                       + sqrt(square(sigma_team[team_tier]) + square(b_prev * prev_sd)) .* z;
  real sigma_marg = sqrt(square(sigma_m) + square(kappa * sigma_eff));
  // Family parameters as scalars (defaults where the family does not use them)
  real tau_r = 0;
  real sigma_w = sigma_m;
```

```
real p_wide = 0;
real gamma_m = 0;
if (dynamic) tau_r = tau_r_raw[1];
if (is_mix) { sigma_w = sigma_m + sigma_gap_raw[1]; p_wide = p_wide_raw[1]; }
if (is_sct) gamma_m = gamma_raw[1];
real sigma_marg_w = sqrt(square(sigma_w) + square(kappa * sigma_eff));

// Random walk as a cumulative sum of steps within each team's block of states
vector[N_states] r_state = r0[state_team];
if (dynamic && N_states > 0) {
  vector[N_states] step = sqrt_gap .* eps_r;
  vector[N_states] cs = cumulative_sum(step);
  r_state += tau_r * (cs - cs[state_first] + step[state_first]);
}

// Total model (empty when use_total = 0)
vector[use_total ? N_teams : 0] t0;
vector[use_total ? N_states : 0] t_state;
if (use_total) {
  vector[3] t_tier = append_row(0, t_tier_raw);
  vector[N_conf] t_conf = t_tier[conf_tier] + conf_level * sigma_conf_t_raw[1] * conf_t_raw;
  t0 = t_conf[team_conf] + b_prev_t_raw[1] * prev_dev_t
       + sqrt(square(sigma_t_team_raw[1]) + square(b_prev_t_raw[1] * prev_sd_t)) .* z_t;
  t_state = t0[state_team];
  if (dyn_t && N_states > 0) {
    vector[N_states] step = sqrt_gap .* eps_t;
    vector[N_states] cs = cumulative_sum(step);
    t_state += tau_t_raw[1] * (cs - cs[state_first] + step[state_first]);
  }
}

// Log prior of the population-level parameters (used by priorsense power-scaling)
real lprior = normal_lpdf(mu_tier_raw | tier_mean, tier_sd)
            + normal_lpdf(sigma_conf | sigma_conf_prior[1], sigma_conf_prior[2])
            + normal_lpdf(sigma_team | sigma_team_mean, sigma_team_sd)
            + normal_lpdf(b_prev | b_prev_mean, b_prev_sd)
            + normal_lpdf(hfa | hfa_mean, hfa_sd)
            + normal_lpdf(lambda | lambda_prior[1], lambda_prior[2])
            + normal_lpdf(sigma_eff | sigma_eff_prior[1], sigma_eff_prior[2])
            + normal_lpdf(kappa | kappa_prior[1], kappa_prior[2])
            + normal_lpdf(sigma_m | sigma_m_prior[1], sigma_m_prior[2])
       + gamma_lpdf(nu | nu_prior[1], nu_prior[2]) - gamma_lccdf(nu_lower | nu_prior[1], nu_prior[2]);
if (dynamic) lprior += normal_lpdf(tau_r_raw | tau_r_prior[1], tau_r_prior[2]);
if (is_mix) lprior += normal_lpdf(sigma_gap_raw | sigma_gap_prior[1], sigma_gap_prior[2])
                    + beta_lpdf(p_wide_raw | p_wide_prior[1], p_wide_prior[2]);
if (is_sct) lprior += normal_lpdf(gamma_raw | gamma_prior[1], gamma_prior[2]);
if (use_total) {
  lprior += normal_lpdf(base_raw | base_prior[1], base_prior[2])
          + normal_lpdf(t_tier_raw | t_tier_mean, t_tier_sd)
          + normal_lpdf(sigma_conf_t_raw | sigma_conf_t_prior[1], sigma_conf_t_prior[2])
          + normal_lpdf(sigma_t_team_raw | sigma_t_team_prior[1], sigma_t_team_prior[2])
          + normal_lpdf(b_prev_t_raw | b_prev_t_prior[1], b_prev_t_prior[2])
          + normal_lpdf(lambda_t_raw | lambda_t_prior[1], lambda_t_prior[2])
          + normal_lpdf(alpha_es_raw | alpha_es_prior[1], alpha_es_prior[2])
```

```
            + normal_lpdf(kappa_t_raw | kappa_t_prior[1], kappa_t_prior[2])
            + normal_lpdf(sigma_es_raw | sigma_es_prior[1], sigma_es_prior[2])
            + normal_lpdf(sigma_tot_raw | sigma_tot_prior[1], sigma_tot_prior[2]);
    if (dyn_t) lprior += normal_lpdf(tau_t_raw | tau_t_prior[1], tau_t_prior[2]);
  }
}

model {
  target += lprior;
  conf_raw ~ std_normal();
  z ~ std_normal();
  eps_r ~ std_normal();
  conf_t_raw ~ std_normal();
  z_t ~ std_normal();
  eps_t ~ std_normal();

  if (N_games > 0) {
    vector[N_games] mu = r_state[home_state] - r_state[away_state] + hfa * home_adv;
    vector[N_eff] mu_e = mu[idx_eff];
    vector[N_games - N_eff] mu_n = mu[idx_noeff];
    vector[N_eff] res = eff[idx_eff] - lambda * mu_e;
    target += normal_lpdf(eff[idx_eff] | lambda * mu_e, sigma_eff);
   target += margins_lpdf(margin[idx_eff] | margin_family, mu_e + kappa * res, sigma_m, sigma_w, p_wide, nu, gamma
   target += margins_lpdf(margin[idx_noeff] | margin_family, mu_n, sigma_marg, sigma_marg_w, p_wide, nu, gamma_m

    if (use_total) {
      vector[N_games] tt = t_state[home_state] + t_state[away_state];
      vector[N_games] es_mu = alpha_es_raw[1] + lambda_t_raw[1] * tt;
     real tot_marg = sqrt(square(sigma_tot_raw[1]) + square(kappa_t_raw[1] * sigma_es_raw[1]));
      target += normal_lpdf(eff_sum[idx_eff] | es_mu[idx_eff], sigma_es_raw[1]);
      target += normal_lpdf(total[idx_es_tot] | 2 * base_raw[1] + tt[idx_es_tot]
                  + kappa_t_raw[1] * (eff_sum[idx_es_tot] - es_mu[idx_es_tot]), sigma_tot_raw[1]);
     target += normal_lpdf(total[idx_tot_only] | 2 * base_raw[1] + tt[idx_tot_only], tot_marg);
    }
  }
}

generated quantities {
  // Ratings at the snapshot week: the latest state plus the random-walk drift since
  vector[N_teams] r;
  vector[use_total ? N_teams : 0] t;
  for (i in 1:N_teams) {
    if (now_state[i] == 0) r[i] = r0[i];
    else r[i] = r_state[now_state[i]];
    if (dynamic) r[i] += tau_r * sqrt(now_gap[i]) * normal_rng(0, 1);
  }
  if (use_total) {
    for (i in 1:N_teams) {
      if (now_state[i] == 0) t[i] = t0[i];
      else t[i] = t_state[now_state[i]];
      if (dyn_t) t[i] += tau_t_raw[1] * sqrt(now_gap[i]) * normal_rng(0, 1);
    }
  }
```

```
  // log_lik: joint p(e, y) where e is observed, p(y) otherwise (the fitted margin likelihood).
  // log_lik_margin: p(y) alone for every game, the marginal law of the margin; comparable
  // across models whether or not they use the efficiency margin.
  // log_lik_total: the total's contribution (joint with eff_sum where observed).
  vector[compute_ll ? N_games : 0] log_lik;
  vector[compute_ll ? N_games : 0] log_lik_margin;
  vector[compute_ll * use_total ? N_games : 0] log_lik_total;
  if (compute_ll) {
    vector[N_games] mu = r_state[home_state] - r_state[away_state] + hfa * home_adv;
    for (k in 1:N_games) {
    log_lik_margin[k] = margin_lpdf(margin[k] | margin_family, mu[k], sigma_marg, sigma_marg_w, p_wide, nu, gamm
      if (has_eff[k] == 1) {
        real res = eff[k] - lambda * mu[k];
        log_lik[k] = normal_lpdf(eff[k] | lambda * mu[k], sigma_eff)
             + margin_lpdf(margin[k] | margin_family, mu[k] + kappa * res, sigma_m, sigma_w, p_wide, nu, gamma_m,
      } else {
        log_lik[k] = log_lik_margin[k];
      }
    }
    if (use_total) {
      vector[N_games] tt = t_state[home_state] + t_state[away_state];
     real tot_marg = sqrt(square(sigma_tot_raw[1]) + square(kappa_t_raw[1] * sigma_es_raw[1]));
      for (k in 1:N_games) {
        log_lik_total[k] = 0;
        if (has_eff[k] == 1) {
          real es_mu = alpha_es_raw[1] + lambda_t_raw[1] * tt[k];
          log_lik_total[k] = normal_lpdf(eff_sum[k] | es_mu, sigma_es_raw[1]);
          if (has_total[k] == 1)
        log_lik_total[k] += normal_lpdf(total[k] | 2 * base_raw[1] + tt[k] + kappa_t_raw[1] * (eff_sum[k] - es_mu
        } else if (has_total[k] == 1) {
          log_lik_total[k] = normal_lpdf(total[k] | 2 * base_raw[1] + tt[k], tot_marg);
        }
      }
    }
  }
}
```

# 13 Appendix: Reproducibility

The pipeline is run with `targets::tar_make()` from the project root, after setting an API key for CollegeFootballData.com; the first full build fits all 226 weekly posteriors and the analyses of this paper and takes several hours on eight cores, and later runs fit only new weeks. `Rscript check_build.R` then verifies the store. This paper is itself a target, rendered by `tar_make()` whenever a target it reads changes; every number and figure is read from the store and nothing is refitted at render time. The analyses beyond the weekly fits (the margin-family comparison, vague-prior refits, leave-one-season-out priors, ablations, baselines, benchmark and Monte Carlo checks, the margin refits and simulation-based calibration) are targets defined in `R/07_analysis.R`, so they are rerun whenever their inputs change.

**Randomness.** Every fit uses a fixed seed (2026 for the published fits and the analyses, $1000 + i$

Table 34: Software versions used for this render.

| Component | Version |
|---|---|
| R | 4.6.1 |
| CmdStan | 2.37.0 |
| targets | 1.12.0 |
| tarchetypes | 0.14.1 |
| cmdstanr | 0.9.0.9000 |
| posterior | 1.7.0 |
| loo | 2.10.1 |
| priorsense | 1.4.0 |
| dplyr | 1.2.1 |
| ggplot2 | 4.0.3 |
| igraph | 2.3.3 |

for the $i$-th simulation-based calibration season, and a seed per step for the replicated data and bootstraps), so a rebuild on the same data and software reproduces every number.

**Software.** R package versions are recorded in `renv.lock` at the project root; Table 34 lists the main ones and CmdStan.

**Data.** The provider revises past data occasionally (late score corrections, reclassified games, new betting lines), so a rebuild months later need not reproduce this paper exactly. Every raw API response is cached as JSON under `data/raw/`, which is the snapshot the paper was built from; Table 35 records, for each season, what the models saw and a hash of it, so a later build can be checked against this one. This build read the data on 2026-10-09.

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Table 35: The data behind this build, by season: games, completed games, games with an efficiency margin and with a closing line, and an xxHash64 hash of the season's data as the models read it.

| Season | Games | Completed | With eff. | With line | Hash |
|---|---|---|---|---|---|
| 2014 | 1,584 | 1,584 | 848 | 865 | `0234befb89987a1e` |
| 2015 | 1,536 | 1,536 | 863 | 832 | `866c714bbf5690e6` |
| 2016 | 1,547 | 1,547 | 857 | 866 | `dc289678caba70c4` |
| 2017 | 1,549 | 1,549 | 869 | 873 | `7bd5a201310168af` |
| 2018 | 1,554 | 1,554 | 881 | 869 | `bcd12f4f0ebaa749` |
| 2019 | 1,621 | 1,621 | 887 | 881 | `83c05fd788c7fcd8` |
| 2020 | 850 | 850 | 568 | 567 | `1aba772c49392adf` |
| 2021 | 1,570 | 1,570 | 887 | 887 | `d70ab3a920d882e7` |
| 2022 | 1,598 | 1,598 | 1,457 | 1,459 | `6eea8f0b578622c2` |
| 2023 | 1,603 | 1,603 | 1,490 | 1,413 | `b74743b4e63ac13d` |
| 2024 | 1,670 | 1,669 | 1,606 | 1,557 | `86e60ae1f2ec7a75` |
| 2025 | 1,687 | 1,687 | 1,658 | 1,597 | `13dc844697aea877` |
| 2026 | 1,609 | 712 | 708 | 802 | `2b73dac80e73866f` |


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